

JOURNAL OF UFO STUDIES

New Series, Vol. 5

1994

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THE J. ALLEN HYNEK CENTER FOR UFO STUDIES

Chicago, Illinois

JOURNAL OF UFO STUDIES

Editor Michael D. Swords
Consulting Editor Mark Rodeghier
Managing Editor George M. Eberhart

SUBSCRIPTIONS The *Journal of UFO Studies* is published annually by the J. Allen Hynek Center for UFO Studies on a subscription basis. Single copies are \$18.00 (subscribers outside the United States add \$3.00 for surface mail).

SUBMISSIONS All manuscripts and editorial correspondence should be addressed to Michael D. Swords, Editor, *Journal of UFO Studies*, Professor of Natural Sciences, Western Michigan University, Kalamazoo, MI 49008, U.S.A. Manuscripts must conform to the appropriate style and format (see instructions on inside back cover). Articles will be refereed by at least two outside reviewers before publication acceptance.

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THE FALCON LAKE CASE: TOO CLOSE AN ENCOUNTER

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ABSTRACT Stefan Michalak was extremely ill after an encounter with a UFO in Canada in 1967. The case was investigated intensely and he was subjected to numerous medical tests. The results of the tests were inconclusive, physicians could find no explanation for Michalak's symptoms. If it was a hoax, it is one of the most contrived on record, involving radiation, ground traces, medical examinations and interrogation by government officials at many levels. The case is still a puzzle to researchers.

Throughout the UFO literature, there are many stories of alleged physiological effects associated with UFO encounters. There are several published reviews of these cases, including Aggen (1969), Crain (1971), Tokarz (1978) and the definitive summary by McCampbell (1987). In addition, reviews of close-encounter cases often include details of medical reactions and effects associated with UFO experiences. The most well-publicized and investigated physiological-effect case in recent ufological history was the Cash-Landrum encounter, in which two women said they were confronted by a diamond-shaped object and suffered various injuries as a result (Schuessler 1984).

One of the most intriguing cases of serious injury attributed to a UFO experience is that of Stefan Michalak*. He returned home from a prospecting expedition with serious ill effects that he claimed were a result of an extraordinary UFO encounter (Rutkowski 1981a).

The incident occurred on May 20, 1967, in an area near Falcon Lake, Manitoba, Canada, approximately 75 kilometers north of the American border in the rocky edge of the great Canadian Shield. Falcon Lake is a resort town at the southern boundary of Whiteshell Provincial Park. The park is largely uninhabited wilderness, about the same size as the state of Rhode Island. The Whiteshell is known for various mineral deposits, and several small mines were established in the surrounding region.

Stefan Michalak was an amateur geologist and had worked the area many times

* Although called Stephen, Steven or Steve in other published accounts, Michalak advises that his proper Christian name is Stefan.



Fig 1 Stefan Michalak standing next to a claim marker in the Falcon Lake area. The claim stake inscription reads 1 VIC, S Michalak, W10053, July 21, 1967, 11 00 A M

Some prospectors had found several quartz veins nearby that were associated with silver deposits, and Michalak had even staked a few claims himself. On May 19, 1967, he traveled from his home in Winnipeg to Falcon Lake, where he spent the night in a motel on the Trans-Canada Highway. He had expected to enjoy a quiet weekend of prospecting.

THE ENCOUNTER

Stefan Michalak left his motel at 5:30 that morning and headed north into the bush. By 9:00 a.m., traveling under a bright, cloudless sky, he had found a quartz

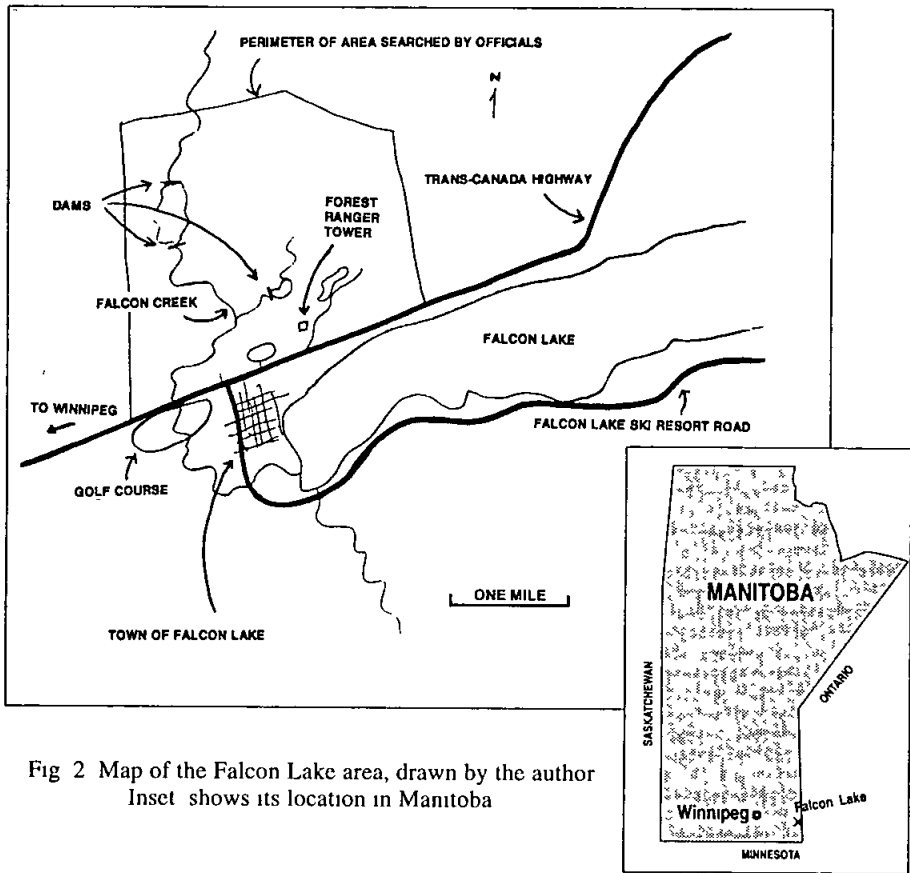


Fig 2 Map of the Falcon Lake area, drawn by the author
Inset shows its location in Manitoba

vein near a marshy area, close to a small stream. At 11 00 a m he had lunch, then went back to his examination of the quartz formation. At 12 15 p m, with the sun high and clouds gathering in the west, Michalak was startled by the cackling of some geese, who were obviously disturbed by something.

He looked up and was surprised to see two cigar-shaped objects with "bumps" on them, an estimated forty-five degrees in altitude, descending and glowing red. As they approached, they appeared more oval and then disc-shaped. Suddenly, the further of the pair stopped in midflight, while the other drew nearer and appeared to land on a large, flat rock which was later determined to be about 160 feet away. The one in the air hovered for a short while, then later departed, changing from red to orange to grey as it flew into the west, where it disappeared behind the clouds.

Focusing his attention to the object on the ground, Michalak saw that it, too, was turning from red to grey, until it finally was the color of "hot stainless steel," surrounded by a golden-hued glow. As he observed the object, he knelt beside a rock where he had been chipping at the quartz. He had been wearing welding goggles to

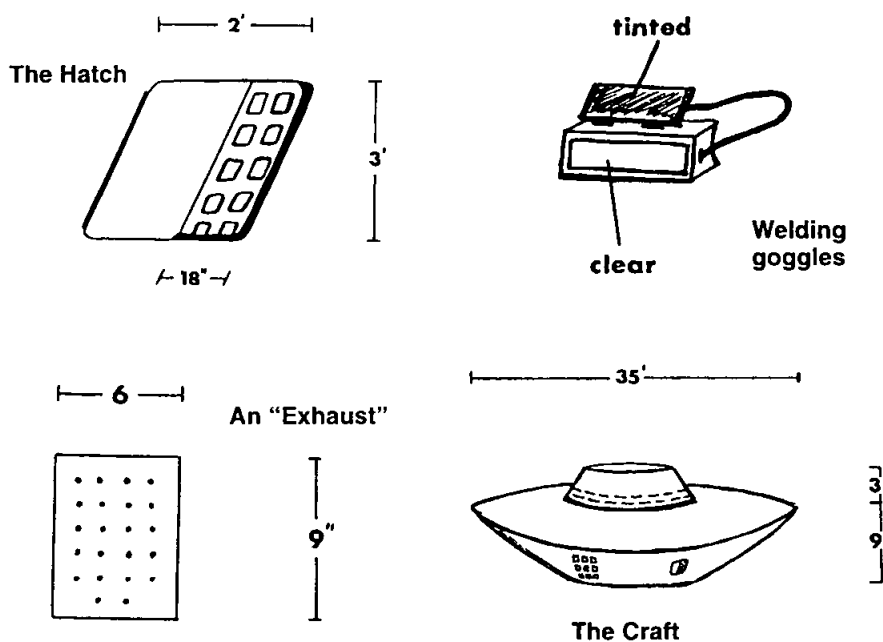


Fig 3 Drawings of various aspects of the Michalak encounter, based on his testimony and sketches

protect his eyes from flying rock fragments. The goggles proved to be very useful, as brilliant light shone from openings in the object, blinding him and creating red afterimages in his eyes. For the next half-hour he stayed near the rock, making a sketch of the object and noting various features. The craft was saucer-shaped, about 40 feet in diameter and approximately 10 feet thick. Its upper cupola or dome was an additional three feet high.

Michalak became aware of waves of warm air radiating from the craft, accompanied by the "smell of sulphur." He also heard the whirring of what sounded like a fast electric motor, and a hissing, as if air were being taken in or expelled.

A door had opened in the side of the craft, revealing some lights inside. This door was about two by three feet in size. Michalak approached to within 60 feet of the craft, and heard two humanlike voices, one with a higher pitch than the other. He was sure that the craft was an American experimental test vehicle, and walked closer to it, sarcastically asking, "Okay, Yankee boys, having trouble? Come on out and we'll see what we can do about it." Getting no response (the voices had subsided), and becoming flustered, he asked cautiously in Russian, "Do you speak Russian?" There was still no answer, so he gave greetings in German, Italian, French and Ukrainian, then once again in English.

At this point, his curiosity got the best of him, and he walked closer to the craft,

ending up directly in front of it. Poking his head into the opening, he saw a maze of lights on what appeared to be a panel, and beams of light in horizontal and diagonal patterns. There was also a cluster of lights flashing in a random sequence "like on a computer."

As Michalak stepped away from the craft, he saw that the wall of the craft was about 18 inches thick. Suddenly, three panels slid over the opening, sealing it "like a camera shutter." He examined the outside of the craft and touched the side of it with his gloved hand. There were no signs of welding or joints, the surface was highly polished, and appeared like colored glass with light reflecting off it, and made "silvery spectra" out of the sunlight. Drawing his glove back, he saw that it had burned and melted when it brushed the side of the object.

Unexpectedly, the craft shifted position, and he was facing a gridlike exhaust vent which he had seen earlier to his left. This vent was about nine inches high by six inches wide, and contained a uniform pattern of round holes, each about $\frac{3}{16}$ inch in diameter. A blast of hot gas shot from these holes onto his chest, setting his shirt and undershirt on fire and causing him severe pain. He tore off his burning garments and threw them to the ground. He looked up in time to see the craft depart like the first, and felt a rush of air as it ascended.

AFTERMATH

When the craft had left, Michalak noticed a strong smell of burning electrical circuits mixed with the original smell of sulphur. Looking down, he saw that some moss had been set on fire by his smouldering shirts, and so he stamped it out. He walked over to where he had left his belongings, and saw that the needle on his compass was spinning erratically, after a few minutes, it became still. He went back over to the landing site and immediately felt nauseous and a surge of pain from a headache.

The landing spot looked as if it had been swept clean (no twigs or stones). However, piled up in a circle 15 feet in diameter was a collection of pine needles, dirt and leaves. As he looked around, his headache became worse, he felt more nauseous and he broke out in a cold sweat. Feeling very weak and dizzy, he vomited. He decided to head back to the motel. On the way back, he vomited several more times and had to stop to regain his strength.

When he finally reached the highway, he was about a mile from where he had entered the woods, so he started off down the road in that direction. He saw an RCMP car coming towards him and tried to flag it down. It passed by, apparently to turn around, because a few minutes later, Michalak heard a voice calling beside him. Michalak told the RCMP officer what had happened. Michalak says the officer listened but recalls the uncooperative constable told him "Sorry, but I have other duties to perform." Then the officer got in his patrol car and left.

After walking for what seemed to be an eternity, Michalak reached the motel.

Thinking he was somehow contaminated, he did not go in but instead remained outside in a clump of trees. He tried to get help at the nearby park headquarters but it was closed. At 4 00 p m , he finally ventured towards the motel and entered the coffee shop to inquire whether or not a doctor was available, as his pain had become considerably worse. He was told that the nearest doctor was in Kenora, Ontario, 45 miles east of Falcon Lake. Not wanting to travel even farther from home, Michalak decided to return to Winnipeg. Thinking that there might be danger for other park visitors and that his encounter was certainly newsworthy, he called the news desk of the *Winnipeg Tribune*. He had expected them to send someone to pick him up and take him a doctor, hearing his story on the way. But as it was a Saturday, the news department was short-staffed and unwilling to comply with his request, especially since he wanted assistance "but no publicity."

He went to his room, where he waited until the next bus to Winnipeg arrived at around 8 45 p m. He called his wife, telling her that he had had an accident and not to worry, but to send their son to meet him at the bus terminal. When he arrived back in Winnipeg around 10 15 p m , his son immediately took him to the Misericordia Hospital.

Throughout the next two years, Michalak was examined by more than one dozen physicians in the United States and Canada. Site investigations were made by members of the RCMP, RCAF, government officials and numerous civilians.

The number of government departments and officials who were involved in this case is staggering. In the ground party which traveled with Michalak to the site there were representatives of the RCAF (Royal Canadian Air Force) Training Command Headquarters, CFB (Canadian Forces Base) Winnipeg, RCMP CID (Criminal Investigations Division), the federal Department of Health and Welfare, and the Manitoba provincial Department of Health and Welfare. In addition, the University of Colorado Condon Committee investigated, *Life* magazine reporters came to Manitoba and two connected but separate civilian groups, APRO (Aerial Phenomena Research Organization) and CAPRO (Canadian APRO), became involved. Furthermore, the federal Department of Mines and Natural Resources took an interest, as did the Whiteshell Nuclear Research Establishment (WNRE), the Manitoba Cancer Institute, the Mayo Clinic, and a host of other medical establishments. The scope of this intense investigation cannot be understated. The Falcon Lake case may well be one of the most intensely investigated well-documented on record.

The case presents a number of elements of particular interest to researchers

- 1 Michalak's burns and other physiological effects,
- 2 the ground traces found at the site,
- 3 radioactivity allegedly associated with the site, and
- 4 mysterious metal fragments found at the site

All of the available data associated with each of these elements will be examined in turn.

CHARACTER OF THE WITNESS

At the time of the incident, Stefan Michalak was an employee of an industrial facility in Winnipeg. He was an industrial mechanic, with knowledge of automotive machinery, welding, and metalwork.

With regard to Michalak's mental state, an examining psychiatrist at the Mayo Clinic noted

I found no evidence of dreams [or] hallucinations. The MMPI was not extraordinary. I can find no overt evidence of significant mental or emotional illness. [Mayo Clinic 1968: 2]

Michalak had never before reported observing anything like the UFO he encountered in 1967. During World War II, he had been an intelligence officer and was very familiar with the appearance and behavior of military vehicles.

In the RCAF report on the incident, much was made of Michalak's association with a man named Gerald Hart (RCAF 1967a). Hart was known to the RCMP as a "subversive" individual, so when Michalak told them that Hart had assisted him in his quest for the site of his encounter, officials became suspicious. (Among other eccentricities, Hart refused to pay income tax and in fact wrote a popular book on how to avoid paying the government anything.)

Furthermore, the actions of civilian UFO investigators were cause for concern in the minds of officials. In particular, Barry Thompson, described by a former CAPRO member as a "liaison between CAPRO and APRO," was

a constant companion of Mr. Michalak and he appeared to be the spokesman for Mr. Michalak during some of the interviews. Both the investigating officer and Professor Craig [of the Condon Committee] agreed there appeared to be monetary gain intentions associated with this relationship. [RCAF 1967a: 4]

However, there was never any "monetary gain" from the incident. Michalak's own narrative account was privately published in late 1967. His manuscript, written in Polish, was translated and printed as a 40-page booklet which quickly sold out. But Michalak saw little of the money recovered after publication costs, and to this day is bitter that "others have made money from my experience, but not me" (Michalak 1980). Actually, because of the small run and limited circulation, it is thought that the publisher lost money. (In the early 1970s a civilian investigator of the case wanted to have the manuscript retranslated and the booklet reprinted, but this never occurred.)

Squadron Leader Paul Bissky of the Royal Canadian Air Force was the investigating officer on the case. His reports to Canadian Forces Headquarters are remarkable in their detail and candid comments about his investigations. But complicating his

reports is the fact that Bissky was a devout skeptic, and told researchers he "didn't believe in that stuff [UFOs]" (Bissky 1980) How much his personal opinions may have influenced his reports is not known

That Bissky thought Michalak was a liar is an understatement At one point during his investigation, he bluntly asked Michalak if he had been drinking on the day of the experience He believed that Michalak was hallucinating because of some alcoholic stupor Bissky appeared sure that alcohol was somehow involved with the case, and he cleverly manipulated Michalak into proving he was not a teetotaler In his first report, Bissky noted

Although the authenticity of Mr Michalak's report had not been questioned up to the second search, he had appeared genuinely sincere and his story was convincing to those who heard it for the first time, some doubts arose during the last search Mr Michalak had staunchly denied having consumed alcoholic beverages at any time while at Falcon Lake Yet a reliable witness at Falcon Lake advised that he personally had served Mr Michalak 4 or 5 bottles of beer the night prior to the trip into the bush [RCAF 1967a 2]

This is an odd note, since the "reliable witness" was obviously a bartender, and Bissky did not state how his source's own reliability had been established Bissky seemed to want to prove there was at least one inconsistency or lie in Michalak's testimony, and the issue of alcohol consumption seemed to be a choice target (In retrospect, even if Michalak admitted drinking several beers, there would still remain the problem of the other physical and physiological evidence, Michalak's drinking probably had no bearing on the case itself)

In his later report, Bissky noted several "discrepancies" in Michalak's story, including this following detailed passage which gives us some insight into Bissky's RCAF investigation

it was proven that Mr Michalak does consume alcoholic beverages, in fact to some considerable extent After coming out from the alleged site, Mr Michalak was purposely taken to the same bar where he had consumed the five beers the night prior to his initial encounter on the 20 May 67

Although he declined the offer of beer (it could not be determined whether this was done because he did not like beer or not), he did accept the offer of several rye "Presbyterians" (rye and half water/ginger ale mix), in fact he even went so far as to purchase a round of drinks for the group He appeared to hold his alcohol reasonably well, although it was noticed that he did loosen up after the third drink and become quite gay, telling numerous stories When questioned about

the vast discrepancy in direction the object departed, he just shrugged his shoulders and laughed it off. When last seen, he was in a jovial mood, remaining at the bar, presumably to await the arrival of his two assistants from Winnipeg. Hence it is very possible that Mr. Michalak may have had a private party on his own the night of 19 May, which in turn could have caused hallucinations the following day [RCAF 1967b 4]

Bissky was convinced that Michalak had drunk heavily the night before his alleged experience, causing him to have imagined the entire UFO encounter ten to twelve hours later. However, Bissky described an alternative theory to the author: Michalak had been drinking and burned himself on a hot barbecue grill. Support for this new theory came from another of Bissky's unnamed but "reliable sources," in this case a woman who was awakened by Michalak pounding on her cabin door at 2:00 a.m. Unfortunately, Bissky said that the woman was at Falcon Lake for a "tryst" and could not come forward publicly. It is interesting that this explanation was never mentioned in any official report, and has no other evidence to support it (Bissky 1980).

Despite all of his attempts to find flaws in Michalak's story, Bissky was forced to concede that

there are certain facts, such as Mr. Michalak's illness and burns and the very evident circle remaining at the site, which are unexplainable [RCAF 1967b. 5]

Even under the intense scrutiny of biased military investigators, the case appeared sound.

As of the time of this writing (January 1994), Michalak still stands by his original story and insists that his experience occurred as he described.

PHYSIOLOGICAL EFFECTS

When asked by the examining doctor how he had been injured, Michalak said he had been "hit by exhaust coming out of an aeroplane." In the narrative of his account, Michalak explained that he had not told the doctor about the UFO because neither he nor the (Chinese) physician were fluent enough in English to make the fantastic story understood, and because Michalak was very tired and wanted to go home. He was given a sedative and went home, where he took a bath to cleanse his wounds then went to bed (Michalak 1967: 23).

The next morning, Michalak was still in some pain and his family noted he had extremely bad body odor and halitosis. He could not hold any food down, but he was not hungry anyway. He said that according to the bathroom scale he had lost six

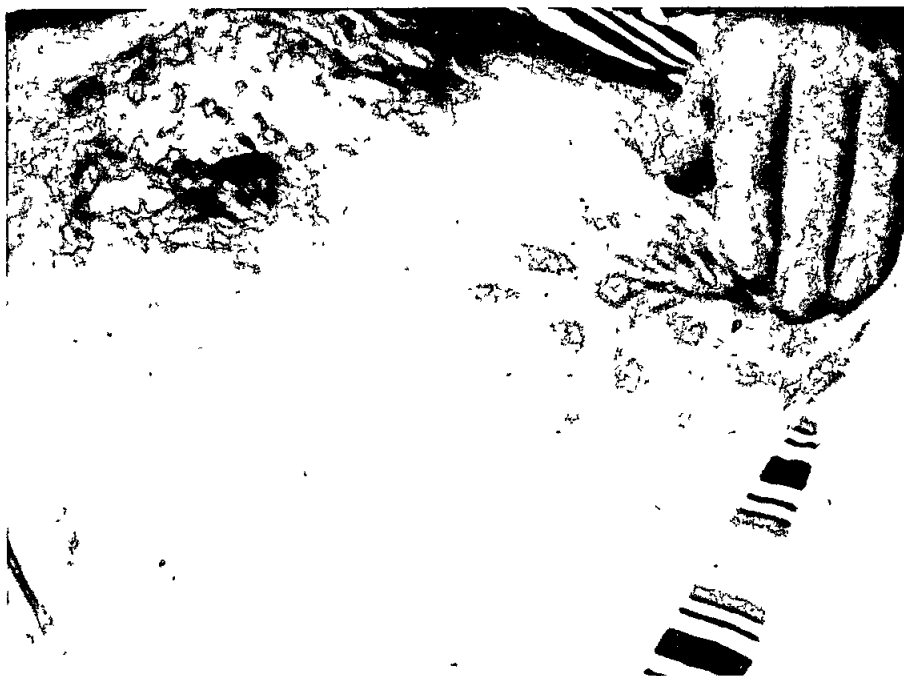


Fig 4 Michalak's chest and abdomen, showing his charred upper body and the "checkers" pattern of red chemical burns on his stomach
The photo was taken a few days after his encounter

pounds during the previous two days, and he became concerned

It was not until that evening that his own physician, Dr R D Oatway, examined Michalak and was told the saucer story Michalak noted "He looked at me with what one may call a professional discretion" (Michalak 1967 24)

Oatway's detailed report, prepared for APRO consultant Dr Horace Dudley, describes Michalak's physical condition at that time

He complained of band-like headache, hot forehead, anorexia and nausea, feeling of blacking out On examination, he appeared rather depressed, dazed, apathetic, but rational and coherent There was singeing of the hair on the forehead at the hairline and over the lower sternal and upper abdominal region Over the upper abdomen, in the mid-portion and especially to the left of the midline, there were numerous reddish, slightly irregular, oval-shaped, slightly raised lesions, arranged with their long axes mainly in a transverse direction These lesions seemed to be consistent with a first degree burn As I recall they were painful and tender but not severely I also observed the burnt under-

shirt which had holes with charred (or blackened) edges corresponding to the site of the burn [Oatway 1968]

Oatway examined the burns and prescribed antinausea tablets and codeine painkillers. Later, he referred Michalak to a dermatologist who gave him some antibacterial skin cleanser for the burns.

During the next two weeks, Michalak's condition improved gradually. He kept a diary of his health during this time. He noted that his weight decreased from 180 lbs before his UFO encounter to a low of 158 lbs on May 27, one week later. Unfortunately, since Michalak had not seen his personal physician for more than one year before his UFO encounter, there was no official record of his preencounter weight. During this time, he also experienced several fainting spells, which he had never had before in his life. He continued to vomit occasionally, but his appetite slowly returned to normal.

As a result of prompting by civilian UFO investigators, Michalak went to a radiologist on May 23. No evidence of radiation trauma was found. On May 30, Michalak was taken by a UFO investigator to the Whiteshell Nuclear Research Establishment, where he was given a whole-body count. Again, nothing above normal background readings was found.

During the period immediately following his encounter, Michalak had a slight drop in blood lymphocyte count, from 25% to 16%. As noted by one investigator, the specific values and corresponding times were

May 24, 1967	16%	
May 30, 1967	21%	
January 15, 1968	31%	[Cannon 1970].

After four weeks, the white-cell count was reportedly back to a normal level. During this time the platelet counts were consistently normal. If Michalak had been affected by radiation, as some have suggested, the counts would have varied more significantly. Brian Cannon, a founder of CAPRO, reported to that group's membership that the healing of Michalak's burns was "a characteristic trait of radiation burns" (Cannon 1968). One hematologist's report, however, indicated "no abnormal physical findings," although Michalak had "some atypical lymphoid cells in the marrow plus a moderate increase in the number of plasma cells" (Oatway 1968). These minor variations do not support some published accounts that claim Michalak had impurities in his blood (Naud 1978).

But Dr. Horace Dudley, a radiologist and APRO advisor at the University of Southern Mississippi, observed that Michalak's

nausea and vomiting followed by diarrhea and loss of weight is a classical picture of severe whole body [exposure to] radiation with x- or gamma rays. I would guess that Mr. Michalak received on the order

of 100–200 roentgens It is very fortunate that this dose of radiation only lasted a very short time or he would certainly have received a lethal dose [Lorenzen and Lorenzen 1968 40–41]

Others did not believe that symptoms of radiation poisoning were present, and the issue has never been fully resolved (Michalak 1967 27–28, Rutkowski 1981b)

Michalak's skin problems also had different interpretations His upper chest, having been diagnosed as thermally burned, healed fairly rapidly His abdomen, where the grid pattern appeared, went through periods of fading and recurrence It had been suggested that these welts were radiation burns

Michalak also had a rash which broke out on his upper torso One investigator said this was due to insect bites, and this is supported by the fact that investigators were indeed bitten by large numbers of black flies at the site However, it does appear that Michalak had more than just a simple patch of bites Medical records noted he had skin infections that were "hive-like areas with impetiginous centers" (Oatway 1968) In another report, he had "generalized urticaria" (Oatway 1967) Along with the recurrence of the rash and urticaria, Michalak reported feeling weak, dizzy and nauseous, and he experienced numbness and swelling in his joints (Michalak 1967 35–36)

It is possible that Michalak had an allergic reaction to something in the environment On September 21, 1967, four months after his UFO experience, Michalak was at work when he became very ill He felt a burning sensation on his chest and neck, his throat constricted and he became very flushed ("turned violet") His hands swelled "like a balloon," he became dizzy, then fainted (Michalak 1967 35) Upon examination, doctors concluded that Michalak had had an allergic reaction of some sort However, considering that he had never had such reactions before his encounter, one might wonder what had triggered the episodes

Michalak gradually recovered from his injuries and stopped having his recurring allergic reactions To this day, however, the strange array of burn scars can still be felt underneath the skin of his lower abdomen

THE MAYO CLINIC

In August 1968, Michalak went to the Mayo Clinic in Rochester, Minnesota The purpose of his visit was to undergo tests in order to determine exactly what was ailing him, since the doctors in Winnipeg appeared to be unhelpful It is worthy to note that Michalak paid for the Mayo tests entirely on his own, as Canadian medical insurance would not cover such a trip

He traveled south and stayed for two weeks at a motel near the hospital, walking across each day and entering as an outpatient He reported that he was given a thorough physical and psychological examination by various doctors, then sent home

Michalak waited for several weeks, but received no word on his results He com-

plained to his own doctor and told CAPRO representatives of the lack of results CAPRO investigators appealed to APRO for help Eventually, ufologist John Keel intervened and as a result, Dr Berthold Schwarz, a psychiatrist and ufologist, assisted by sending a letter of inquiry to the Mayo Clinic, asking about the medical reports In reply, he was sent what has been referred to as the "letter of denial " Dated (perhaps significantly) April 1, 1969, the letter bluntly stated

I have checked through our registration desk and I find that we have never had a patient by that name registered at the Mayo Clinic If he had been a patient I suspect that I would not have been able to send you information without a release from him, state laws being what they are, but I can tell you we don't know anything about him [Barry 1969]

This immediately spurred shouts of "cover-up!" from some individuals who learned of the letter, and rightly so, as this appeared to be a deliberate attempt to mislead the investigation However, Schwarz tried again, with a different tactic, he asked Michalak to sign a simple medical records release form and forwarded that to the Mayo Clinic in January 1970 The reports came immediately

Michalak had been found to be in good health but with neurodermatitis and simple syncope (fainting spells due to sudden cerebral blood pressure losses) The syncope was suggested as having to do with hyperventilation or impaired cardiac output (Rovelstad 1970) This is interesting, as Michalak has indeed had heart problems during the past decade

Furthermore, the Mayo Clinic report described Michalak's physiological problems in more detail

Since May, 1967, he has had repeated clearing-up and recrudescence of the erythematous and pruritic lesions on his chest and legs Sometimes these occurred at intervals of approximately 112 days, but this has not been consistent Since January, 1968, reoccurrences have been more frequent but the symptoms were briefer in duration Generally, the chest lesions appear as minute points or "grains," enlarge progressively to the size of a quarter or a half dollar, and are very pruritic The time between initial appearance and disappearance has ranged from a few days to several weeks Various medications have not been helpful [Mayo Clinic 1968 1]

His condition was viewed as being quite serious, especially given the following information

Mr Mechallack's [sic] main reason for coming to the Mayo Clinic now is because of headaches and "blackout spells" which have attended

the other symptoms since he was severely ill in January, 1968. Headaches are mainly bitemporal, steady, and excruciating. Skin problems occur at the same time. Blackout spells are not sudden but cannot be predicted accurately enough to permit him to drive during symptomatic periods (he is fearful of hurting himself and/or others). Gradually, his eyesight begins to dim until everything goes black. He has time to sit down but is unconscious for a few minutes or more. Allegedly, his wife has viewed him during these spells and he recounts no symptoms suggestive of seizures. He declares that he is unable to hear during the spells. [Mayo Clinic 1968: 1]

RADIATION

On June 30, 1967, Michalak traveled to Falcon Lake with Gerald Hart. Hart had offered to help in the search, and told Michalak he visited the area frequently on his own. Michalak took him up on his offer.

He not only took me there but helped me in the search. I looked throughout the woods checking rocks I had seen before, examining chips I had made in the stone and finally, after six hours, we came upon the spot. [Michalak 1967: 31]

They found a ring of debris, thought to have been made when the object lifted off. In addition, they found bits of Michalak's shirt and his tape measure which he had left behind.

When they returned to Winnipeg, they informed the RCAF of their findings. On July 2, RCMP, RCAF and CAPRO investigators accompanied Michalak to the site, gathering samples and taking photographs. The RCMP analyses of the samples showed significantly high radiation readings. On their recommendation, consideration was given to cordoning off the area due to a possible health hazard. This was noted in an *Incident Report* from an inspector with the RCMP Crime Laboratory in Ottawa, the result of tests on some samples sent by the RCMP to the Department of National Health and Welfare. They found

a radiation value of 3 microcuries in the soil sample. The radiation is from a radium source and is a possible serious health hazard. [RCMP 1967]

Samples taken from the site by Michalak and Hart were eventually tested by the Radiation Protection Division of the Canadian Department of National Health and Welfare. They examined samples of "soil, burned shirt and steel tape for possible radioactive contamination." The initial gamma analysis showed significant levels

of "Ra 226 or its equivalent "

In a report prepared by Stewart Hunt for A K DasGupta, the head of the Safety Assessment and Control Section of the Division, details of the radioactivity assessment were described. Hunt outlined very clearly and cautiously the reasons his Division was involved

The task of assessing the radiation hazard to the general public and the landing site did not initially indicate that it would be necessary for the Division to involve itself in the RCMP and DND investigation. However, it became obvious that collection and coordination of all background information was essential due to the following

- a The apparent lack of cooperation by the principals toward the military and police
- b The exact location of the sighting had not been established at the time of the Division's initial involvement [Hunt 1967: 1]

Hunt flew to Winnipeg and met with various other officials on July 26, 1967. They had a brainstorming session, during which they arrived at a complicated plan of action that included a thorough investigation of the site, analyses of the physical effects and a detailed background check not only of the principal witness but also of the civilian investigators.

As part of their investigation, the team traveled to East Braintree, Manitoba, near the Whiteshell, where hazardous waste from the Manitoba Cancer Institute was buried. The reason for this was the suggestion that someone had "seeded" the landing site with commercially produced radium. If it had not come from a radium watch dial, they reasoned that it must have come from a nuclear waste disposal site. Again, the seriousness with which the case was regarded is evident. Tampering with a nuclear waste disposal site is a very grave matter. No indication of tampering was found, however. The team also visited Michalak's place of employment to see if it used radium in any of its products.

They then visited Michalak¹ at his home to check for radiation in the samples he had in his basement. Hunt noted

A survey of the basement showed no evidence of any loose contamination. Radiation fields were detectable coming from the plastic bags containing soil and the remnants of Michalak's burned shirt.

Mr Michalak has no respect for contaminated materials. He handled the contaminated soil with his bare hands, and was made to wash his hands with great difficulty [Hunt 1967: 2].

Hunt was both horrified and suspicious. How could an amateur geologist and trained engineer not be concerned by radioactive debris?

Hunt's visit was what helped convince Michalak to visit the site with Bissky and

the others. Hunt went along to verify that radioactivity was present. He found that

One small area was found to be contaminated. This was located across the crown of the rock. There was a smear of contamination about 0.5×8.0 inches on one side of the crack. There was also some lichen and ground vegetation contaminated just beyond the smear. The whole contaminated area was no larger than 100 square inches. All water run off areas were checked for possible contamination, but nothing was found [Hunt 1967: 2].

After examining the site, he felt there was "no serious health hazard involved."

The fact that only a small area was radioactive conflicted with the fact that Michalak's steel tape measure was radioactive, yet it was found "40 paces" from the site. The explanation offered for this was that everything taken from the site had been left together in a pile in Michalak's basement, so that unaffected items could have become contaminated after the fact.

In an undated Department of National Defence *Minute Sheet*, an official in the office of the Chief of Defence Staff in Ottawa noted

There is some doubt that the soil samples did in fact contain "226" or pure radium. This question of doubt is a scientific evaluation beyond this investigation.

The quantity of "226," namely 0.5, is equal to approximately $\frac{1}{3}$ of that associated with an average wrist watch.

However, the quantity, although pertinent, does not explain how this "smear" got on the rock at the alleged landing site. This is what is bothering the scientific people [Canadian Department of National Defence 1967?].

Perhaps the most interesting aspect of the "radium seeding" scenario was Hunt's visit to the provincial Environmental Sanitation Laboratory in Winnipeg.

Samples from [the] UFO site and those taken from Michalak's residence were checked under a UV light. The samples taken from the UFO site gave an indication that they were contaminated with Radium luminous paint. The samples from Michalak's house did not respond to the UV light [Hunt 1967: 3].

This was puzzling. If Michalak or Hart had seeded the radium, then they would surely have had some luminous paint in their own radioactive samples. Yet this was present only in the later samples.

Further confusing details were found when Hunt visited the home of Barry Thompson, the APRO investigator. Hunt checked some soil and vegetation samples

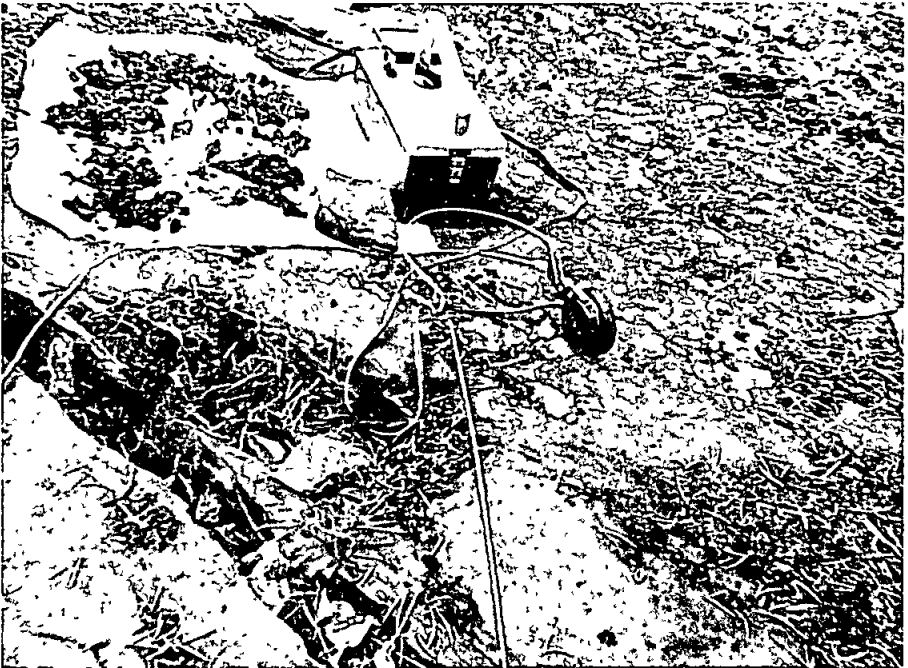


Fig 5 The appearance of the rock slab over which the UFO was said to descend The crack in the foreground is where chunks of metal were dug out by Michalak in May 1968 The geiger counter was used to check for radioactivity at the site

Thompson had in his possession Thompson had been given the samples by Michalak when he accompanied him during a separate visit to the site on July 17, 1967 Hunt noted that one sample

proved to be radioactive Levels up to 1mR/hr were detected The sample was sealed in a plastic bag A contamination check was made of the area where the samples were, using the UV light The area was extremely cluttered with photographic equipment and a great deal of junk Several areas responded to the UV light, but these did not prove to be areas of contamination, probably photographic emulsion splashes Thompson appears to be a very sloppy worker [Hunt 1967 5]

This raises the possibility that "emulsion splashes" were also the cause of the luminosity found in the RCAF samples from the landing site

Hunt learned that Thompson had given some samples to George Dyck, a technician at the Nuclear Medicine Department of the Winnipeg General Hospital On his

own time, Dyck had tested the samples using his department's standard laboratory equipment. One sample was said to have shown a "1.4 MeV peak" and two other weaker peaks. Hunt visited Dyck and was introduced to nuclear medicine specialist Dr. F. Helmuth, who had examined Michalak's burns when he had been brought in following his other medical tests. Hunt also interviewed other nuclear medicine specialists at the hospital. Dr. R. Walton, executive director of the Manitoba Cancer Clinic, was apparently embarrassed to have his organization involved in such tests because "they weren't particularly interested in becoming involved in work of this nature." The stigma of UFO investigation was too much for the medical establishment.

THE METAL

Between July 1967 and May 1968, the landing site was visited by a variety of individuals. One of these was Mr. E. J. Epp, who searched the area for radioactivity as part of a check by the provincial Department of Mines and Natural Resources. They were concerned that Michalak had misdirected the earlier searchers to protect his claims. However, not only did Epp not find any radioactivity, but Michalak didn't file any claims until the fall of 1967.

The lack of radioactivity at the time is important, because on May 19, 1968, Michalak again visited the site with a friend. In his report to the Condon Committee, Roy Craig said Michalak found

... massive pieces of radioactive material in a fissure of the rock within the "landing circle." This ... consisted of two W-shaped bars of metal, each about 4.5 in. long, and several smaller pieces of irregular shape. These items were said to have been found about 2 in. below a layer of lichen in the rock fissure. The two fragments each consisted of a central massive metal portion which was not radioactive. One of these was 93% and the other 96% silver. Both contained copper and cadmium, and had a composition similar to that found in commercially available sterling silver or sheet silver. The metal was coated with a tightly-adhering layer of quartz sand, similar to that used as a foundry sand. This also was not radioactive. The radioactivity was contained in a loosely-adhering layer of fine-grained minerals containing uranium. This layer could be removed readily by washing and brushing. The minerals were uranophane and thorium-free pitchblende, characteristically found in vein deposits. [Condon 1969: 323]

In his own teletype to headquarters, Bissky said that when he examined the metal pieces at Michalak's home

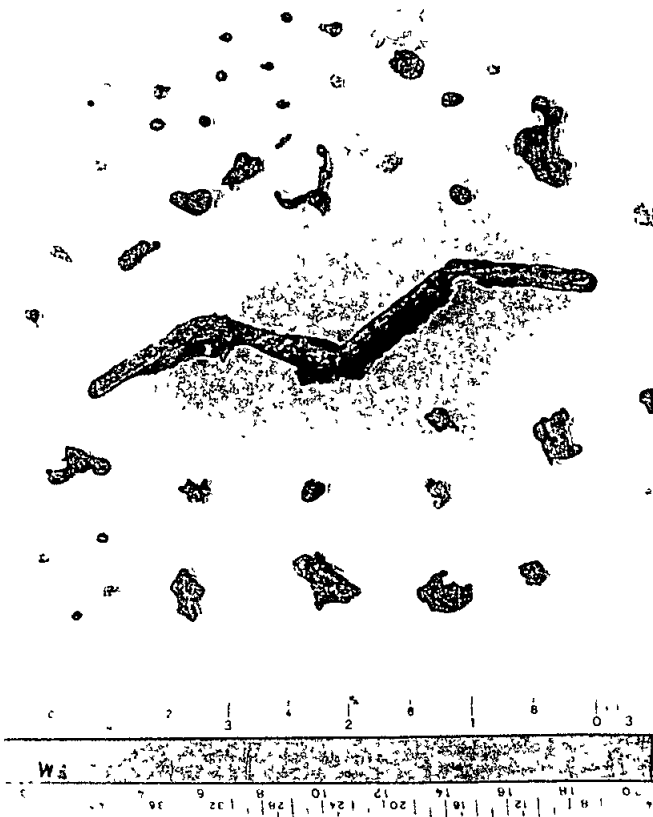


Fig 6 Metal samples retrieved from the crack in the rock

All shown objects were subjected to civilian Geiger counter and majority of readings at same level as that of the dial face of a service wrist watch in same counter [Bissky 1968]

Again, there was speculation that radium was implicated in the radioactivity of the metal. It is ironic that it came from Bissky's own watch.

Bissky also observed that

Larger objects appeared to have been cast for a specific design and Mr M indicated he had considerably more in his possession but would not allow viewing or indicate the exact numbers although he did allow that some were in a reverse "S" design.

He then

attempted knicking one of the larger pieces with a knife and found metal very resistant to knife although there was no normal metallic ring when struck by another metallic object or against a stone. [Bissky 1968]

This last observation is curious, since a four-inch chunk of dense metal would not necessarily ring when struck

Bissky's concern that the matter still was very suspicious was obvious as he concluded

should it be found metal is of unusual raw material, feel that NRC should be brought into picture for expert investigation. It may be that metals are normal large deposits of silver or other composite materials found in this area

And, most importantly

it is interesting that items have been located at exact point of alleged landing of UFO. Should this be a hoax, someone is going to considerable effort to perpetrate same [Bissky 1968]

An understatement, indeed!

A number of institutions performed analyses on the metal pieces. Biospace Associates apparently had some samples tested through Colorado State University. They noted that

This particular UFOlogy is made of silver, with no metallic impurities detectable by the x-ray fluorescence analysis [Kachur 1968]

In a note from the UFO Research Institute, located in Pittsburgh, Dr. J. Roesner reported that

The gamma spectra were complex, 15 distinct energies ranging from 0.11 MeV to 2.57 MeV could be resolved. The three major contributors to the total gamma radioactivity had energies of 0.61 MeV, 1.10 MeV and 1.53 MeV and decayed with half-lives of ~14 days, 8 days and 21 days, respectively. A semiquantitative chemical analysis showed that 95 percent of the specimen is silver. The amount of copper in the specimen was determined to be 0.5 percent. The energies and half-lives of the gamma rays emitted by the specimen do not agree with the expected decay of silver activation products formed in an (n, γ) reaction on natural silver [Weitzel 1968]

In his report to the Condon Committee, Craig quoted the conclusion of R J Traill, head of the mineralogy branch of the National Research Council of Canada, who reviewed the WNRE findings

I would interpret the specimen as pieces of thin sheet silver that have been twisted, crumpled, partly melted and dropped into or otherwise placed in contact with nearly pure quartz sand while still hot They have subsequently been covered with loosely adhering radioactive material which consists of crushed pitchblende ore, much altered to uranophane and containing associated hematite [Traill 1968 2]

Craig further noted

In view of the thoroughness of earlier searches of the site for radioactive material, it is improbable that the particles discovered a year later would have been missed had they been present when the earlier searches were made [Condon 1969 323]

However, CAPRO insisted otherwise

The fact is that there is overwhelming evidence that the metal was there, at least since July 1967 When the metal was located, and since it was taken from the middle of the rock, we examined the soil samples removed from the sight [sic] in July 1967 These samples also contained tiny fragments of the same metal and no doubt the samples taken by the authorities contain pieces as well [CAPRO 1969 6]

Brian Cannon, a civilian investigator for CAPRO, was rightly concerned that the metal samples seemed to undermine the credibility of the case It looked as if the metal bars were unrelated to Michalak's experience To check this, CAPRO had the metal bars tested for the presence of radium According to their report

Analyses confirmed the presence of Radium 226, the same source as was found in the soil specimens The luminous watch dial paint theory dulled considerably [Cannon 1969c]

This latter point is particularly disturbing Could experienced nuclear technicians have made such a mistake? The matter becomes more curious when one considers the results of reanalyses by the Ufology Research of Manitoba (UFOROM) during 1977 to 1983 Soil samples allegedly from the Falcon Lake site were provided by a former CAPRO representative and tested for UFOROM at the University of Manitoba The samples showed natural uranium activity but no radium signatures This suggested that earlier indications of the presence of radium were in error

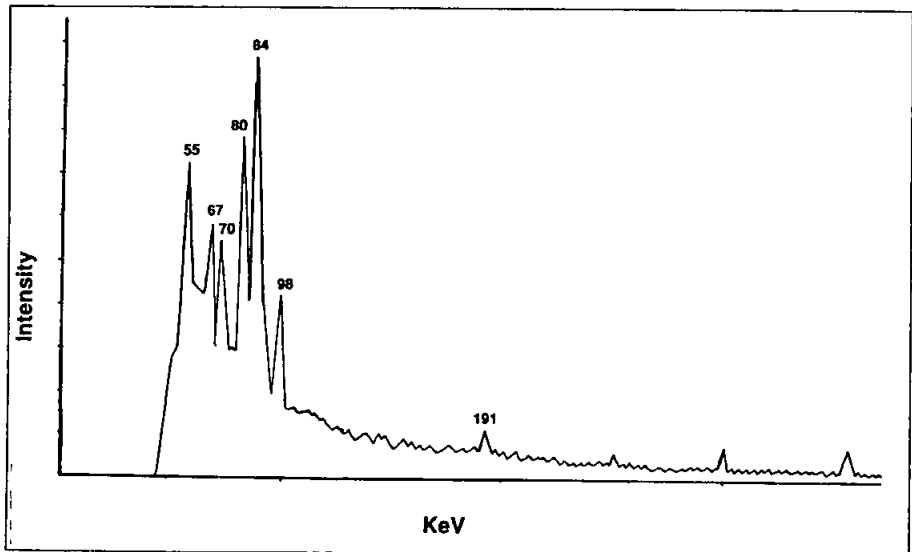


Fig 7 Results of a 1980 analysis of soil samples from the Michalak site. The samples were obtained in 1967. The graph shows identified peaks of radioactivity that can be shown to be indicative of natural uranium.

In an internal Whiteshell Nuclear Research Establishment memorandum, lab analyst J.D. Chien reported on the analyses of "chared [sic] fabric," native silver, mineral fragments, twig fragments and jack pine needles. He wrote that

The gamma spectra showed an abnormally large 190 KeV photopeak which was thought to be due to enrichment of U 235. However, chemical extraction for uranium of the 100 mesh sample and subsequent mass spectrometric analysis indicated the samples containing uranium were of natural U 235 content. Experiments are being continued to identify the cause of the large 190 KeV photopeak. [Chien 1968]

A further analysis of a soil sample was done in 1994 by UFOROM associate Greg Kennedy of Montreal. Gamma-ray spectroscopy found four radionuclides: U235, Pb214, Bi214 and Cs137. The cesium was probably due to fallout from nuclear weapons tests. Again, no enriched uranium was found, and no metal particles. The soil was simply naturally high in uranium, a typical finding in the Whiteshell region.

The original soil samples retrieved from the site contained only natural radioactivity. However, radium 226 was detected by some investigators. It is not clear whether or not this was an error. The metal samples, on the other hand, are definitely mysterious and do not appear natural. To solve the puzzle, it would be most useful to obtain a small metal sample for reanalysis.

SITE INVESTIGATION

The most scientific report on the case was published by the infamous Condon Committee Dr Roy Craig and Mary Lou Armstrong of the University of Colorado both visited Michalak in June 1967 Accompanying them was John Fried of *Life* magazine, which wanted to do a feature on the incident Unfortunately, when Michalak tried to lead them to the site on June 4th, he was unsuccessful As he explained

Nature changes quickly at that time of the year and the forest was very much different . . . than it had been when I was there before The same thing happened later when the members of the RCAF and the RCMP went with me to search for the spot

We even used a helicopter, but it was no use It seemed as if the place had disappeared from the face of the earth [Michalak 1967 29–30]

Michalak noted that the investigators were “disenchanted” with his inability to find the site Who could blame them? Their first visit was only two weeks after the incident, and already the story had a serious flaw (It is probably because of this that the case was not regarded very highly by the Condon Committee and the USAF Indeed, if *Life* magazine had found the site, the story would have attracted much more attention and might have been more carefully documented)

Access to Information requests by several ufologists (including this writer) have uncovered documents which provide more insight into the incident but also raise many more questions In some cases, there are outright contradictions between civilian and government or military records of the investigations

The official RCAF report is undated but is thought to have been submitted in mid-June 1967 by Squadron Leader Paul Bissky Bissky noted he first led a search party to Falcon Lake on May 25, but could not find the site On May 30, he visited Michalak to see if he felt well enough to accompany an investigation team Michalak declined, but he drew a sketch of the area, described the site in detail and identified the likely area on an aerial photograph On May 31, four RCMP, two RCAF and one other set of investigators in an H112 helicopter searched for the site, with no success

In his report, Bissky wrote

The RCMP returned to Winnipeg that evening and persuaded Mr Michalak to accompany them to Falcon Lake the next day Thinking he may be able to quickly locate the site from the air, Mr Michalak was taken over the area by helicopter but he found no recognizable features He stated he could probably do better on the ground The search then proceeded with him leading the ground party, with the helicopter monitoring the proceedings from the air With the aid of RCMP

portable radios, an air/ground link was possible and this greatly assisted in directing the ground party to the most likely-looking areas as described by Mr Michalak. Following a frustrating afternoon and evening search, Mr Michalak insisted the ground party had been very very close to the sought-after location as he recognized several physical features and areas where he had chipped rocks during his last prospecting visit [RCAF 1967a 2]

It should be remembered that this was a joint investigation by military and police officers, armed with sophisticated rescue and detection equipment. Their search on June 2 was also unsuccessful.

The RCAF team did not return to the area until July 28, when they had convinced Michalak to lead them to the site that he and Gerald Hart had found on June 25. In his supplemental report, dated September 1, 1967, Bissky wrote

Following an informal report from the RCMP to the effect that the samples submitted to CFHQ for analysis had proven to contain considerable radioactivity, and that a representative from the Dept of National Health and Welfare was being sent out to investigate further, contact was made with all those concerned and a decision was made to

- a Interrogate [sic] Mr Michalak again with the view to ascertaining the possibilities of his home and himself being contaminated by the materials which he brought out as samples, and
- b Investigate the alleged landing site of the object, since Mr Michalak had now been to the sight [sic] twice since his initial encounter [Author's note with Hart and Thompson]

Although Mr Michalak had previously been most uncooperative regarding taking either the military or RCMP into the site, once the possible gravity and implications were made clear to him, he volunteered to lead a party into the area [RCAF 1967b 2]

In other words, it was Bissky's opinion that Michalak did not want to take the officials to the site until they convinced him that radioactive contamination was a serious possibility. Michalak's distrust of officials and hesitancy was noted in Bissky's earlier report as well.

When asked to provide the location of the site, Mr Michalak objected very strongly on the basis that during his 25 June hunt he had in fact located what he had searched for originally and until such time as he could stake his claim, he had no intention of having anyone go near this area. It was pointed out to him that there was no intention of jumping his claim and that it was in the interest of the Canadian Public that he show the site to S/L Bissky. Mr Michalak stated that no matter what

anyone thought of him, he would not cooperate until his claim was filed. It turns out, however, that he now has to share his claim with his new-found partner, Mr Hart, whom he had never met before. This was particularly surprising since he was very noticeably suspicious of the RCMP during the visit, so much so that he did not invite the RCMP into the basement to examine some [of] the latest samples brought back from the latest "hot find" [RCAF 1967a Addendum]

However, when they finally did convince Michalak to lead them to the site on July 28, they were baffled by its appearance and location

Mr Michalak successfully led the party to the sight [sic], in approximately 45 minutes going through considerable heavy bush, brush and undergrowth. The site was within 50 yards from where the ground party had searched on 1 June 67, but was not visible at that time because of the heavy bush located between the two areas [RCAF 1967b 2]

In other words, there was a reasonable explanation why the site had not been found by earlier searchers. This seemed to quell some doubts based solely on the inability to locate the site. They found

the outline of an approximate 15 foot diameter circle on the rock surface where the moss and earth covering has been cleared to the rock surface by a force such as made by air at very high velocity [RCAF 1967b 2]

The investigators took samples of the dirt, rock and vegetation and also examined trees within a few yards of the circle. This latter point was because there was some concern that the alleged size of the UFO (35 to 40 feet in diameter) would infringe upon several trees and saplings. There was no indication of burns or other "disturbances." This in itself was curious, since Michalak said he brushed against the craft, burning his glove. It seems logical that, at the very least, some leaves or limbs would have been seared or broken.

Considering the size of the alleged object (35-40 ft across), it is highly unlikely that it could have made a descent into and ascent from the area searched without having made contact with some of the surrounding trees. Even if it was not completely heated on the surface, there should have been some evidence of having brushed a tree or some bark removed. Yet there was no indication at all [RCAF 1967b 3]

One possible explanation is that the burned leaves fell and decayed during the two months since the event. None of the investigators were plant pathologists, so some

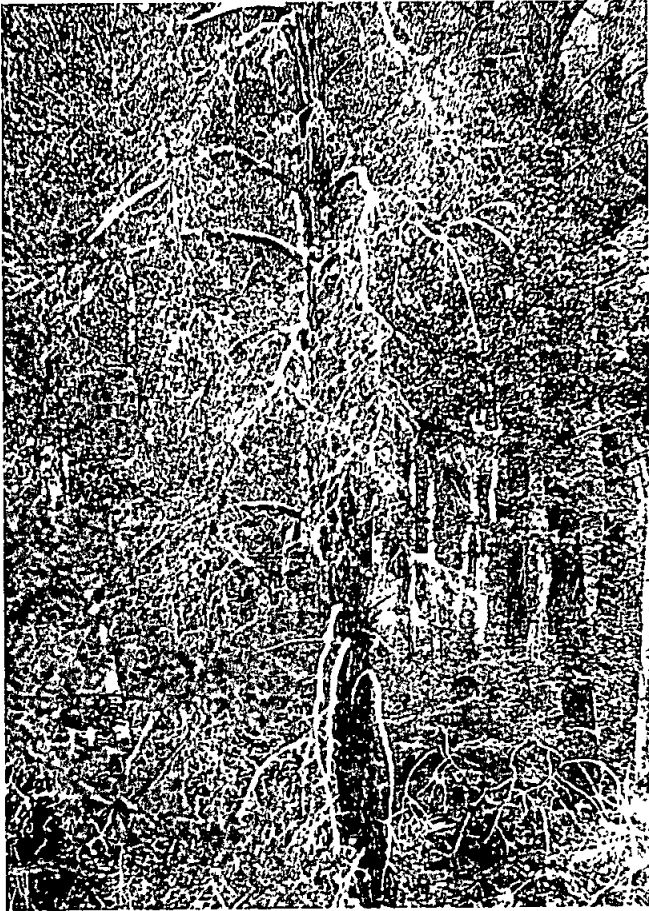


Fig 8 A tree next to the site, showing what was thought to be signs of "radiation" or "burning" caused by the landing and takeoff of the craft Photo was taken in 1967

evidence might have escaped them. However, some investigators (and Michalak) insist that some trees were damaged and the evidence ignored. This is borne out by the following description of the site in the *CAPRO Bulletin*:

The clearing consists of three main outcroppings of rock covered with the usual covering of lichens and moss. One of these outcroppings, the one over which the object hovered, was pitted extensively and had slight radioactivity. The lichen and moss that covered the other rocks in the group was blown clear and was heaped in a ring around the edges of the rock over which the object was said to have hovered.

A small tree which was growing through a crack in the rock had been bent and broken at the base and now lay on its side. The leaves of this tree discoloured in the following manner: on each leaf there was a round circle of brown within which was an area of red in the centre of which there was a hole. This sample was confiscated by the investigator sent by the National Research Council [Cannon 1968: 4].

As this tree was not mentioned in the official report, this issue has not yet been resolved.

But Craig had noted that there were no signs of any disturbances on the trees or other flora. When Michalak visited the site again in the fall of 1967, he found ample evidence that something had occurred there.

On September 30th I returned to the place where the craft had landed—to see if there were any other changes. I could not believe what I saw.

The leaves in the area looked as if they had been sprayed with some killing chemical. All were withered and dead. No vegetation grew within a 50 foot radius of the landing site. The area where the craft had touched down was still visible. [Michalak 1967: 36].

If someone had tried to make the site look “good,” the embellishment of a circle of affected vegetation would certainly have been an excellent idea.

What didn’t escape the investigators was the complete absence of physical evidence besides the ring of debris.

...the complete removal of all evidence from the site makes it highly suspect that it was never there. One would have thought that some shreds or burned shreds or burned shirt particles would have remained, even after the two months elapsed time. [RCAF 1967b: 4].

Hart and Michalak appeared to have taken confirmatory evidence that might have bolstered the credibility of the story. However, had “burned shirt particles” been found, they would likely have been thought to be suspicious as well, perhaps planted by someone to support the case. One can ask whether or not any evidence would have been considered bonafide, given Bissky’s personal conviction that the case was a hoax.

GOVERNMENT INTEREST

The Canadian government seemed to refuse access to information about the Falcon Lake incident when a question was raised in the House of Commons in 1967. On June 29, 1967, it was reported that a member of parliament, Edward Schreyer, asked about UFO investigations, specifically with regard to the Michalak case. The

Speaker of the House immediately "cut off the subject without government reply" On November 6, 1967, Defence Minister Leo Cadieux stated

it is not the intention of the Department of National Defence to make public the report of the alleged sighting [House of Commons 1967]

This was in response to requests by several cabinet members to obtain information on the incident On November 11, 1967, Schreyer formally placed a written question on the Commons order paper seeking information on UFOs However, the question did not solicit a useful response

On October 14, 1968, House Leader Donald MacDonald again refused an MP, this time Barry Mather, access to reports on the Michalak case However, on February 6, 1969, Mather was given permission by a member of the Privy Council to examine their file on UFOs "from which a few pages have simply been removed" It was reported that outright release of the file "would not be in the public interest," and could create a dangerous precedent that would not "contribute to the good administration of the country's business" (House of Commons 1969)

Bondarchuk (1979) reported that "portions of the complete government report are available for public scrutiny" at the National Research Council in Ottawa However, "noticeably missing are the RCMP study of the burned items, as well as the government's final conclusion, if indeed one exists"

POSSIBLE CORROBORATIVE REPORTS

When Michalak's experience was covered by the local media, many people reported their own UFO sightings from around the same time and area On May 19, 1967, residents of Lockport, near Winnipeg, reported a UFO with a "glowing ring of heat" moving at "indescribable speed" On May 21 three people watched a "round reddish glowing object at treetop level" from their cottage at Eleanor Lake in the Whiteshell They reported it to the RCMP on May 23, after Michalak's story was published Two men watched a "large, cigar-shaped object travel across the horizon at a tremendous speed" on May 25 That same night, a large, orange, egg-shaped object was seen near Souris, and two other witnesses saw "two very brilliant stars in close proximity to each other" over Winnipeg (All cases in UFOROM files)

In 1978, a man contacted UFOROM with information about his encounter at West Hawk Lake one night in May 1967 Although he could not recall the exact date, he claimed it had occurred "the same time as Michalak" He and a companion were walking from West Hawk Lake to Caddy Lake along a highway They had walked three of the four miles to their destination when his companion stopped to tie his shoelace The man, looking straight ahead, was startled to see a large, disc-shaped object glide silently into view above the trees and move across the highway It flew

over the trees on the other side of the highway and was lost to sight. Needless to say, the man's companion did not look up in time to see the object. The man's detailed sketch of the object had some resemblance to the Michalak's drawings, with some discrepancies.

In July 1992, a woman called UFOFORUM with information that she and her daughter had observed a UFO as they were traveling home from Falcon Lake along the Trans-Canada Highway the same weekend as the Michalak case. Around 4:00 p.m., they had left their cottage at Star Lake and were heading west when they saw a "perfect flying saucer" over the trees on the north side of the road. It was hat-shaped, with windows on its upper surface which were giving off "pinkish-mauve" light. The rest of the object was silver, and it appeared to be spinning counterclockwise. Her daughter sketched the object, and as they watched, it "disappeared into thin air." Independent sketches of the object by both witnesses agree in detail and seem to show a craft similar to that encountered by Michalak.

UNSOLVED MYSTERIES

In 1989, a producer of the NBC program *Unsolved Mysteries* contacted the Michalaks and other individuals (including the author) for information relating to the Falcon Lake case. In June 1992, Mr. and Mrs. Michalak, their son Stan and the author were flown by NBC to a remote set in South Dakota for a re-creation of Michalak's UFO encounter.

The segment aired on November 4, 1992. On the air date, 22 calls were received by UFOFORUM and 20 were received by the NBC operators in California. None of the local calls provided any information directly relating to the Falcon Lake case, but seven callers reported their own UFO experiences. One caller reported seeing a bright orange light over Winnipeg "the same night as Michalak did."

The next day, November 5, 1992, the author was interviewed on a radio talk show devoted to the case. Only one caller provided any relevant information. This woman claimed that she was a former employee at the Whiteshell Nuclear Research Establishment at the time of the incident. She said that her supervisors candidly noted that the Falcon Lake site was "very radioactive."

The *Unsolved Mysteries* segment was rerun on March 17, 1993, with similar results. No new solid leads were received.

CONCLUSIONS

In the report of the United States government-sponsored UFO project, Michalak's experience was described as "unknown," meaning there was no explanation. Their concluding remarks were impressive.

if [Michalak's UFO encounter] were physically real, it would show the existence of alien flying vehicles in our environment [Condon 1969 323]

What really happened at Falcon Lake? There is no question that some level of radiation was found at the site where Michalak said he had his experience. As well, he did exhibit some very unusual ailments, including reported weight loss, peculiar burn marks on his chest and stomach, charred hair, an odd rash and recurrent dizziness. Some UFO investigators have said he had met alien beings; some say he stumbled upon a secret government or military craft. Skeptics have proposed the only other explanation that would explain many of the facts: a hoax.

The only published comment by a skeptic about the incident was that by Donald Menzel and Ernest Taves, who dismissed the case by noting

First, the project investigator and the prospector together were unable to locate the area of the happening. Second, the "illness" appeared to have been caused by nothing more exotic than insect bites. And third, the project's attempts to establish the reality event revealed "many inconsistencies and incongruities." [Therefore] we regard the case as a badly executed hoax [Menzel and Taves 1977: 104]

Since the RCAF conceded that the site was difficult to find and since Michalak's illness was substantially more than insect bites, this dismissal appears somewhat abrupt. And, while there are certainly many incongruities in the case, these tend to heighten the mystery and not force dismissal by themselves. Even if the case was a hoax, it is not clear as to who might have been responsible, what was the motive, or the gain. In fact, even Bissky conceded the hoax was of a high caliber. If it was a hoax, its execution was quite elaborate, as it fooled several different levels of investigation, and there are still many unanswered questions today.

If the incident is a hoax, there are several possibilities for the identity of the perpetrator. If Stefan Michalak alone was the hoaxer, he would have needed many resources in addition to the stamina to stick to his story under intense questioning for more than 25 years.

The silver pieces found at the site are definitely suspicious. The hoaxer would have needed to have visited the site at least once prior to the placement of the items in order to locate a suitable position to hide them. The hoaxer would also have had to have had access to pitchblende ore and a way to cast the silver bars in order to give the appearance of "found" objects. (For its re-creation of the incident, *Unsolved Mysteries* used an amalgam of solder and other materials to make convincing replicas of the metal pieces.) In addition, the hoaxer would have known that someone with a Geiger counter would return to the site so that the items could be discovered. Since the metal bars were not discovered after a thorough search of the area by the official investigators in 1967, the hoaxer had several months to prepare the site.

for discovery by Michalak in 1968. During this period, dozens of persons likely had access to the site.

Other inconsistencies make the story problematic. It is interesting that the location of the encounter was within view of a forest ranger tower. Craig reports that the forest ranger on duty at the time of the incident did not observe either the landing or flight of the UFOs, or the smoke which resulted from the ignition of vegetation. This would seem to flaw Michalak's story effectively, although the individual in the tower might not have been looking in the direction of the site constantly. But since the object was landed for at least 45 minutes, and if it gleamed in the sun or emitted an "intense purple light" (as it was said to have done), it is puzzling as to why the individual in the tower did not see it.

Another problem that the Condon report noted was the direction in which the object departed. This direction was 255 degrees, which would have the object pass within a mile of the local golf course. No objects were reported by anyone from the golf course, although if the speed of the UFO had been great, this is not necessarily unusual. Craig noted that a northward-opening gap in the trees was inconsistent with the 255 degree bearing. However, Michalak clearly stated that the object rose vertically before departing.

If Michalak made the story up, why would he have gone to so much trouble to make it appear authentic? The Mayo Clinic psychiatric report concluded that Michalak was not the type of person to fabricate stories of this nature. If he had "fallen on a barbecue" (suggested by one skeptic) and burned himself while partying at Falcon Lake, why would it be necessary to claim it was the result of a UFO encounter? If he wanted to make it look good, would he not have made sure the site was found easily, especially since *Life* magazine was going to give the story publicity? A hoaxer of this caliber would certainly have wanted that kind of attention.

Another possibility is that Michalak had a real encounter with something, but someone else decided to improve the evidence so that the case seemed better. Given the number of people involved in both the military and civilian investigations, this is much more likely than a solo hoax attempt on Michalak's part. The problem is in the elimination of suspects.

Finally, a note should be made about the use of hypnosis in the investigation of the case. CAPRO investigators located two clinicians who used hypnosis in their practices. In one of the earliest examples of hypnosis employed in ufology, Michalak underwent at least one hypnosis session in the late 1960s. Recently, a tape recording of a session was made available for study. Unfortunately, little information that Michalak had not already recalled consciously was uncovered during the session. (The issue is complicated by the fact that one of the hypnotists now denies that Michalak ever underwent hypnosis, yet the tape recording clearly indicates such a session took place.)

If we assume that Michalak's story is truthful, then we have a solid report of a landed UFO, complete with physical and physiological effects. Personal interviews with the Michalaks have shown them to be sincere people. They are intelligent,

levelheaded individuals, and well-read on many subjects. Their annoyance at their notoriety is apparent, and their defensiveness at further proddings show that they have been subjected to severe ridicule and criticism since the incident first hit the media in 1967.

It is likely that much more information about the case is still in various files—lost, hidden or otherwise yet unavailable to researchers piecing together the puzzle of that afternoon in 1967. It is hoped that anyone reading this report will reexamine their files and perhaps make researchers with UFOROM, CUFOs or other bodies aware of the relevant material.

Investigations are continuing. If officials were convinced Michalak was a hoaxer, why was he not prosecuted for public mischief? There was definitely evidence towards this end. It certainly would have been an interesting court case, arguing about the existence of UFOs. It is even possible that Michalak would have won.

ACKNOWLEDGMENTS

It has proven very challenging to recover useful documents and piece together the Falcon Lake investigations. I would like to thank the following people for their kind cooperation and assistance in the preparation of this article: Roy Bauer, George Eberhart, Greg Kennedy, Maria Michalak, Stanley Michalak, Stefan Michalak, Mark Rodeghier, Berthold Schwarz, Vladimir Simosko, and Michael Swords.

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THEMATIC CONTENT ANALYSES OF THE REPORTS OF UFO ABDUCTEES AND CLOSE ENCOUNTER WITNESSES: INDICATIONS OF REPRESSED SEXUAL ABUSE

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ABSTRACT Twenty alleged extraterrestrial abductees participated in a study exploring the presence of dissociative and posttraumatic stress disorder symptoms. Two comparison groups of 20 sightees and 20 sexually abused persons provided comparative data. Interviews and questionnaires elicited detailed accounts from the abductee group, and thematic content analysis rendered three predominant themes of 1) being chosen, 2) missing time, and 3) bodily intrusions. Results showed that 45% of the abductee population manifested PTSD symptoms and 70% of the abductees manifested dissociative symptomatology. The three themes in the thematic analysis suggest that sexual abuse may be at the root of some of the abduction stories.

In 1989, while writing a predoctoral essay for my psychology program, I turned my attention to the purported nonfiction accounts of extraterrestrial abductions. As I read two well-known extraterrestrial abduction books, *Communion* by Whitley Strieber (1987) and *Intruders* by Budd Hopkins (1987), I noticed that three themes consistently emerged in the narratives: 1) amnesia, 2) bodily intrusions, and 3) being chosen. Through my doctoral work in psychology, I had learned that amnesia is commonly related to the trauma of sexual abuse. The theme of bodily intrusions was well-represented in these abduction stories through accounts of cell-sampling techniques and nasal probes. Being physically overpowered was also suggestive of sexual abuse. The third theme I noticed in these nonfiction accounts, being chosen, is a theme that occurs in the stories children tell when a parent or other significant adult singles him or her out for special sexual attention.

These themes indicated to me that the extraterrestrial abduction stories merited further attention. I had the uneasy feeling that there might be another story disguised within the extraterrestrial story. Although I could not completely account for the thought, it seemed to me that these frightening stories were true. I believed that the people telling these stories had been assaulted. Yet I found myself questioning whether these assaults had been perpetrated by alien forces.

Other researchers (Ring and Rosing 1990) have suggested that sexual abuse may play a part in the proliferation of extraterrestrial abduction stories. In particular, through cluster analysis, Rodeghier, Goodpaster, and Blatterbauer (1991) revealed a subgroup of abductees who report greater incidence of childhood sexual abuse as

well as elevated scores on posttraumatic stress disorder and fantasy prone inventories

In dissociative disorders such as multiple personality disorder, abuse has been known to surface as fantasy (Ganaway 1989, Noll 1989) Noll discusses the possibility that both the UFO abductee/contactee population and satanic cult ritual survivors are using screen memories to disguise awareness of childhood sexual abuse

The many reports of abductions by extraterrestrials ("UFO abductions") are also very similar to reports of "satanists" and may also be indicative of dissociative disorders, especially since one of their overriding characteristics seems to be "missing time " They likewise involve the "recalling" of childhood traumata in which patients claim to have been kidnapped and then sexually and physically abused by more powerful, adultlike beings. Thousands of people apparently now claim to have had such UFO abduction experiences—but are these extraterrestrials and satanists one and the same? [Noll 1989 7]

A significant percentage of the alleged abductees and satanic cult survivors may be using their narrative memories (embodied in the multiple personality disorder syndrome) to screen unacceptable historical memories If so, the following question needs to be asked Why this particular fantasy? A useful approach to the etiology of this belief system is provided by the third factor in Kluff's Four-Factor Theory (Kluff 1984) for understanding the origins of the multiple personality disorder syndrome

Kluff's factors include 1) inherited dissociation potential, 2) traumatic life experiences that overwhelm nondissociative adaptive ego defenses, 3) certain shaping influences and substrates that determine the form taken by the dissociative defense, and 4) inadequate restorative experiences by significant others Ganaway expands upon the role of this third factor

No doubt there are many reasons why specific themes are chosen for development by a patient in shaping fantasy-based trauma memories as a means of coping with fact-based trauma Impinging on this psychodynamic formulation would be extrinsic religious, socio-cultural and interpersonal (object relational) influences, as well as perhaps neuroanatomical and neurophysiological factors [Ganaway 1989 214]

The media, including such sources as magazines, books, and television shows, may supply material that persons can assimilate while in a dissociated state In other words, the media may function as Kluff's third factor and shape the dissociative defense into memories of ritual abuse and UFO abductions Later this material may be recalled as personal traumatic memories under certain circumstances such as hypnosis or other dissociated states

RESEARCH PROCEDURES

I decided to investigate the possibility that sexual abuse perpetrated by humans may be the source of *some* of the extraterrestrial abduction stories. Detailed results of this study are currently in press (1994). There were 60 subjects in my study, and one subject group consisted of 20 alleged abductees/contactees. Preliminary packets were sent to prospective abductee/contactee subjects who came from four sources: 1) ten subjects responded to a letter sent to members of a national UFO organization, 2) four subjects volunteered to tell me their stories when I attended an East Coast UFO conference, 3) four subjects were individuals referred to the study by local therapists affiliated with the Mutual UFO Network (MUFON), and 4) two subjects were individuals who responded to an advertisement placed in the magazine *UFO*.

I gathered information about the abduction or contact experiences through interviews and questionnaires which explored the details of the reported occurrences. Then I applied qualitative thematic content analysis (Lofland and Lofland 1984) to the data gathered by the intensive interviews and questionnaires.

The subject population in this study consisted of three groups: 20 abductee/contactees, 20 sightees, and 20 sexually abused subjects. There were 15 women and 5 men in the abductee/contactee group whose ages ranged from 22 to 73. The sightee group was similarly constituted as there were 13 women and 7 men in that group whose ages ranged from 25 to 61. The sightee group was composed of 1) three individuals who attended an East Coast conference, 2) eleven subjects who responded to a MUFON newsletter advertisement, and six individuals referred to me by local therapists affiliated with MUFON.

Clinicians in the Connecticut-Rhode Island area selected certain patients to participate in the study as members of the sexually abused group. It should be noted that only six of the 20 subjects in the sexually abused group chose to fill out the data sheets. Due to reasons of confidentiality, and because the clinicians were responsible for determining whether or not these were appropriate subjects for the study (that is, they were individuals who had memories of early childhood sexual abuse but no memories of extraterrestrial contact) this omission of demographic data was

Table 1. Abduction Narrative Sequence

-
- | | |
|---|----------------------|
| 1 | Capture |
| 2 | Examination |
| 3 | Conference |
| 4 | Tour |
| 5 | Otherworldly Journey |
| 6 | Theophany |
| 7 | Return |
| 8 | Aftermath |
-

Adapted from Bullard (1987)

Table 2. Close Encounters of the Third Kind: UFOs and Entities

Type A	— Entity is observed inside the object only (the true occupant), through doors, ports, etc The association is <i>explicit</i>
Type B	— Entity is observed getting into and/or out of the object (the egressed occupant) The association is <i>explicit</i>
Type C	— Entity is seen only in the vicinity of the object, not entering or leaving it The association is <i>implicit</i>
Type D	— Entity is observed but no object, although there is general UFO activity in the area at the time The association is circumstantial
Type E	— Entity is observed but no object, and there is no record of UFO activity at the time The association is negative
Type F	— A close encounter with an object but no entity is seen, however, voices are heard or messages received The association is implicit
Type G	— Abduction or on-board experience by the witness, with entities usually present or at least implicit

Adapted from Bloecher (1980)

accepted The six subjects who did fill out data sheets indicated that they were women between the ages of 21 and 33

The abductee/contactee group in this study included individuals who either asserted that they had had "abduction experiences" or who fit the Close Encounters of the Third Kind (CE3) category Table 1 summarizes several characteristics of the typical abduction narrative (Bullard 1987) while Table 2 (Bloecher 1980) presents CE3 categories Subject selection for the abductee/contactee group was conducted by comparing the initial descriptions of extraterrestrial or UFO encounters (detailed on the data sheets) to the patterns established by Bullard and Bloecher The prospective subject did not have to describe every component of the abduction sequence specified by Bullard or the CE3 sequence, however, it was imperative that the subject indicated that he or she recalled contact with extraterrestrial forces

All subjects in the abductee/contactee group in this study are individuals who claim to have had close encounters with extraterrestrials Twelve of the 20 subjects state that they were involuntarily abducted from a natural environment, forced to go aboard spacecraft or into some otherworldly place, and then they were returned to earth The other eight subjects in my study report that they had direct contact with extraterrestrials

Study results did not reveal any significant differences between the thematic analysis and scores of the abductees and contactees, and in this article the two types of experiencers will hereafter be referred to as simply abductees The similarities between persons reporting encounters with extraterrestrials and persons reporting abductions by extraterrestrials echoes Ring and Rosing's findings in the Omega Project These authors state that "those reporting abduction episodes are not a psychologically distinctive group compared to others who have had different [significant] UFO-related experiences" (Ring and Rosing 1990: 69) Parnell and Sprinkle (1990) also found few statistical differences between abductees and contactees in their work on

personality characteristics of persons claiming UFO encounters

I administered two psychological inventories to the 20 abductee subjects. These included the 49-item MMPI Subscale for the Assessment of Combat-Related Post-Traumatic Stress Disorder (Keane, Malloy, and Fairbank 1984) and the Perceptual Alteration Scale (Sanders 1986) which measures dissociation.

To yield comparative data which could be analyzed for its statistical significance, two other groups of 20 subjects each received these psychological inventories. The first group consisted of "sightees" (Davis 1988), that is people who report witnessing unidentified flying objects but who do not recall extraterrestrial abduction or contact experiences. The second group was constituted by individuals who report good recall of early childhood sexual abuse experiences that involve penetration. The members of this second group were people who are currently in therapy and who do not recall any extraterrestrial contact.

Why these two groups? To begin with, I needed a control group with whom I could compare the results of the tests administered to the abductee group. Previous research (Davis 1988) revealed that individuals reporting UFO sightings devoid of extraterrestrial beings or communication tend to produce scores within the normal range on psychological tests. In addition, I suspected there would be interesting differences between the abductees and the sightees. Stories about mysterious sightings are quite different from stories about being abducted and overpowered like a laboratory animal. Likewise, stories about mysterious sightings are quite different from the stories contactees tell of inexplicable telepathic communications and the frequently concomitant missing time. Yet my impression is that the general public often lumps all UFO and alien stories together. Therefore I wondered if my research would turn up differences between the groups known as sightees and abductees.

The other comparison group, the 20 subjects in the group composed of sexually abused individuals, were all people referred to me by clinicians. I included this group so that I could compare the scores of people who recall early childhood sexual abuse to the scores of alleged abductees. Whatever the results might be, they could shed light on similarities or dissimilarities. This, in turn, might inform my theory that there could be a link between allegations of extraterrestrial abduction and sexual abuse perpetrated by humans.

STATISTICAL ANALYSIS OF THE PSYCHOLOGICAL INVENTORIES

An extensive explanation of the statistical results of my study is in press (Powers 1994). Here I will briefly summarize the results of the PTSD subscale (Keane, Malloy, and Fairbank 1984) and the PAS (Sanders 1986). These scores were submitted to a one-way ANOVA which indicated significant differences between the groups on both the PAS and the PTSD inventories. Table 3 presents means and standard deviations for the three subject groups.

For the MMPI PTSD subscale, the optimal cut-off score for inpatient populations

Table 3. Results of One-Way Analysis of Variance on the PTSD and PAS Scores of the Three Subject Groups

Group	N	PTSD		PAS	
		Mean	Std Dev	Mean	Std Dev
Abductee/ Contactee	20	19 050	9 997	116 200	20 531
Abused	20	31 200	11 386	146 050	30 732
Sightee	20	6 000	3 524	97 100	9 323
Tukey's Studentized Range (HSD) Test for Variable PTSD $\alpha = 0.05$ $df = 57$ $MSE = 80.669$					
Critical Value of Studentized Range = 3.403					
Minimum Significant Difference = 6.834					

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is 21. The sexually abused population in this study has a mean score of 31 which is significantly above the score of 21 (Orr, et al 1990: 333) required to determine PTSD symptomatology.

The abductee group has a mean score of 19 which is very close to the cut-off point of 21. Table 4 presents individual scores on the PTSD inventory for all three subject groups. Since nine of the abductee scores are 21 or above, according to the criteria established by Orr, et al (1990), 45% of the abductee subject population have scores which indicate PTSD tendencies. The abductee population mean of 19 is lower than the mean of 25 produced by Vietnam combat veterans in Orr's study, and this lower score could reflect less severe PTSD symptomatology in the abductee group.

The sightee group effectively functioned as a control group as evidenced by the mean score of 6.0. These scores demonstrate that there may be a very important difference between those persons who report UFO sightings, and those who recall extraterrestrial contact and/or abduction.

On the PAS (Sanders 1986), the results parallel the PTSD scores. The sexually abused group produced the highest scores, the abductee scores indicate dissociative tendencies, and normal scores were produced by the sightee group.

The PAS scores for the individuals in all three subject groups are presented in Table 5. Use of the 108 cut-off point (Sanders 1986) indicates that 70% (N=14) of the abductee individuals, 100% (N=20) of the abused subjects, and 10% (N=2) of the sightees present dissociative symptoms.

Research by Ring and Rosing (1990) also indicates that persons who claim to be extraterrestrial contactees or abductees often have a childhood history of physical and/or sexual abuse. The implications of the dissociative symptomatology demonstrated by the abductee subject group merit further exploration. The qualitative data

Table 4. Individual Scores of Three Subject Groups on the PTSD Inventory

	Abductee/Contactee	Sexually Abused	Sightee
1	16	19	1
2	14	19	8
3	3	38	3
4	17	44	6
5	22	20	0
6	19	22	6
7	8	40	4
8	21	45	5
9	21	21	3
10	4	19	11
11	15	39	6
12	14	45	8
13	20	21	12
14	25	19	11
15	40	38	7
16	22	45	2
17	37	19	10
18	24	25	9
19	33	40	2
20	6	46	6

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Table 5. Individual Scores of Three Subject Groups on the PAS Inventory

	Abductee/Contactee	Sexually Abused	Sightee
1	125	121	101
2	118	108	110
3	80	187	94
4	114	172	105
5	125	120	102
6	129	110	104
7	97	175	93
8	101	165	100
9	122	170	99
10	95	108	101
11	112	129	89
12	86	168	103
13	118	124	103
14	129	112	97
15	158	179	109
16	124	169	101
17	141	119	92
18	121	119	84
19	144	190	74
20	85	176	81

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gathered through questionnaires and interviews bring to light some key issues in the abduction/contact accounts

THE QUESTIONS I ASKED

As I designed a survey to elicit these stories of contact with extraterrestrials, I did my best to keep the wording neutral. Leading questions distort information, and a major aim of my study was to convey the subjects' stories in their own words. I had already established through the subject-selection process that all 20 subjects reported contact with extraterrestrials, and so I felt that I could be bold about asking them directly about these experiences. Yet I did not want to unduly influence their descriptions of these contacts. I finally settled upon the following 16 questions which pursue the details of the experience.

Extraterrestrial Contact Survey

- 1 How old were you the first time you saw an Unidentified Flying Object (UFO)?
- 2 Where did the event take place?
- 3 Please describe what this UFO looked like and how you reacted when you saw it?
- 4 How old were you the first time you saw an extraterrestrial?
- 5 Where did the event take place?
- 6 Please describe what the extraterrestrial looked like and how you reacted when you saw it.
- 7 Do you have memories of being taken aboard a spacecraft? Yes/No. If yes, please describe what you remember.
- 8 Some people report "missing time" after seeing a UFO or an extraterrestrial. Have you had this experience? Yes/No. If yes, please describe this missing time experience.
- 9 How many times have you had contact with extraterrestrials?
- 10 Please list the approximate ages at which these contacts occurred.
- 11 What kinds of feelings do you associate with this extraterrestrial contact?
- 12 Did you have any idea that an extraterrestrial contact was going to happen prior to the event? Yes/No.
- 13 How did you feel after the extraterrestrial contact?
- 14 Why do you think the extraterrestrials chose to contact you? Did they tell you why?
- 15 There are people who report that they have the sense that extraterrestrials are watching them before and after a contact? Has this been your experience?
- 16 If someone were to make a movie about your contacts with extraterrestrials,

what would the storyline be? Please describe your overall history with UFOs and extraterrestrials in as much detail as possible. Your emotions, your hopes, fears, and theories about these experiences are all very important.

I then applied qualitative thematic analysis (Lofland and Lofland 1984) to the data gathered by the intensive interviews and questionnaires. The three themes that I had previously delineated in the abduction accounts (Strieber 1987, Hopkins 1987) also appeared in this study's data.

There is an unmistakable connection between extraterrestrial abduction and sexual violations. It would be best to let these stories speak for themselves, so I will present excerpts from my subjects' accounts to illustrate the presence of the three themes of being chosen, amnesia, and bodily intrusions. Some details have been changed to protect the identities of these individuals.

THEMATIC ANALYSIS OF THE QUESTIONNAIRES/INTERVIEWS

(a) Being Chosen, Missions, and Intergenerational Contacts

Fourteen of the 20 abductee subjects (70%) indicated that they had been specifically chosen by the extraterrestrials due to their special qualities and/or to perform a mission. The following statements are representative of general comments made by subjects concerning their selection.

They told me they were glad to see me again. Also that I have a mission which will be revealed to me at the appropriate time. And, they said that I will see them again in the future.

When I was little, I asked them why they were coming to me all the time. They said it was because I was the Chosen One. I told my mother about it and she looked at me sort of perplexed. She said, "When do you see them?" I said, "At night when I am in bed."

A tall, dark one looked into my eyes, and she said, "You are mine." She didn't exactly say it. She *thought* it at me, and I knew through telepathy what she was saying. She let me know that I had been chosen to work for the aliens. They had picked me out while my mother was pregnant with me. I have special sensitivities. I am very intuitive, and I often have dreams that come true.

Three of these 14 subjects discuss how, in their cases, the extraterrestrial abductions are intergenerational. This could be relevant because sexual abuse frequently spans generations (Jiles 1980, Kempe and Kempe 1984). Perhaps the extraterres-

trial abduction stories produced by several generations of one family are a means by which the ongoing sexual abuse can be kept a secret

Whether or not sexual abuse is occurring in a family, the very fact that a certain family may pass down tales of alien abduction is significant. To feel like a bona fide member of that family, a child might need to convince him or herself that such contacts are happening. Otherwise, that child would not fit in with the other special people in the family. Even if the family does not hold open discussions about extra-terrestrial contact, the children may overhear relatives talking, and they may be very sensitive to adult reactions to certain topics. The issues of truth and reality could get very confusing in a family in which certain members claim a history of alien abductions.

The following three excerpts demonstrate this intergenerational theme.

I do believe they came after my daughter because they contacted me earlier. I believe that there's something she meant. There is something in our family that is special to the aliens. [This subject also stated that her daughter knew her mother had been contacted.]

UFOs have been coming to my sister and me since we were young. I don't remember the first time, but they were with my Mom and my aunt. We all have scars on our legs in the same place, the same position. I was proud to have a scar just like my mom's. They took my sister first, and I wondered why they didn't come after me. Later, they took both of us.

When I finally told my uncle what had been happening to me, he shocked me by telling me his own story. They came to him when he was 12, and he was out in the woods camping. This was similar to my own experience. I first saw them when I was 11, and I was sleeping out on the back porch. He had been sworn to secrecy, just like me. It felt so good to finally tell someone about the weird operations they performed.

In response to the question, "Why do you think the extraterrestrials chose to contact you?" three of the subjects state that this selection was based upon the unique psychic powers these individuals displayed as young children. One of the three major traits of fantasy proneness (Wilson and Barber 1983) is a history of self-reported paranormal experiences. Furthermore, Wilson and Barber suggest that these paranormal experiences are due to the extraordinary propensity these fantasy-prone individuals have to fantasize with hallucinatory vividness. In other words, these people visualize their fantasies so clearly that they experience the perception of spirits, other worlds, and magical experiences. Although there are contradictory results concerning the applicability of fantasy proneness to the extraterrestrial abduction/contact phenomenon (Bartholomew and Basterfield 1991, Conroy 1989, Ring

and Rosing 1990, Rodeghier, Goodpaster, Blatterbauer 1991), the results of this study suggest that this variable warrants further scrutiny. The following excerpts illustrate the presence of self-reported early childhood paranormal experiences in the abductee/contactee population.

When I was a kid, I did some psychic things. I knew where things were. I could find anything for my mother. I think the aliens were interested in me because I had these abilities.

Around 6 years old I had a strange thing happen to me. My grandmother locked me out and I sat on the steps for about two hours wanting in. We had a green door with a white knob and I stared at it so long, determined to get in. When I stood up, my whole body seemed to blend into it. I was on the other side. My grandmother screamed and turned the knob. It was still locked. She looked at me and said, "Don't you ever speak of this to anyone." I never did until I told my husband.

In the second grade I had a Guardian Angel come to me. I watched as the Angel slowly approached my bed. She was all-white and a *good* entity [emphasis added by subject on written answer]. The Guardian Angel appeared because I was sick and crying in the night with an earache and my mother didn't wake up.

Another reason for selection by the extraterrestrials, which was given by four of the 20 subjects, is the upgrading of either the extraterrestrial species or the human species through creating hybrids: crossbreeding human beings and extraterrestrials. Specifically those who were abducted were to be used in these hybrid experiments. The following excerpt is from a subject's account concerning the purpose of her abduction.

[The extraterrestrials] said they were trying to "upgrade" the World races. Took ova and sperm and tried to breed a better, higher (spiritually better) prettier and healthier races among the people of this planet. Wanted me to have a baby for them.

The qualities extraterrestrials are looking for in candidates for their genetic experiments were described by one subject:

The "Greens" [a type of extraterrestrial] told me they are looking for certain people with certain qualities to upgrade the races [both human and extraterrestrial races].

- 1 High auras—psychic—"Truth Student" types
- 2 Healthy, strength, stamina

- 3 Intelligence
- 4 Beauty
- 5 General "good" natured

Sometimes they leave the woman to carry the child, other times they use surrogate mothers, and other times they use their own people or breed in incubator types. They are experimenting with us.

If these stories of being chosen by extraterrestrials are screen memories for being chosen, during early childhood, for sexual abuse, then recasting the experience as a selection with such a grand purpose might restore meaning to lives threatened by traumatic memories.

The following lengthy excerpt illustrates the theme of being chosen for a sexual contact which is a rape. The woman recounting this story explains that the extraterrestrial forces orchestrated this encounter for the purpose of upgrading the *human* race, as an act of benevolence and caretaking of the human species. She further explains that this act of procreation occurred between herself and a boy who later turned out to be her daughter's friend in high school. The subject describes how extraterrestrials can manipulate time so that persons can be "brought back in time" and made to be various ages. When the sexual encounter takes place, the man is 24 years old. However, when this woman first meets this person in everyday life (two years later), he is only five years old. At the time that this narrative was submitted to this research study, this young man is alleged to be 19 years old.

I was pulled out of the room and totally undressed. I was shoved back into the empty room and the door was closed. There, on the floor, was a double-size mattress, not too clean. The young prince-like man who had seemed to recognize me entered from another door on the opposite side. He, too, was naked and with horror I realized what was going to happen. Never very thrilled with sex, I stood shocked, trying to cover my body with my hands and hunched over, but he only laughed. "I can make this pleasant, or I can be rough. It's all up to you, because I will do what I am supposed to do with you. If you cooperate and let me, you will not be hurt. If you fight, you will be."

I lay still and submissive, but I hated him and I let him know it. He was neither upset nor distressed. Actually, he rather ignored me and seemed to feel that it was only a job to be done and finished with.

A bit later on, angry and in pain, I lay nervously shaking. He rubbed my body all over and answered some questions for me. He explained that I knew him very well as we were heavily "soul and karmically linked." He said that he was twenty-four years old and that we would soon meet, although I would not recognize him as he was being brought "back in time" to meet me. I would have a child by him and it would be raised in a marvelous setting [by the extraterrestrials].

The conclusion of this story is that the subject gave birth to a baby girl and was returned to her everyday life. Although the story's duration is one year, the woman had this experience in one night. She accounts for this time discrepancy by referring to the extraterrestrials' ability to manipulate time.

One interpretation of this story is that the subject has recalled a rape wherein she behaved submissively due to threatening circumstances. Her recollection of the discussion in which the young man tells her they are karmically linked might indicate that this rape was perpetrated by someone she knows. Also, the fact that she identifies the man as being her daughter's friend adds an interesting dimension. The subject describes how she told the young man about this memory one night. She states that he reacted with extreme astonishment, in her words, "He stared at me in shocked and baffled surprise." It is significant that this woman chose to tell the young man about her memory, and this intimate disclosure calls into question her feelings toward her daughter's friend.

In addition to the purpose of genetically upgrading extraterrestrial or human races, eight of the 20 subjects specifically stated that they were chosen by the extraterrestrials to perform missions. Of these eight subjects, three stated that their special purpose had not yet been revealed but that it would be when the time was right. The remaining five subjects described a variety of purposes including designing superior low-income housing, saving the sasquatch creatures, performing various humanitarian acts in the tradition of Christ (for example, raising the dead), teaching people not to pollute the earth, and preparing the human race to accept the forthcoming extraterrestrial visits.

They told me pollution was destroying our Planet Earth and that I must speak to the people. The extraterrestrials were here to warn us that Planet Earth was to be destroyed.

I was educated by the little people. I was shown a movie of Earth rising out of water. Then I landed on a mountain in China. I saw a crowd of maybe 25 to 30 people around a small house. The crowd parted for me to enter and I went in. I saw a very sick dying child. I knelt down and motioned for water. I began to bathe the boy. He was limp and wet with sweat. He then opened his eyes and sat up and asked for food. His fever had broke. Then I went outside and then I was home.

The Star People would have me read the newspaper and they would read through my eyes. They would see property that had Sasquatch [Bigfoot, otherwise known as the Abominable Snowman] on it that they would want me to buy.

The subject who refers to extraterrestrials looking through her eyes to read the

newspaper introduces the theme of bodily intrusions since the alleged extraterrestrials are occupying her consciousness. This theme is also apparent in reports wherein subjects state that they have been used in genetic experiments. The next section of this thematic analysis will examine accounts of bodily intrusions as they occur in the subjects' narratives.

(b) Reports of Extraterrestrials Physically Intruding upon Human Subjects and Unelicted Reports of Sexual Abuse by Humans

Discussion of various bodily intrusions must begin with a review of the gender distribution in the abductee population: there were 15 women and 5 men ranging in age from 22 to 73 years of age. Three of the 15 women described surgical procedures the extraterrestrials used to remove ova from their bodies for hybridization purposes. These 3 women also asserted that they have been informed that their hybrid children are being raised by the extraterrestrials. No men reported sperm being removed, although one man did describe scars in his groin area that he suspects are due to extraterrestrial surgical procedures. The following excerpt is the most detailed description of a gynecological procedure obtained in this research project:

Another operation I recalled was something going in through my naval. I don't recall what went in, but I do recall a snake-like feeling squirming around in my womb. It cramped me up, nauseated me, and there was pain. They said they were counting my babies and told me I would only have one (to remember). Apparently they can see which eggs are destined to be fertilized. I had my tubes tied a few years ago and had the body aches from the gas the doctor filled my womb with.

The subject appears to be describing a medical procedure known as a laparoscopy wherein a tubelike instrument with special optic attachments is placed through an incision in woman's umbilical area to examine the female organs and/or perform surgery. Ova may be obtained from the ovaries during this procedure.

This same subject also mentions that she has been told by the extraterrestrials that she has at least one child with them. The subject states that she has been promised that the extraterrestrials will come and get her "once and for all" so that she can join these hybrid children. These invisible and unattainable children pose some interesting questions. Perhaps the idea of hybrid children is a fantasy that tends to be spun by women who have had sexual relations that do not allow for children. In other words, in incestuous relations, conception would be 1) an exposure of the illicit sexual contact and 2) condemned by society.

The subject cited above spontaneously informed me that she had been molested as a child.

They [the extraterrestrials] came to check me after I was molested as a child to see how serious it was. They said they saw my light (aura) had changed

Another subject also mentions that she was sexually abused as a child

I wonder if you are one of those researchers who thinks that this whole thing [extraterrestrial abduction/contact] is due to sexual abuse? I can tell you that it isn't because I was abused when I was a kid, and I've been abducted, and I know the difference

These unexpected revelations from these two subjects add a provocative dimension to the research results. More research needs to be conducted which specifically examines the extraterrestrial abduction/contact stories told by individuals who *also* recollect sexual abuse perpetrated by humans. This research would be very helpful because it would lead toward differentiation between 1) screen memories of sexual abuse (perpetrated by humans) which account for extraterrestrial abduction memories, and 2) cases where an individual has been sexually abused by another human *and* he or she recalls an alien abduction which is unrelated to human abuse.

Another point which must be made is that if, indeed, extraterrestrial assaults are taking place, then the victims of these encounters would demonstrate dissociative phenomena such as posttraumatic stress disorder (Laibow 1992, Neal 1991, Wilson 1990). Laibow, Neal, and Wilson have in common their willingness to allow for the veracity of the abduction/contact narratives, despite lack of empirical evidence. Laibow indicates this openness with the following cautionary note to clinicians:

The therapist must remember that while he may have strong convictions pro or con the abduction actually having occurred, it is not within either his capability or expertise to make such a judgement with total certainty.

The development of PTSD symptoms could be a natural response to little understood traumatic events. Wilson states that the purported abductee's attempts to master states of helplessness leads to "states of hypervigilance and ego-defensiveness to maintain psychic integration" (Wilson 1990: 13). The development of nonempirical belief systems, such as belief in an elaborate typology of extraterrestrial entities, can be viewed as part of the adaptational dilemma. Wilson concludes that effective therapeutic interventions should address stress symptoms:

Since victims of experienced anomalous trauma are likely to suffer from PTSD or other forms of distress, proper counseling can do a great deal to alleviate the stress response syndrome and its maladaptive features. [Wilson 1990: 14]

Five subjects described traumatic and invasive encounters which resulted in nosebleeds and/or recollections of procedures involving probes in their nasal cavities. The following citations represent this recurring theme:

The first operation that always comes to mind was the "brain" operation. I was laid on my back and held down either by their will power or some magnetic force that paralyzed me. They came at me with a *long* needle-like thing which was inserted up my nose into the brain cavity. At some point when it breaks or penetrates something in there, there is a *loud* firecracker bang that goes off in the head and makes you jump.

The night after I saw the UFO, my boys developed sudden bloody noses. All of us felt watched and there were "Bigfoot" tracks in the snow outside the house.

I liked them [the extraterrestrials] even though they hurt me. When they put the needle in my head, in my nose, there was a loud bang. Then I couldn't hear anything for awhile. It was like I went to sleep, but I wasn't really asleep. I know I had my eyes wide open the whole time.

The theme of unwanted bodily invasions is evident in these accounts. The subjects are describing the overpowering of their will through physical violations.

Another type of bodily intrusion is described by one subject who refers to extraterrestrial abductions which she remembers occurring on several evenings. She suspects that blood samples were being drawn from her body for experimental purposes.

After these encounters, I would have bruise marks and be sick the next morning. I would always go on to work and I would have the round burn marks on my legs. Always over a vein or artery.

This subject alleges that she was once abducted by the extraterrestrials and taken with them on a mission to collect blood from cattle.

We would fly low, lasso a cow, and take her off. Then they would take a big needle into her neck vein. I watched blood went into a tube and then into a big tank. The cow's eyes would glaze over. Then I knew she was dead. We would fly back and drop her in the pasture with other cows. The little people do not eat meat. They take the blood home with them.

According to this account, the woman is being treated like an animal subject. a

cow She is a specimen, just as cows are specimens, to be used by the extraterrestrials for their mysterious purposes A little child who is manipulated and overpowered by an adult during sexual abuse also might feel like a specimen being used

The last type of physical invasion described by three subjects in this research project is the production of scars caused by experimental incisions

I have these strange scars in my groin area It almost looks like I had a vasectomy, but I didn't

This subject did not state whether or not anyone else had examined these purported scars However, allegations of these scars underscores the idea that humans are being used as research animals

Descriptions of bodily intrusions perpetrated by extraterrestrials need to be examined Thematic analysis of these descriptions suggests that many of these alleged intrusions are sexual in nature Furthermore, they frequently take place in asexual, scientific, and frightening circumstances These accounts may cloak more prosaic memories of frightening and abusive situations

(c) Missing Time

Sixteen out of 20 subjects (80%) reported periods of missing time or amnesia during alleged extraterrestrial abductions and contacts The following five excerpts are representative of these reports

I was taken somewhere I don't know if it was aboard a spacecraft, or where The next thing I knew, it was six hours later

When I first saw the UFO, I felt as if I observed it for about five minutes When I went inside, my first reaction was to look at the clock (I don't know why) I was amazed to find that an hour had passed by

Yes, I lost forty-five minutes one time And a hour and a half another time [during extraterrestrial contacts]

I don't know how I got inside [a UFO] or was placed outside I know that one hour passed that I can't account for at all

On a motorcycle ride once we saw an ET, grey, bald, black eyes, 4' tall, at the edge of a field we went by The driver asked if I saw "that thing" and I said, "Yes " He wanted to go back for a second look and I told him he'd probably not like him and prefer going on I knew what the ET's reaction would be [not to like the driver] The next thing I know, I was walking down the country road at night toward the motorcycle

which was parked by the roadside. My friend was waking up from a "nap" in the grass alongside the road. He was embarrassed to have fallen asleep there for no reason.

Four of the 16 subjects reporting amnesia mentioned that the extraterrestrials may be causing people to have amnesia in order to either protect the subjects and/or to protect the extraterrestrials. Since the alleged abductions/contacts were imposed upon the human subjects, this type of benevolent explanation for the events suggests what Terr (1991) describes as a "tendency to identify with aggressor." In other words, what has been recalled as an involuntary and frequently frightening experience is being reconstituted as an act of kindness.

This phenomenon is associated with situations in which hostages take the perspective of their captors and thereby justify the acts of violence of which they are victims (Terr 1991).

The following two citations illustrate this benevolent interpretation of the purported amnesias which occurred after extraterrestrial abductions.

They couldn't let me remember these events until now because I wasn't spiritually ready. I needed to become pure, first. I didn't know why I became a vegetarian, but that was necessary. Now I can remember it, as much as they want me to, and now I can tell the world that the little people [extraterrestrials] are here to help.

They have this way of blocking your mind, wiping the slate clean. People wouldn't understand. You heard about the saucer crash out west and they [the U.S. government] covered it up. There were aliens who were still living in that crash, but they were stuffed into research laboratories where they died. The aliens have to protect themselves from us, because we hurt them.

CONCLUSION

The three themes of being chosen, bodily intrusions, and missing time should make us question what is really going on with the extraterrestrial abduction phenomenon. No one knows what the truth is, but clearly something important is happening here. The psychological inventories I employed indicate that, in this preliminary study, a significant percentage of the alleged abductees manifest dissociative (70%) and posttraumatic stress disorder (45%) symptoms. These results coupled with the suggestive themes which emerge in the thematic analysis indicate that the abduction phenomenon is, for many people, a very serious matter.

Given the prevalence of childhood sexual abuse, and the sexually violating nature of many of the alien abduction stories, it is imperative that the psychological com-

munity investigates the possibility I am suggesting. Many of these stories about the aliens among us could be about trauma which has been inflicted upon children by adults. If this is true for some percentage of the reported abductions, then this underlying trauma needs to be addressed in therapeutic settings.

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ETHICS CODE FOR ABDUCTION EXPERIENCE INVESTIGATION AND TREATMENT

THE ABDUCTION STUDY CONFERENCE ETHICS COMMITTEE*

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ABSTRACT A comprehensive set of ethical guidelines for investigators and mental health professionals who have reported abduction experiences is presented. The Code and accompanying comments discuss competence and responsibility, minimal and reasonable risk, assessment, informed consent, treatment issues, ethical protocol in investigations (including a discussion of the use of hypnosis), anonymity and confidentiality, dissemination, and approaches to ethical issues. A discussion of symptomatology similar to the abduction experience, and a sample informed consent form are included as appendices.

Note The board of directors of the Center for UFO Studies voted unanimously at its February 1994 meeting to approve this Ethics Code and adopt it for immediate use by CUFOS. Its adoption entails the use of the Code to guide and regulate the conduct of CUFOS-affiliated investigators and researchers of the abduction phenomenon. Furthermore, CUFOS will encourage other investigators and organizations to adhere to the Code, and we will use the Code as one criterion in publication decisions for the *Journal of UFO Studies* and other CUFOS publications. The CUFOS board wishes to express its heartfelt thanks for the splendid and exacting work of the Committee and to all others who contributed to its creation. We believe the Code is an important step forward that will provide the necessary and proper protection to witnesses in our investigations.

By definition, ethical problems remain unresolved. By their unresolved quality, they provoke a continuous anxiety in the practicing [scientist] and concomitantly a desire to search, to oppose, to think, and to re-search [Bernal y del Rio 1975: 2546].

The *Ethics Code for Abduction Experience Investigation and Treatment* is a guide for investigators and mental health professionals (MHPs) working with the increasing population of individuals who have reported "abduction experiences."

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BACKGROUND

The Code was developed by a group which came together during the Abduction Study Conference (ASC) held at MIT in June 1992 to study ethical issues in abduction investigation and treatment. This group initially consisted of seven individuals attending the ASC (four mental health professionals, a research psychologist, a ufologist, and an experiencer) who responded to an open invitation. Attrition due to other obligations ultimately shrunk the Committee to the following four individuals, each of whom represents a different and complementary area of interest and expertise: Stuart Appelle (research psychologist), Georgia Flamburis (experiencer), David Gotlib (physician and psychotherapist) and Mark Rodeghier (ufologist). This group formed the Abduction Study Conference Ethics Committee, which decided that the most constructive way to address these ethical issues was by developing a Code of Ethics, a comprehensive statement of principles for ethical conduct in abduction investigation and treatment.

In developing the Code, the Committee studied ethical codes of other professional organizations, and the unique features of the abduction experience. The Committee sought guidance from experiencers, investigators, mental health professionals, social scientists, bioethicists and others. In discussions during and after the ASC, it became clear to the committee that investigators and MHPs working with the abduction experience come from diverse backgrounds with varying degrees of familiarity with existing codes of conduct for human subject research and therapy. It also became apparent that many of the ethical issues involved with the abduction experience were not dealt with adequately in the existing codes.

For example, because the abduction experience is nearly always reported as disturbing or traumatic (at least initially), and because the cause(s) of the experience are not fully understood, investigation and treatment often interact. Abduction experiencers may approach investigators not only to provide or obtain information, but also for help in coping with the experience. Similarly, MHPs may be asked not only for help in coping, but for explanations and interpretations of the phenomenon. When the goals of the experiencer are in conflict with the goals or expertise of the investigator or MHP, ethical problems arise.

In April 1993 the Committee distributed a draft version of a proposed Code of Ethics for public comment. Copies were mailed to the 150+ attendees of the ASC, 150+ subscribers to *Bulletin of Anomalous Experience*, and other interested individuals. Over thirty written responses were received from investigators, MHPs and experiencers. Many additional responses were received by members of the Committee through personal communication. These comments guided the Committee in revising the initial draft and in developing the document presented here.

These guidelines will not eliminate risk. They will not resolve all ethical dilemmas regarding risk. They cannot satisfy every investigator's or MHP's personal interpretation of risk. But they will help in achieving minimal and reasonable risk in any investigative or treatment intervention.

PREMISES

A number of premises also guided the development of this Code

- 1 This Code takes the position that not only the experiencer, but also science and clinical practice, are best served when the rights and welfare of the abduction experiencer are considered paramount
- 2 Mental health professionals working with abduction experiencers are not only ethically guided by the codes of their professional organization, but also legally bound by their licensing bodies. Mainstream researchers working with human subjects are also bound by the ethical standards of their professional organizations. This Code cannot supersede those existing codes. Rather, it complements and supplements those existing codes in regard to activities specifically associated with the abduction experiencer population, and it serves as a code of conduct for investigators not already bound by the ethics code of a professional organization
- 3 This is a code of ethics, not a manual on methodology or technique. It is recognized that inappropriate behavior, poor research methodology or dubious treatment approaches may be criticized on many grounds. However, this Code addresses methodology and technique only in regard to ethical criteria
- 4 Clinical competence is critical to the provision of ethical treatment, and research competence is critical to ethical conduct in an investigation. Although competence is an ethical issue, the criteria of competence is a professional issue and not the purview of this Code
- 5 Although theories as to the cause or causes of the abduction experience abound within the abduction community (investigators, MHPs and experiencers), mental health professions, and academic disciplines, no theory has been established as correct according to conventional standards of science or clinical practice. Therefore, this Code assumes no particular theory of causality

BENEFITS OF ADOPTING THE CODE

Adoption of this Ethics Code will provide the following benefits

- 1 *Protection of Experiencer* It is the primary purpose of this document to protect the rights and welfare of the individuals with whom abduction investigators and MHPs work. Ethical investigation and treatment will serve to protect the experiencer from unreasonable risk created by participation in investigation or treatment

- 2 *Benefits to science* Research conducted along these ethical guidelines will facilitate open and unbiased responses from experiencers, and reduce the chances for misinterpretation or misapplication of the findings
- 3 *Benefits to treatment* Therapeutic outcome depends on the ethical behavior of the MHP Treatment for the abduction experience that follows ethical guidelines will best serve the experiencer
- 4 *Protection of MHP and investigator* Adherence to these guidelines provides a safeguard against behaviors inconsistent with legal, professional, and ethical conduct, especially for those not members of professional societies or whose professional conduct is not regulated by a licensing authority They do not supersede the ethical guidelines of individual societies or the law
- 5 *Public and media relations* Adherence to these guidelines will promote professional behavior, which will increase public support for this work and will help ensure that information disseminated to the media will be based on practices appropriate for investigation and treatment

ACKNOWLEDGMENTS

The Committee benefited greatly from the many investigators, MHPs, and experiencers who provided commentary on an earlier draft The Committee thanks all those whose concern, contributions and support guided the development of this document

PART A: ETHICAL GUIDELINES FOR INVESTIGATION AND TREATMENT

I Introduction

1. Application of Ethics Code

The *Ethics Code for Abduction Experience Investigation and Treatment* is a statement of principles It cannot specify all the practices that are considered ethical and unethical The Code can prove useful in resolving a wide variety of ethical dilemmas in investigation and treatment, regardless of what new information about the abduction experience comes to light, but only if each individual in this field accepts full responsibility for maintenance of ethical conduct in the work they have chosen This responsibility requires a constant awareness and examination of one's attitudes, emotions, behaviors, expectations, and needs

2. Definition of Terms

2.1 Abduction Experiencer As there exist a number of possible causes for a reported abduction experience, investigators and MHPs may work with individuals

whose reported experiences stem from a variety of factors. This Code applies to work with all of these individuals. Therefore, it is necessary that the terminology and definitions used in this Code imply no particular theory of causality.

Any label used to describe or classify someone has connotations and limitations. This Code has adopted the term *abduction experiencer* because of its widespread (albeit not universally accepted) use and recognition as a term intended to be objectively neutral in regard to causality, neither implying that an abduction has actually taken place nor that it has not. As used in this Code, it simply indicates someone who reports experiences in their life which are consistent with, suggestive of, or thought to be associated with, being "abducted" (i.e., "carried or led away in secret or by force," Random House 1980) by apparently nonhuman entities.

2.2 Investigator An *investigator* is someone whose objective is the collection, analysis, and interpretation of data pertaining to an abduction experience.

2.3 Mental Health Professional (MHP) A *mental health professional (MHP)* is someone whose objective is the diagnosis of, and treatment (therapy or counselling) for, disorders of emotion, cognition, behavior, and adjustment associated with an abduction experience. A MHP has licensing or certification reflecting training appropriate for providing mental health services.

II Guidelines

1. Competence and Responsibility

To maintain the integrity of investigation or treatment in regard to the abduction experience, investigators and MHPs have a responsibility to use only those skills and techniques they have acquired through training, education, or experience, and to maintain strict supervision of the assistants with whom they work. Investigators and MHPs should not misrepresent themselves in terms of competencies or credentials they do not have.

Although investigation and treatment are different activities with different objectives, they can interact. What an investigator does may indeed be therapeutic, but treatment, including supervising a "support" group, is not a function of an investigation. What a MHP does may indeed produce data valuable to an investigation, but an investigation is not a function of treatment. Because of these interactions, treatment and investigation may complement each other and be mutually beneficial, but they may also interfere with each other to the detriment of both science and the abduction experiencer. Therefore, it is important that investigators and MHPs are careful to practice only the skills for which they are qualified, and to practice them in a way that goals and obligations are clear and well-defined.

2. Minimal and Reasonable Risk

Investigators and MHPs working with abduction experiencers must strive to avoid causing them any mental or physical harm, and where such harm is foreseeable or unavoidable, they should strive to minimize it. The concept of "reasonable" or "minimal" risk is central to resolving ethical dilemmas in this regard. As defined in the

"Federal Policy for the Protection of Human Subjects" (Federal Register 1992) "*minimal risk* means that the probability and magnitude of harm or discomfort anticipated are not greater in and of themselves than those ordinarily encountered in daily life or during the performance of routine physical or psychological examinations or tests" (p 28013-4) In addition, whether or not risks are "reasonable" should be considered "in relation to anticipated benefits, if any, to the subject and the importance of the knowledge that may reasonably be expected to result" (p 28015)

In deciding whether the criteria of minimal and reasonable risk will be met, investigators and MHPs dealing with abduction experiencers should bear in mind that (a) the emergence of disturbing or traumatic experiences, while not inevitable, must always be anticipated, and (b) change in behavior or attitude, while not inevitable, must always be anticipated The eliciting of an abduction experience, or the bringing about of change in behavior or attitude, may be either harmful or beneficial But they will always represent a potential risk

3. Assessment

3.1 Assessment as "Abduction Experiencer" Investigators and MHPs should be very conservative in classifying an individual as an "abduction experiencer" This is not just an issue of semantics It is an issue of ethics and good science The personal consequences of false labeling or diagnosis, and the potential for methods of investigation and treatment to create artifactual abduction memories, require a conservative approach in identifying the abduction experiencer

The decision regarding whether or not an individual is an abduction experiencer must be based on the characteristics of the reported experience, and should initially be based on conscious recall as reported through interview and/or questionnaire responses

Although there is not yet consensus regarding what contextual or experiential elements are necessary or sufficient to define the abduction experience, the literature suggests certain elements as most characteristic These include (but are not limited to)

- recall of an abduction or encounter with apparently non-human entities,
- missing time related to recall of unidentified lights, objects, or apparently non-human entities,
- unusually realistic and emotionally intense dreams or dream-like experiences of UFOs or apparently nonhuman entities

Paranormal experiences, sleep disturbance experiences, UFO sightings, or a person's concern, belief or suspicion that they have been abducted, often accompany abduction experiences, but have not been indicated by the literature to be, *in and of themselves*, defining criteria of an abduction experience In these and other cases where a determination of "abduction experiencer" is marginal or uncertain (i.e., where "core" experiences are absent), a decision to explore for abduction expe-

riences, or treat *for the abduction experience*, should be made only if consultation among the investigator, a MHP, and the subject or client suggests proceeding is clearly in the best interest of the subject/client

3 2 Assessment for Risk To insure that the criteria of minimal and reasonable risk are satisfied, following assessment to determine if an individual is an abduction experiencer, assessment for risk should precede the decision to explore with that individual their known or suspected abduction experiences, or to treat *for the abduction experience*

3 3 Assessment for Risk in Investigations Prior to any investigation with a subject of their known or suspected abduction experiences, assessment should determine that no organic, mental, or behavioral disorder exists that would put the experiencer beyond a reasonable risk if an investigation were to proceed. Because the administration and interpretation of assessment tests or interviews requires special training, these should be administered by an appropriately trained MHP, or an investigator or collaborator acting under the supervision of a MHP. If, on the basis of such assessment, the MHP determines that investigatory procedures will put the subject beyond reasonable risk, an investigation should not be carried out.

3 4 Assessment for Risk in Treatment Any program of treatment to deal with presenting symptoms associated with the abduction experience should be preceded by professional assessment to determine that no organic, mental or behavioral disorder exists that should be treated directly.

4. Informed Consent

4 1 Nature of Informed Consent As part of any ethical program of investigation or treatment with abduction experiencers, the experiencer's participation must be voluntary and must be based on adequate knowledge of the investigation or treatment protocol and possible consequences. To obtain informed consent, the abduction experiencer must receive, in understandable language, enough information about the proposed study or course of treatment to enable them to decide whether or not to participate.

4 2 Elements of Informed Consent in Treatment In the case of treatment, informed consent should be obtained according to the ethical guidelines established by the MHP's professional organization. Particular attention should be paid to the nature and course of treatment (including any theoretical biases about the nature of the abduction experience that will impact the direction of treatment), risks and discomforts, fees, and issues of confidentiality.

4 3 Elements of Informed Consent in Investigations In the case of an investigation, the protocol should be provided in a written statement, which should include

- 1 The reasons for the study
- 2 The investigation protocol
- 3 Anticipated benefits
- 4 Anticipated risks, discomforts, and inconveniences
- 5 A description of procedures for confidentiality

- 6 A statement describing conditions under which the abduction experiencer's participation may be terminated
- 7 A description of fees or expenses, if any

This document should be signed by the abduction experiencer to signify the experiencer's acceptance of the protocol (A Sample Informed Consent Form is included as Appendix 2)

4.4 Informed Consent in the Use of Hypnosis Where hypnosis is involved, in either investigation or treatment, the abduction experiencer should be made aware of the particular characteristics of hypnotic recall including the inability to determine the veracity of the memory, increased conviction as to the veracity of the memory, and the potential effect of recall on perception and belief

5. Theoretical Bias in Treatment

Clients have a right to know the nature of the treatment to be provided by MHPs whose services they seek. MHPs whose theoretical position regarding the abduction experience requires adherence to, or prohibits them from considering, any particular theory of causality in guiding the initial or subsequent direction of treatment, should make this clear to their client

6. Ethical Protocol in Investigations

The abduction experience is often disturbing or traumatic. It has the potential to bring about lasting changes in belief, behavior, or attitude. Neither dealing with emotional distress, nor producing changes in belief, behavior or attitude, are proper objectives of an investigation, and they should be avoided. Mindful of these considerations, ethical investigation protocols will include

- 1 Procedures which minimize the leading or judgmental effects of any investigative interventions (e.g., interviews, questionnaires, hypnosis, etc.)
- 2 Termination of an investigation if risk to the experiencer's welfare is suspected
- 3 In explorations with a subject of their known or suspected abduction experiences, the availability of a MHP should the experiencer require or request their services
- 4 Instructions for contacting the investigator should questions or concerns arise subsequent to participation in the investigation
- 5 Hypnosis, when used, should be carried out only by an individual qualified in the procedure (see Part A, section II 1), and only after a MHP has determined that this investigative protocol is not contraindicated (see Part A, section II 3 3). Informed consent should be obtained (see Part A, section II 4 4)
- 6 Peer review and approval of the investigation protocol. Such a process is, according to U.S. and Canadian Federal policy, required practice for individuals affiliated with nearly all academic or medical in-

situations Investigators not so affiliated should seek review and approval from competent peers

- 7 Conclusions, when discussed with the subject, should be presented as the opinion or interpretation of the investigator and not as established fact, except where there is unambiguous corroboration from objective data Conclusions should be appropriate to the data, and the basis or rationale for such conclusions should be provided

7. Anonymity and Confidentiality

Both the information from and the identity of abduction experiencers are subject to the principle of confidentiality Investigators and MHPs have an obligation to protect the confidentiality of the abduction experiencers with whom they work This means maintaining confidentiality in data collection, record keeping, and data dissemination whether by writing, lecture, through the media, or conversation The maintenance and limitations of confidentiality should be provided to the abduction experiencer as a matter of informed consent

8. Dissemination

8.1 Access to Findings Ethical research allows critical assessment of one's work, and the opportunity to build upon the findings of others Therefore, ethical reporting of investigations will involve dissemination of or availability of the methodology and data from the investigation Similarly, MHPs who feel that they can advance knowledge through dissemination of the outcomes or insights derived in the course of treatment for the abduction experience are behooved to do so Dissemination of findings by both investigators and MHPs is subject to the restrictions as outlined in Part A, section II 7, Anonymity and Confidentiality

8.2 Dealing with the Media Investigators and MHPs dealing with the media or any public forum must be careful to ensure that the statements, commentary, and information they provide, are accurate, stem from appropriate clinical or investigative procedures, and are consistent with this Ethics Code

8.3 Participation of Abduction Experiencers The direct involvement of abduction experiencers with the media, at conferences and panels, or in any other public forum, is a decision that must be left to the abduction experiencer However, investigators and MHPs working with the experiencer should advise them about the potential hazards of public recognition and scrutiny Where the investigator or MHP is to be a participant, every precaution must be taken to avoid coercion or exploitation of the experiencer

9. Approaches to Ethical Issues

9.1 Familiarity with Ethics Code Investigators and MHPs working with the abduction experiencer population have an obligation to be familiar with this Ethics Code, and to apply it in their activities with abduction experiencers

9.2 Conflicts in Applying Ethics Code Because the abduction experience is not fully understood, and investigation and treatment with this population so rudimentary, particular situations may arise for which the application of these ethical stan-

dards of conduct is unclear. Also, investigators working strictly with databases, and investigators collecting data from abduction experiencers other than that pertaining to the experience itself, may be uncertain as to the applicability of sections of this Code to their investigations. In such cases, the investigator or MHP should seek guidance from other investigators or MHPs, and should try to resolve any ethical conflict in a manner as consistent as possible with this Ethics Code.

PART B. EXPLICATION OF ETHICAL GUIDELINES

I Comments on Definitions

1. Abduction Experiencer

The definition of "abduction experience," and even the choice of this term, concerns the investigator as a matter of science (it should correctly describe the phenomenon under study), concerns the MHP as a matter of diagnosis (it should correctly describe the event apparently responsible for the presenting symptoms), and concerns the individual reporting the experience as a matter of identity and self-image (it should correctly describe the characteristics of the experience as understood by the experiencer). However, the literature reflects a lack of consensus among MHPs, investigators, or experiencers regarding which contextual, emotional, or situational elements must be present to qualify as the target experience. Convictions are strong but agreement is not.

Nor is there consensus regarding the terminology to be used in reference to the target experience. The most widely used term in this regard is "abduction." Some object to this term because it implies an occurrence in event-level reality. Even among those who do not object to the term on these grounds, some object to it because they feel it precludes the possibility of experiencer compliance, or that because of its pejorative tone, it precludes the possibility of the experience being positive or beneficial.

The term "experiencer" is widely used to convey neutrality regarding the objective reality of the reported experience. Yet some object to this word for this very reason, or because they feel it conveys an insufficient sense of horror or victimization to the reported events.

Given this lack of consensus among investigators, MHPs, and experiencers, and the fact that any term will convey different meanings to different people, no definition or terminology used in this Code will satisfy everyone. Such a goal would be unrealistic. Rather, the position taken by this Code is that at this stage of empirical and theoretical development it is inappropriate to define or label the experiences under study in a way that assumes any particular conceptualization. The term and concept of "abduction experiencer" was selected for use in this Code because, despite the objections mentioned above, it is consistent with this position.

The use of "abduction experiencer," in this Code and in the literature, is intended to avoid injustice to any particular theory of causality. And it has the additional

advantage of immediate recognition. Nevertheless, the term does not perfectly describe the experience, nor has it always succeeded in its objective to convey neutrality. Although differences will continue as to how to best define these reported experiences, the investigative and clinical communities may well regard the development of better terminology as an ethical issue to add to their agendas.

2. Investigator

The term "investigator" is widely used in reference to those engaged in the study of the abduction experience. The term usually refers to a person who gathers and checks the facts and circumstances relating to the events and persons involved in a reported abduction experience. For the purposes of this Code the term also applies to individuals working strictly with databases, and those conducting experimental studies (systematic hypothesis testing) related to the abduction experience.

3. Mental Health Professional (MHP)

Professionals who provide treatment for emotional, cognitive, behavioral, or adjustment problems, whose training and experience are recognized by licensure or certification to practice, are known by many different titles (psychologist, psychiatrist, social worker, counselor, etc.). MHP is the umbrella term for all such professionals.

II Comments on Guidelines

1. Competence and Responsibility

1.A MHPs Competency for mental health providers is usually determined by governing boards within the profession, and/or by licensing or certification bodies. Prerequisite training and education is typically reflected in advanced degrees in the discipline. However, the abduction experience and its concomitant symptomatology are not typically part of a MHPs formal education and training. Therefore, MHPs working with abduction experiencers have a responsibility to familiarize themselves with the characteristics of the abduction experience, and with approaches to treatment that might be most productive in working with abduction experiencers.

1.B Investigators Although mainstream researchers have extensive training in research design and data analysis, "investigation" of an abduction experience does not necessarily require expertise at this level. However, investigators should be familiar with appropriate methods of data collection and interpretation. These include procedures to minimize the possibility of biasing testimony, or of generating data that are artifactual (i.e., that are a product of the investigative protocol itself, rather than the phenomenon under study). Investigators should also be familiar with appropriate procedures for interpreting the results of an investigation, so that interpretations are neither inconsistent with the results, nor go beyond what the results dictate. Investigators who do not have formal training appropriate for conducting an investigation, should seek guidance from those who do.

The Code specifies (Part A, section II 6) that "neither dealing with emotional distress, nor producing changes in belief, behavior, or attitude, are proper objectives of

an investigation.” This remains true for investigators who interact with their subjects outside of the formal investigation, and may be particularly problematic for investigators who are involved in group gatherings of abduction experiencers. Such gatherings, often referred to as “support groups,” may have a social function or may exist only to provide a forum for intellectual and philosophical exchange. However, they will always represent opportunities for affecting emotional distress, and for changing belief, behavior, and attitude. Because such outcomes may be inevitable, investigators should closely examine the role they play at such gatherings. In any case, investigators should not organize, supervise, or otherwise participate in support groups for the express purpose of bringing about such changes.

I C Hypnotists The practice of hypnosis for either research or treatment purposes is not licensed or regulated in most of North America. There exists no widely accepted licensing body or official standards for practicing hypnosis. However, some guidelines are provided by hypnosis societies which exist to provide education and training for professionals who use hypnosis in their work.

Requirements for using hypnosis in treatment are generally more rigorous than they are for using hypnosis in research. This is because in research settings the assessment for risk is not necessarily done by the individual doing the hypnosis, whereas in the clinical setting the hypnotist (the MHP) is expected to perform the assessment for risk. Also, the research hypnotist follows a defined experimental protocol in order to obtain meaningful data, whereas the MHP, whose overall objective is to produce clinical improvement, must have the clinical skills necessary to treat for presenting symptoms.

Individuals performing hypnosis for investigation of the abduction experience should do so within a protocol consistent with this Code. Such a protocol must include an assessment for risk before any investigation proceeds. Beyond that, such an individual should be (a) competent in the basic techniques of inducing and removing the hypnotic condition, (b) able to recognize and manage abreaction during the hypnosis session, and (c) competent in techniques appropriate to obtaining the desired experiential data during the hypnosis session.

Generally accepted guidelines for the use of hypnosis in treatment include the above, plus overall clinical competency in diagnosis and treatment of the presenting problem. MHPs should only use hypnosis to facilitate treatment with patients they are competent to treat without hypnosis.

2. Minimal and Reasonable Risk

The principle of minimal and reasonable risk is central to this Code, and to all professional codes dealing with ethical treatment of human subjects. However, it is not the objective of this Code to suggest guidelines that will *eliminate* risk. That is not a realistic goal. Almost any activity has the potential of harming its target audience. A book, a movie, a lecture, a conversation, even a “look,” all have the potential to cause distress. So too do routine interactions with peers, family, coworkers, and service and product providers. This is why minimal risk is defined in terms of the harm or discomfort that one may normally expect in everyday life.

The related notion of "reasonable" risk goes somewhat further. It suggests that some risk, even if minimal, may not be reasonable. Even minimal risk should be avoided if it is not a necessary consequence of the investigative protocol. The concept also acknowledges that some risk, even if greater than that expected in everyday life, may be reasonable if the potential benefits to subject and science are sufficient to justify the risk.

However, it is not always clear which risks qualify as reasonable. One purpose of this Code is to provide guidelines to aid investigators and MHPs in addressing this. Abduction experiencers are at risk if a preexisting disorder may be reinforced or exacerbated by an investigation. Abduction experiencers are at risk if they do not fully understand the vagaries of memory and hypnotic retrieval. They are at risk for abreaction to the memories. And they are at risk for the consequences of public disclosure.

Adherence to the guidelines in this Code will reduce risk from preexisting disorders through assessment. It will reduce risk from investigative protocols through proper informed consent, risk from abreaction through the availability of clinical intervention, and risk from public scrutiny by adherence to the principles of anonymity and confidentiality.

3. Assessment

3 A Assessment as "Abduction Experiencer" Prior to conducting an investigation with, or providing treatment for an abduction experiencer, it should first be determined that the individual is an abduction experiencer. This prohibition on treatment and investigation prior to assessment does not preclude the collection of demographic data about an individual, obtaining a psychological and medical history, or administering questionnaires or interviews to gain a record of an individual's conscious recall of significant life events and experiences. In fact, such activity is required to make the assessment of "abduction experiencer," and to determine whether or not to work with an individual as an abduction experiencer.

Investigators and MHPs should bear in mind that *it is not the purpose of this assessment to determine whether or not an actual abduction has taken place, or to determine any other causative agent to the abduction experience*. A determination of causality is an interpretation of the abduction experience, and may be a goal or outcome of investigation or treatment. This assessment only determines whether an individual's experiences qualify for study or treatment for the abduction experience.

As a guide in making such a determination, the Code lists a number of characteristic elements of the abduction experience (based on the abduction literature). Unfortunately, there are no standardized diagnostic criteria for identifying an individual as an abduction experiencer. Because of this, and because of the risks associated with false labeling and misdiagnosis, the Code prescribes a conservative approach in making the assessment of "abduction experiencer."

This is especially important in regard to experiences which are only suggestive of an abduction experience. The examples cited in the Code (paranormal experiences, sleep disturbance experiences, UFO sightings, or a person's concern, belief or suspi-

cion that they have been abducted) have all been reported as associated with abductions, but are also frequently encountered in individuals who have not had an abduction experience. In these marginal cases, the potential for false labeling or diagnosis is even greater, and so greater caution is indicated. The Code does not preclude investigation or treatment in such marginal cases. However, investigation of or treatment for the abduction experience should proceed only if consultation among the investigator, a MHP, and the subject or client suggests it is in the subject's or client's best interest (see below).

3.A.1 Determining "Best Interest" Determining what is "in the best interest" of the subject/client is different from assessing for risk. The assessment for risk asks the question, "Is there any way that the techniques of investigation or treatment for the abduction experience would put the subject at unreasonable or unnecessary risk of harm?" The determination of "best interest" asks, "Will it benefit the subject/client to engage in investigation or treatment for the abduction experience?" Such a determination includes elements of assessment for risk and informed consent, but goes further. Where the subject's experiences are believed to be marginal or uncertain, it may include a frank discussion with the individual that for them, the potential of creating artifactual experiences may be especially great.

The determination of "best interest" provides the subject/client an opportunity to express expectations and needs, and to obtain counseling, reassurance, and information without being subjected to a formal course of investigation or treatment. Because diagnosis and the provision of counseling are the purview of a MHP, the Code specifies that a MHP should be included in any discussion to determine "best interest."

3.A.2 Using Hypnosis for Assessment The issue of using hypnosis as a method of assessment, rather than as a method of investigation, deserves special comment here. It may be fairly argued that emergence of abduction experiences during hypnosis is a core characteristic of the phenomenon, and may therefore be a critical procedure in assessment for the abduction experience. However, investigators have only demonstrated the ease with which abduction experiences can be elicited, not that the elicited experiences existed (albeit in "hidden" form) independent of the hypnotic procedure. On the other hand, the propensity of hypnosis to generate false memory is well established (see Part B, section II 6 B). Therefore, the concept of "reasonable risk" precludes this Code from advocating the use of hypnosis in the assessment phase. To use hypnosis to "search" for abduction experiences in order to identify someone as an abduction experiencer, would begin an investigation not only prior to assessment as an abduction experiencer, but also prior to assessment for risk.

3.A.3 Communicating the Results of Assessment as Abduction Experiencer If an assessment of "abduction experiencer" is to be communicated to the individual, the investigator or MHP should make it clear that this is not an assessment that the individual has been abducted, but only that their experience is consistent with those that define the abduction experience. In making these assessments, the same courtesy, respect and compassion investigators extend to abduction experiencers should

be offered to individuals who are deemed not to be abduction experiencers. Investigators should validate those individuals' subjective experiences, but not offer an analysis of the nature or source of those experiences.

3 B Assessment for Risk In order to avoid risk, one must be aware of risk. For this reason, after the assessment for being an abduction experiencer, assessment for risk is necessary.

3 B 1 Sources of Risk Assessment for risk is primarily concerned with the following potential sources of harm:

(1) Ignoring a psychiatric or medical disorder because of assessment as an abduction experiencer Some known conditions have symptomatology which resembles features of the abduction experience (see Appendix 1). Individuals suffering from such disorders are usually highly motivated to find an explanation for their discomfort, to put a name on what has been troubling them. Most often, these disorders are "hidden" in the sense that patients may deny, minimize or rationalize their symptoms. Sufferers of such disorders may recognize their own experiences in accounts of the abduction experience, and present for investigation or treatment of their own suspected abduction experience. When such individuals receive an explanation from a trusted individual, whether investigator or MHP, that their symptoms are those of an abduction experience, it is far less likely that their condition will ever be properly diagnosed or that they will seek treatment for their disorder. The persistence of some symptoms may be further rationalized by the belief in abductions and that they are a life-long and uncontrollable process.

(2) Attributing independent symptomatology to the abduction experience Coexistence of an abduction experience with physical or psychological complaints does not necessarily mean that the abduction experience is the cause of the complaints. For instance, while some proportion of abduction experiencers may suffer from anxiety or depression, it does not follow that the anxiety or depression is necessarily caused by the abduction experience, or that investigation or treatment for the abduction experience is indicated.

In both situations noted above, the potential risks to the subject or client are (a) depriving them of timely diagnosis and effective treatment, and (b) exacerbating the symptomatology through investigation or treatment for the abduction experience. Failure to screen for such confounding conditions in investigations also decreases the meaningfulness of the conclusions of the investigations.

3 B 2 Responsibility for Assessment The Code specifies that a MHP, or an associate working under the supervision of a MHP (this person may be an "investigator") is required to make this assessment because only a MHP has the qualifications to (a) make the determination (diagnosis) of whether disorders exist, and (b) make a professional judgment as to the extent to which they represent a risk in investigation or treatment for the abduction experience. For example, the presence of dissociative phenomena is not necessarily pathological, and may not preclude an investigation. Also, the presence of pathology does not in and of itself preclude investigation.

Investigators who have not established a collaborative arrangement with a MHP

should restrict their activities to data collection as appropriate for assessing whether an individual is an abduction experiencer. These initial contacts, if done appropriately, should furnish a considerable amount of information and may be suitable for the purposes of both investigator and experiencer. However, this will effectively preclude an investigator in such a situation from fuller investigations, including those using hypnotic regression.

This is regrettable, but unavoidable. Minimizing risk (and therefore assessing for risk) is a *sine qua non* of ethical investigation, and this is true regardless of the availability of assessment resources. Increasing the number of MHPs knowledgeable about the abduction experience and willing to provide risk assessment services should, therefore, be a top priority of the investigation community.

3 B 3 MHPs' Function in Assessment The role of the MHP in making an assessment is to identify contraindications to an investigation process which the experiencer is already interested in pursuing. The MHP is not making a referral to the investigator. This is an important distinction, because making a referral implies the MHP is assuming some responsibility for the investigator's competency and the outcome of the process. Furthermore, when the MHP is a psychiatrist, such a referral may be in contravention of the following guideline of the American Medical Association:

A psychiatrist should neither lend the endorsement of the psychiatric specialty nor refer patients to persons, groups, or treatment programs with which he/she is not familiar, especially if their work is based only on dogma and authority and not on scientific validation and replication [Karasu 1980]

MHPs may be reluctant to become involved in the assessment process out of concern that simply by performing the assessment, they are tacitly endorsing the investigator. To safeguard the MHP and make the MHP's responsibilities clear to all concerned, the MHP needs to specify whether the assessment is based on specific knowledge of the investigator and the procedures to be followed, or simply a general screening for contraindications to abduction investigation (and hypnosis, if relevant). In either case, common sense dictates that a MHP agreeing to perform such assessments should have some degree of familiarity with the referring investigator and his or her practices.

A determination by a MHP that investigation will not put the subject beyond reasonable risk is not a guarantee against an unfavorable outcome. Reasonable risk is not an absence of risk. The MHP's only responsibility in the case of an unfavorable outcome is if such an outcome was due to a preexisting disorder or risk factor that the MHP ought to have detected during the assessment.

3 B 4 Assessment for Risk in Treatment The principles outlined above apply equally to MHPs engaged in treatment of experiencers. A MHP engaged in treating an individual for the abduction experience is obliged to do a complete assessment, as they would for any new client.

4. Informed Consent

The concept of informed consent means much more than simply obtaining an abduction experiencer's agreement to cooperate with an investigation. It is critical that the experiencer understand the nature of the procedures to be used, and the possible consequences of participation in the investigation. This is especially important when the investigative protocol calls for use of an intervention that may put the experiencer at risk, or the effect of which is uncertain. It is the responsibility of the investigator to obtain informed consent under circumstances that allow the experiencer to make an educated decision regarding participation in the investigation, and in a way that minimizes coercion or social pressure to participate.

To accomplish this, investigators should provide in a clearly written statement, a description of the investigative protocol and an explanation of the potential risks and benefits of participation. They should not exaggerate potential benefits, misrepresent the potential for risk, or present unjustified promises regarding the efficacy of the investigative method (see Sample Informed Consent Form, Appendix 2). To convey such information properly, informed consent requires that investigators be well informed themselves (see Part B, section II 1 B on Competency).

Informed consent should always be obtained in a way that provides the experiencer every opportunity, during as well as prior to the investigation, to decide whether or not to participate. Investigators should also be aware that no language incorporated into the informed consent statement can waive the experiencer's legal rights or release the investigator from liability for negligence. This is not just a matter of ethics. It is also a matter of law.

There is one situation in which some elements of informed consent may be waived. That situation is when the investigative protocol would be severely compromised by obtaining such consent. For example, in studying the effect of expectancy on hypnotically retrieved memory, proper informed consent regarding hypnosis (see Part A, section II 4 4) might interfere with the ability to assess expectancy.

In a case such as this, it would be the investigator's responsibility to demonstrate that the potential benefits to science outweigh the potential consequences to the subject. The investigator should also be able to demonstrate that there is no alternative protocol that would answer the experimental question without creating unnecessary risk. In any case, at the conclusion of the study the subject should be properly informed (debriefed), about the true purpose and design of the experiment.

Ethical dilemmas of this sort are commonplace in the behavioral sciences. The difficulty in resolving them objectively is one of the main reasons human subject research is expected to undergo peer review, and why this Code indicates peer review should be a standard component of "ethical protocol in investigations" (see Part A, section II 6).

5. Treatment

Because the cause of the abduction experience has not been satisfactorily established by either scientific or clinical standards, the MHP should not present his or her beliefs, or conclusions derived from his or her personal or professional experi-

ence, as verified *fact*. This does not preclude the MHP from informing the client about the MHP's own personal or professional *opinion*. Indeed, the Code directs the MHP to provide such information to the client if such opinion is to guide the course of treatment. This allows the client to make an informed decision about whether or not to proceed based on knowledge of which possible theoretical approaches (and hence treatment strategies) will and will not be considered.

In informing clients that treatment may not establish the cause of the abduction experience, MHPs should be sensitive to experiencers who feel certain about the origin or explanation of their experiences. Such clients may feel they, their experience, or their interpretations are being devalued. The MHP should make it clear that uncertainty as to the cause of the experience is a limitation of the clinical and scientific methods and not a reflection of the MHP's perception of the client's beliefs.

MHPs may be interested in a client's abduction experience not only as a possible cause of presenting symptoms for which to treat, but as a source of information from which to advance knowledge. However, the MHP should not act as both investigator and clinician. Investigation of an abduction experience focuses on the details of the experience. Treatment focuses on the meaning and effect of that experience for the individual. When the focus on the client is diluted by the interests of investigation, research, publication, or other agendas, the client's interests are not served. A MHP working with an experiencer as an investigator should refer the experiencer elsewhere for treatment services.

6. Ethical Protocols

6 A Ethical Protocol in Investigations Good research is also ethical research. First, ethical research provides for better data. When an investigation is carried out along ethical guidelines (informed consent, minimized risk, assured confidentiality and anonymity) subjects are more cooperative and more willing to communicate candidly and honestly. Data, therefore, are less likely to be misinterpreted. Data are also less likely to be misapplied, either in directing future research or in guiding treatment.

Second, ethical research serves science as an institution. It fosters respect for the activity of science and for its practitioners. This generates support from the public, the media, and from science's governmental and private benefactors.

Third, ethical research promotes education. Because ethical research is more likely to provide a correct understanding than is research not conducted along ethical guidelines, and because such research is more likely to receive support both from within and without the institution of science, it increases the likelihood that accurate knowledge will be disseminated into the public domain.

By following the ethical guidelines in this Code, investigators of the abduction experience will not only be protecting the experiencers with whom they work from harm, they will also be strengthening the perception and reality of abduction research as a legitimate endeavor.

6 B Ethical Protocols and Hypnosis The use of hypnosis in abduction research is so widespread that "investigation" of an abduction is often used synonymously with

hypnotic regression. Because of the prevalence of this particular investigation protocol, its use deserves a close examination

6 B 1 General Risks Associated with Hypnosis General risks associated with hypnosis in almost any setting (Kleinhauz and Eli 1987) include overtly distressing effects such as spontaneous abreactions, acute anxiety states, and decompensation of latent pathology, including depression, dissociative disorders and latent psychotic states. Incidence of these can be reduced (but not eliminated) by appropriate screening and diagnosis beforehand (see Part B, section II 3 B, "Assessment for Risk"), and managed when they occur by techniques familiar to the appropriately trained practitioner. For this reason the Code requires assessment for risk prior to investigation or treatment, and suggests that investigators and MHPs using hypnosis be competent to recognize and manage abreaction.

6 B 2 Risk for False Memories Many abduction investigators and MHPs, as well as the public at large, assume that memories retrieved under hypnosis are substantially accurate and reliable. However, it is generally understood within the mental health profession that it is impossible to determine the historical accuracy of memories obtained through hypnotic regression without reference to objective records, regardless of the intensity of affect expressed during recall or the percipient's conviction of accuracy.

Moreover, within the mainstream scientific literature it is well-established and widely acknowledged that hypnotically retrieved memory is often not reliable. The content and acceptance of hypnotically recalled memory is affected by the inherent suggestibility of the hypnotic condition, the set (internal factors, i.e., expectation, intention, personality, mood, values, attitudes, beliefs), and the setting (external factors, i.e., context, physical and social environment, expectations and behavior of others present). Although the applicability of these studies to the abduction experience has not been determined, they suggest the potential for "creating" abduction experiences. The existence or scope of this problem is not known, but its potential puts subjects and clients at risk.

These limitations do not preclude hypnosis from being used in an ethical manner in both investigation and treatment of the abduction experience, just as it is regularly used in investigation and treatment in other areas. However, they do require that abduction experiencers be properly informed regarding the characteristics and inherent limitations of hypnotic memory, and that results be interpreted in light of these characteristics and limitations. As with more conventional kinds of memory, hypnotic recall may be accurate or inaccurate. But as with more conventional memory, in the absence of compelling independent evidence, no hypnotically retrieved abduction experience can be *assumed* to be accurate, and it is unethical to not make this clear to those considering undergoing hypnotic regression.

6 B 3 Risk for Changes in Cognition and Behavior Abduction experiencers are at risk because of the potential of hypnotic recall to change their perception of self, personal history, and the nature of reality itself. This kind of change may sometimes be a goal of treatment, but it does not fall under the purview of investigation. The

situation is further complicated because experiencers who actively seek out investigators are not just interested in contributing to the knowledge base of abduction research, but also in undergoing a personal process of exploration

Because hypnotic regression may bring about long-term changes in a subject's behavior or attitudes, abduction experiencers are at risk for effects that should not be a consequence of an investigation. While such changes are not an inevitable outcome of using hypnosis, the possibility of such an outcome dictates that abduction experiencers be made aware through proper informed consent that such change is not an objective of the investigation, but that the possibility of such change exists.

6 C Discussing Conclusions with Subject Conclusions from an investigation may be considered as fact only if there is unambiguous corroboration from objective data. In the absence of such corroborative data, investigators should adhere to the same ethical guideline regarding interpretation as described for MHPs in section 5. Do not present opinion or belief as fact. Subjects are entitled to (and expect to) discuss the results of the investigation with the investigator at the end of the process. The results may be presented as consistent with a particular hypothesis, or as consistent with the results of other cases. But the interpretation of results should be presented as interpretation, not as fact. The investigator should exercise tact and sensitivity in doing this, particularly where their interpretation is different from that of the subject.

7. Anonymity and Confidentiality.

Conducting investigation, research or treatment with abduction experiencers will, as a matter of course, collect sensitive, personal information. Such data should normally never be linked to or associated with any identified individual when findings are discussed with or disseminated to the media, the public, or other colleagues. If an experiencer has explicitly waived confidentiality, then this restriction can be relaxed.

Sensitive information must also be safeguarded when it is collected and recorded by such means as ID numbers and restrictions on physical access to the records. Provisions for confidentiality should be described in the informed consent form (see Appendix 2).

MHPs are typically mandated by law not to disclose information collected in their work with a client. Although this constraint should not prevent MHPs from providing assessment services to investigators, it should clearly be understood by investigators that much information will not normally be disclosed by a MHP, unless absolutely necessary or when consent has been given by the experiencer.

8. Dissemination

8 A Access to Findings Critical assessment of the work of others is a key to scientific advancement. To do this, scientists must be able to study not only the information collected but also how it was collected (the methodology). Findings should be reported fully and without omission of significant data. Details of theories, methods, and research designs that might bear upon the interpretation of findings should be disclosed. Investigators should cooperate in efforts to make raw data

and associated documentation available to others, always insuring that confidentiality not be violated

8 B Dealing with the Media As stated in the Premises, no theory as to the cause of the abduction experience has been scientifically or clinically established. While not meant to discourage either vigorous debate or the adoption of any particular position vis-a-vis the ultimate cause of the abduction experience, this section of the Code directs investigators and MHPs to present evaluations of the meaning and significance of the abduction experience, and their own findings, with caution and due regard for the limits of current knowledge.

8 C Participation of Abduction Experiencers Public appearances for experiencers are not without risk. Scientific meetings are not support sessions but are for critical discussion and debate. Intensely personal experiences, as well as one's own mental stability, personality and character, can come under acute and sometimes unsympathetic scrutiny. The same is true of media presentations, especially television and radio interview programs, many of which are fundamentally entertainment vehicles designed to produce dramatic conflict.

Such situations can be especially difficult for someone who has not come to terms with his or her own experience. Given that treatment and investigation are confidential, and in practice the MHP or investigator is usually responsible for presenting a request for an appearance to the experiencer (unless this person has already "gone public"), the following points should be carefully considered:

- The nature of the meeting or program should be known and any risks should be made clear to the experiencer
- The benefits to the experiencer, if any, should be discussed
- The MHP or investigator should in no way coerce, encourage, or otherwise influence the experiencer to participate. For conferences, the use of videotaped or audiotaped presentations may be considered as an alternative to a live appearance.

At all times, the MHP or investigator must remember that the decision to participate is solely that of the subject, and that it is best to err on the side of caution.

9. Approaches to Ethical Issues

There will be times in the career of anyone involved in the abduction experience when the ethical choice is difficult to identify and/or to carry out. In situations where the ethical choice is unclear, consultation with other investigators and MHPs on the best way to proceed consistent with the Code is encouraged. Consultation should not be restricted to those within one's usual circle or theoretical orientation, nor even to those familiar with the abduction experience. Receptivity to all points of view, even those uncomfortable to hear, is essential to successful resolution of ethical conflicts.

In assessing one's day-to-day practice, as well as in resolving ethical dilemmas, investigators and MHPs should be guided by the spirit as well as the letter of the

Code *Do no harm* Make sure there are no problems that will be made worse by what you do, and try not to create new problems

APPENDIX 1

SYMPTOMATOLOGY SIMILAR TO THE ABDUCTION EXPERIENCE

There are a number of psychiatric and organic disorders, psychological phenomena, and paranormal phenomena, with elements similar to those found in the abduction experience. Some of these are notoriously difficult to identify unless one is aware of them and actively looking for them. This section presents examples of disorders or phenomena which have characteristics in common with the abduction experience. It is not an exhaustive list.

Dissociation and Dissociative Disorders

Dissociation is a lack of the normal integration of thoughts, feelings, and experiences into the stream of consciousness and memory. Dissociative disorders are common in the general population, with a prevalence rate of dissociative experiences in the adult population estimated at between 3 and 10% (Ross, Joshi, and Currie 1990, Ross 1991). There is no difference in prevalence of dissociative experiences between men and women, and no relation to income, employment status, education, place of birth, or religious affiliation (Vanderhinden, Van Dyck, Vandereycken, Vertommen 1992). The following are dissociative experiences (Steinberg, Rounsaville, Cicchetti 1991):

- **Amnesia** Inability to recall a specific and substantial block of time or important personal information that is too extensive to be explained by ordinary forgetfulness
- **Depersonalization** Detachment from oneself, a sense of "going through the motions," or of looking at oneself as an outsider
- **Derealization** The sense that one's surroundings are unreal
- **Identity confusion** Subjective feelings of confusion, uncertainty, or puzzlement centered on the person's knowledge of one's own identity
- **Identity alteration** Behavioral manifestations of identity confusion, such as exhibiting a special knowledge or skill without knowing how this skill was acquired
- **Fugues** Sudden unexplained travel away from home or one's customary place of work, with inability to recall one's past

Investigation and treatment for the abduction experience may exacerbate dissociative pathology, and may solidify the suspicions of individuals with dissociative

disorders who have identified in the popular abduction literature experiences that match their own, thus "creating" abduction experiencers

Disorders of Sleep and Wakefulness

Many characteristics of the abduction experience are also seen in a variety of sleep disorders which represent admixtures, incomplete declaration, or rapid oscillations of sleep and wakefulness (Mahowald and Schenck 1992) The prevalence of such sleep disorders is unknown The prevalence of narcolepsy, a clearly defined syndrome with specific clinical and laboratory diagnostic criteria, has been cited as up to 1 per 1,000 (Mitler, Hajdukovic, Erman and Koziol 1990)

Three symptoms of sleep disorders which are consistent with symptoms associated with the abduction experience are (Mahowald and Schenck 1991, Mahowald, Schenck 1992, Mitler, Hajdukovic, Erman and Koziol 1990)

- Sleep paralysis—the persistence of REM atonia into wakefulness
- Waking dreams—vivid and bizarre experiential states, associated with the intrusion of REM dream mentation into wakefulness
- Automatic behavior, such as driving long distances with no memory for the journey—the admixture or rapid oscillation of wakefulness and NREM sleep

"Old Hag"—Nocturnal Assault

David Hufford's studies of nocturnal assault experiences are reminiscent of the nocturnal visitations reported by some abduction experiencers Hufford demonstrated that this phenomenon, called being "hagged" or visited by the "Old Hag" in the culture of Newfoundland and Labrador, is not culture-bound, but is found under different names in different cultures Hufford concluded that the "Old Hag" experience is something other than simple suggestion modified by a cultural myth (Hufford 1982)

Hufford defined an "Old Hag" experience in terms of four "primary features "

- Subjective impression of wakefulness,
- Immobility (variously perceived as paralysis, restraint, fear of moving),
- Realistic perception of actual environment,
- Fear

Sense of presence, a numinous quality, and a fear of death were common secondary features Most, if not all, of the above also figure prominently in the abduction experience, although there is much in the abduction experience that is not part of the Old Hag

Paranormal Experiences

Abduction experiences overlap with experiences described in the paranormal literature. Floating and passing through walls have features in common with out-of-body experiences. Visions of lights and unusual entities are reported in both the paranormal and abduction experience literature. Unaccounted-for movement of objects is attributed in the paranormal literature as the effect of psychic forces (psychokinesis or poltergeists), and in ufology as associated with the abduction experience.

A number of recent studies have found a correlation between paranormal experiences and dissociative experiences (Richards 1991, Ross and Joshi 1992, Ring 1992, Schwartz 1983). With specific reference to UFO encounters, Ring (1992) concluded that "individuals who report UFO encounters and near-death experiences appear to have a greater likelihood of showing dissociative tendencies in their psychological functioning."

APPENDIX 2

SAMPLE INFORMED CONSENT FORM

The following is provided as a sample consent form that might be used to obtain informed consent from an abduction experiencer considering participation in an investigation in which hypnosis will be used. It incorporates the seven elements of informed consent described in the Code, and is consistent with the Code's guidelines for minimizing bias or misrepresentation. It is assumed that investigators may want to modify this form in accordance with their own preferences for language, or the specific requirements of their investigation protocol. (A version of this consent form appeared as part a paper presented at the Abduction Study Conference at M I T)

Investigation of Anomalous Experience

The purpose of this study is to learn more about "anomalous" (uncommon, unexpected) experiences that many individuals have reported. An investigation of your own experiences will help researchers in this effort. To advance knowledge most effectively, it is important that information learned in this investigation be shared with other investigators. In doing so, unless otherwise authorized by you, your identity will remain anonymous and any information that would allow identification will be held in confidence.

To learn more about your experiences, we are interested in the details of what you can now remember. We are also interested in what you do not remember at this time. Many of the events that we experience are remembered accurately long after they have occurred, but sometimes we "remember" events inaccurately, and sometimes we do not remember them at all.

Hypnosis has been used in a variety of situations for the purpose of retrieving forgotten memories, and it will be used in this investigation. Although hypnosis is

often portrayed as a very mysterious process, it is simply a procedure which instructs you to re-direct your attention, and in so doing may alter your thinking, perception, or consciousness. However, hypnotic memory—just like regular memory—may be misleading and inaccurate (yet subjectively convincing), or simply remain unretrievable. Therefore, neither the success of hypnosis in retrieving new memories, nor the objective accuracy of these memories, can be assured. Nevertheless, it has been successful in many cases, and may also help you in remembering events that you cannot recall at this time.

Hypnosis is a safe procedure for most people. However, for some individuals, hypnotic regression may not be advisable. For your own protection, a clinical psychologist will determine whether your medical or psychological history suggests any factors for which proceeding with hypnosis would not be advisable.

Although hypnosis itself is safe, any new memory (whether through hypnotic or regular recall) has the potential to affect you. A new memory may provide comfort and it may relieve anxiety or uncertainty. On the other hand, it may be unpleasant or discomforting. Further, new memories may bring about changes in your attitudes, beliefs, or behavior. You should understand and accept these possibilities if you choose to participate in this study.

Also, it is neither the purpose of this study, nor within the investigator's area of expertise, to provide counseling in regard to experiences you already recall, or experiences that may be elicited during the investigation. If this is a concern of yours, you may wish to consider alternative services which are available to you.

Successful hypnosis sometimes requires participation in more than one session, and the duration of any individual session depends on the nature of the material retrieved. However, it would be reasonable to expect that each session will last at least ½ hour although it could be considerably longer. If time is a consideration for you, please let the investigator know. You may, of course, withdraw from participation at any time and are under no obligation to return for additional sessions. Also, your participation in this study does not oblige the investigator to continue beyond the time frame the investigator feels is appropriate. In any case, it is the intent of the investigator to work with you so that the exploration of your experience is mutually satisfactory.

Statement of Consent to Participate in Investigation of Anomalous Experience

1 I have read the attached information describing the purpose and method for this Investigation of Anomalous Experience. I have had an opportunity to ask questions regarding this investigation, and all questions have been answered to my satisfaction. I understand that if I have questions or concerns in the future regarding this investigation, I may contact

2 The availability of professional services has been explained to me. Mindful of

such options, I agree to participate in this investigation as described I understand that I may withdraw from participation at any time I also understand that the results of this investigation may be made available to other investigators, and that my identity will remain anonymous, unless I agree otherwise

Name (please print) _____

Signature _____

Date _____

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AN ASSESSMENT OF THE CROP CIRCLE PHENOMENON

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It was in the 1980s that public attention was drawn to the formation of the so called crop circles in a fairly restricted area of southwestern England. Increasing media attention went hand-in-hand with increasing frequency of observations until in the late 1980s they reached a kind of enthusiastic hysteria. Organizations formed, doing nothing but studying, photographing, describing and interpreting the observed pattern, giving themselves constitutions, publishing journals dedicated exclusively to crop circles, and calling themselves "cereologists". Once the BBC and the Army (which owns some of the ground favored by the crop circles) invested respectable sums in observing equipment and TV reports, crop circles became big business and produced numerous beautifully illustrated books, some of them best sellers. The only steady figure among these cereologists remained Dr Terence Meaden, a careful observer who wrote one of the first books on crop circles *The Circles Effect and Its Mysteries* (Meaden 1989a).

In the meantime, crop circles went through three stages: an early, simple stage consisting of one circular area containing flattened crops in a spiral pattern (Figure 1), followed by multiple circle arrangements such as quintuplets (Figure 2), then its most recent stage in which huge, complex pictograms were found (Figure 3). No reliable numbers are available, but it has been estimated that at the height of the publicity a total of 800 crop circles were found in one year. With the appearance of the two elderly hoaxers Doug and Dave in 1991, public interest declined and so did the number of crop circles, much to the regret of farmers who charged admission to thousands of visitors inspecting the pictograms on their fields.

DESCRIPTION

The simple crop circle of the first stage has remained the basic unit of the later stages and any assessment of their origin has to start with it. The name is misleading. These are not circles but circular areas of depressed crops. They have four characteristics:

- 1 The crop is flattened in the depressed area



Fig 4 Purported UFO landing site near Killaly, Saskatchewan, discovered November 14, 1979, after a white light had been seen the previous night. The witness found a ring of smoldering grass 30 feet in diameter with snow still visible in the center of the circle.

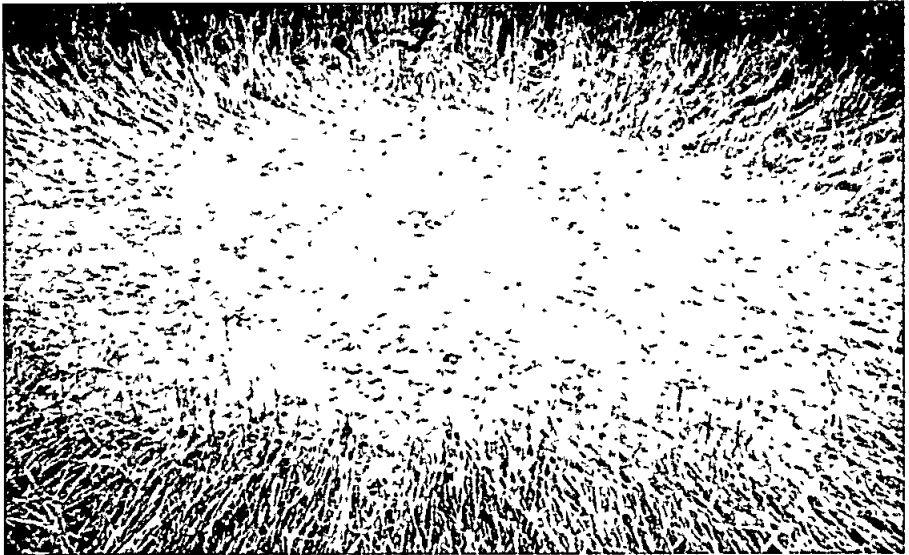


Fig 5 Ground marking found in a wheatfield near Dellroy, Ohio, on June 28, 1965. Wheat was completely missing from the center of the ring. At the edges the stubble was broken outward at an angle of 20° – 30° from the horizontal. No UFOs had been reported.

- 2 There is a sharp, circular wall consisting of vertical, unaffected crop stems surrounding the flattened area
- 3 There is a spiral pattern diverging—in the great majority of cases—outward as evidenced by the direction of the flattened crop stems
- 4 The individual stems are never broken, only bent, even if they are 1 cm thick As a consequence, they keep growing horizontally

It has been stated that crop circles form both day and night, but the vast majority has formed during the night It has been stated that they are related to UFOs, but about 90% of the English observations are not associated with UFO reports It has been stated that they form on the lee slopes of mountains, but there are just as many, if not more, on flat ground

Are crop circles a new phenomenon? Yes and no UFO-related circles have been reported for many decades in various parts of the world, but never in the numbers reported from the U K They were usually called “rings” (another misleading term) In his monumental work Ted Phillips (1975) describes more than 700 trace cases Among these are 107 rings, of which 53 are directly associated with alleged UFO landings

That is one difference from the English crop circles There are two more In the earlier UFO ring cases the grass or crop is not only flattened, but destroyed, dehydrated, or totally burnt (Figure 4) Furthermore, the vertical outer wall is always missing (Figure 5) In rare cases spiral patterns were recognized In fact, there was an instance in an Australian swamp in which water was still swirling immediately after the alleged UFO took off

INTERPRETATIONS

Although the prevailing interpretation of the earlier ring cases was that they were UFO-related, there were other hypotheses, such as outward growth of fungus For the U K crop circles, a long menu of explanations has been offered in the literature and at conferences atmospheric vortices, plasma phenomena, soil anomalies, animal tracks, hoaxes, channeling, dowsing, ley lines (energy paths), alien intelligences trying to communicate, and of course UFOs Most of these purely speculative hypotheses will be ignored here

Originally, Terence Meaden—often quoted or attacked by his fellow cereologists—had proposed atmospheric vortices, especially *dust devil* vortices during the day and *descending vortices* during the night Later on he advanced the so-called *plasma vortex* theory These hypotheses are worth discussing

Stage 1 Simple Circles

The atmospheric boundary layer is probably the best observed part of the earth's atmosphere There have been thorough investigations of dust devils A dust devil forms over heated ground, best visible in the desert, but less or not at all visible over

other types of heated surfaces. Dust devils are highly turbulent and usually leave a trail of debris, not a neat circular imprint. Their wake is hundreds of meters long as they drift along with mostly moderate winds.

But there is more. Figures 6 and 7 show cross sections through dust devils carefully investigated by Sinclair (1973) and others. Notice the pressure drop in the center and the tangential velocity v which reaches its maximum of more than 10 m/s outside the narrow dust core and falls off gradually toward larger radial distances. Nowhere do you see the sharp discontinuity required for a well-defined circle wall. The radial velocity is mostly negative (inwards) outside the narrow core, again falling off gradually outwards. In other words, the air is spiraling towards the central pressure minimum. There is a convergent flow, not a divergent one as observed in most crop circles.

Could it be that the actual core of the dust devil is the crop circle? The inner core contains descending motion, surrounded by a mantle of rapidly rising, dust-carrying air. Inside the core the radial motion is positive (outwards). However, not only is the core much too narrow, the maximum tangential winds lie just outside the core and again the crop would be flattened mostly in the outer, inward spiraling area without any wall discontinuity. In other words, the dust devil does not fill the bill.

What about the nocturnal occurrences? Is there such a thing as a *descending vortex* that leaves a sharply defined imprint? Everyone knows about tornadoes touching the ground. These are gigantic events descending from thunderstorm clouds with preexisting vorticity. There are also the so-called *downbursts* that have caused airline crashes. Again, these are highly turbulent events connected with strong, moist convection. Is there a small, mild, nocturnal counterpart that just touches the ground and retreats again leaving a brief, sharp imprint? The nightly boundary layer is generally stable and calm and decoupled from the warmer and faster flow above. None of the experts on atmospheric boundary layer flow whom I questioned knows of such a phenomenon and I personally am not aware of one either. Though we cannot accept the hypothesis at this time, we cannot really exclude it.

Let us turn to the next hypothesis. *Lee vortices*, vortices with vertical axes forming on the lee (downwind) side of hills. All existing observations indicate that lee vortices are transient, highly turbulent, moving features. Arnt Eliassen—one of the most respected atmospheric scientists alive—suggested in 1991 the following possibility. Often, during cold nights, a stagnating air mass rests in valleys topped by a temperature inversion. This air mass is decoupled from the flow aloft. If a moderate to strong upper flow passes over the hill and creates traveling lee vortices in the layer above the stagnating air mass, their pressure field may induce a rising and rotating motion within the resting stable air mass underneath, which occasionally may touch the ground leaving a vortex-like imprint on the vegetation. Eliassen calls it an analogue to a “centrifugal pump” and intends to model this concept mathematically. Until we learn more about it, we must suspend judgment. Of course such a mechanism could only apply in hilly terrain. Again, one would expect the flow spiraling inward, at variance with the crop circle observations.

Stage 2 Multiple Circles

As mentioned earlier, Stage 2 of the crop circle formations began in the mid 1980s when the circles multiplied in new patterns. With increasing frequency quintuplets, triplets, and other arrangements of multiple crop circles were found. Meaden now advanced the idea of the *plasma vortex*, which was enthusiastically accepted by many cereologists. It derived from the ball-lightning hypothesis of the great Russian physicist P. L. Kapitsa (1955), who made the subject of ball-lightning scientifically acceptable. His ideas originated from his involvement in nuclear explosions. In the presence of a very strong radio frequency source the electromagnetic radiation propagating along and reflecting from a conducting plane (such as the earth's wet surface) can form an interference pattern with nodes and antinodes that may be intense enough to produce local ionization.

In what I consider a serious misunderstanding of this hypothesis, Meaden has interpreted the multiple-satellite crop circles as nodes and antinodes of a central plasma vortex. Furthermore, regarding the required plasma electron density there is a discrepancy of about six orders of magnitude as compared with the highest values observed in the dry atmosphere under nonthunderstorm conditions, that is, at typical crop-circle conditions. There is also nothing like the required enormous radiofrequency source in a quiet night. Finally, his interpretation of the thin circles sometimes connecting or surrounding the multiple, small satellite circles as an "ion racetrack" is at variance with the ion-free pathlength at ground pressure, which is about 10^{-5} cm. (By the way, Kapitsa's ball-lightning theory is not generally accepted. He has not clearly identified the radiofrequency source although he suspects lightning.) The happy acceptance of the plasma vortex by many of Meaden's fellow cereologists is—in my opinion—grossly misplaced.

Stage 3 Pictograms

Just when cereologists thought they had found some explanations, Stage 3 of the crop circles started in the late 1980s with increasingly elaborate patterns. The circles added clean-shaved rings, double rings, triple—even quadruple—rings and then in 1990 formed huge, delicately designed images called pictograms, that can best be seen from aircraft. They are of the order of 100 to 500 feet long. The number of events skyrocketed and the result was what one might call "epidemic" excitement.

So, here we come to the interpretation that many readers must have had in mind all the time: hoaxes. The number of proven hoaxes has strongly increased since 1990, some directly under the eyes of nocturnal observing teams. By leaving marks, such as astrological charts in the circle centers, they gave their hints, or they came out into the open and approached the press, like the aforementioned Doug and Dave. Yet, by inquiries among fellow scientists in England, I learned to my surprise that many of them doubted that a few hoaxsters could have produced the huge number of elaborately designed patterns. Did the hoaxsters hoax their hoaxes? Why such an enormous effort? What would be the motive? Are there mythical societies perform-

ing certain rituals? Can the joy to fool your fellow citizens successfully be so compelling? The suspicion of hoaxes has always been there, especially because of the widespread presence of tramlines (tractor lines) which give easy access to the crop circles. We have no choice but to conclude that the Stage 3 pictograms are manmade and will eventually be proven just that. However, we should not call them hoaxes. One can think of other motives.

A UFO Connection?

This brings us back to the simple (Stage 1) crop circles and their international counterparts, the rings described by Phillips. As mentioned before, these rings seem to be a different phenomenon and are generally UFO-related—about half of them coincide with alleged UFO landings. They don't show the walls, the flat spiral crops, and their inside is dehydrated or burnt. This is not the place to explain the UFO phenomenon, but one must ask the question: Are crop circles UFO-related? The cereologists respond guardedly. The number of UFO-related reports is small. Surprisingly, two exponents (at that time) of the British UFO Research Organization (BUFORA), Jerry Randles and Paul Fuller, in their book *Crop Circles: A Mystery Solved* (1990) argue strongly against UFO involvement and for the plasma vortices of Meaden (which they obviously don't understand). They do so not only as an explanation for the crop circles but also for UFOs in general (as Philip Klass had done before). In this otherwise very informative book they write: "Meaden is editor of the prestigious *Journal of Meteorology* and is widely regarded as a leading authority on atmospheric vortices," which may have contributed to their choice of plasma vortices. It should be mentioned that this journal published by Meaden should not be confused with the *American Journal of Meteorology* (now the *Journal of Atmospheric Sciences*). Even if one does not agree with Meaden's theories, one must respect him as an outstanding observer.

If we accept that crop circles and rings are not the same phenomenon, and if we consider the enormous number of crop circles found in a small area of England, we have little difficulty in denying that crop circles are a UFO phenomenon.

CONCLUSION

The above assessment is a personal one and does not claim to be the final answer, rather it is a "best scientific guess." To summarize:

In their simplest form (Stage 1) crop circles cannot be explained by dust devil-like atmospheric vortices for the reasons given in this report. A small door should be left open for nocturnal crop circles formed by a descending vortex—descending into a cold, nightly boundary layer—although such a phenomenon has not yet been observed in the atmospheric sciences.

Regarding the more complex patterns of Stages 2 and 3, my expectation is that they will prove to be manmade (for whatever motives, not necessarily hoaxes). The

plasma vortex appears scientifically improbable

A UFO connection cannot be established for crop circles, although it exists for the earlier rings with their burnt or dehydrated interiors. So, for the crop circles at least, we may not have to explain one unknown by another one.

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HISTORICAL LINKS BETWEEN THE OCCULT AND FLYING SAUCERS

DAVID W. STUPPLE

EDITOR'S NOTE This posthumous article by Dr. David Stupple, sociologist and popular-culture scholar of Eastern Michigan University, appears here by a bit of good fortune. While conducting research in the Gray Barker archives at the Clarksburg-Harrison Public Library in West Virginia, I discovered this apparently unpublished manuscript among the files. Stupple had worked with Gray Barker in surveying the saucer enthusiasts who subscribed to Barker's newsletter. Because of David Stupple's excellent fieldwork and intellectual skills, it seemed both a shame and an opportunity to have one of his research papers unpublished. I have made several inquiries as to whether the paper has in fact appeared elsewhere, all of which have been negative. Please forgive us if we have failed to locate it in the sea of possibilities. With the permission of the Stupple family, I am pleased to present this insightful essay, written ca. 1980, by a fine scholar deeply missed by serious people interested in the UFO phenomenon and its sociological ramifications.

Astral travel and anti-gravity are common themes in popular culture. Less known is their acceptance as factual theories of air flight by members of the flying saucer social world—the hobbyists, sectarians, and others who share a collective excitement about tales of contact between earthlings and extraterrestrials. This essay is an exploration of how these theories developed and how they function within this little-known subculture.

We begin with the observation that ideas are born, transformed, and abandoned in social interaction. That's sociological common sense. To understand belief systems we must understand the forms of social organization within which they are embedded. We shall start with a sketch of the organizational forms that comprise this social world, then we will have a vantage point from which we can trace the career of astral travel and anti-gravity as explanatory schemes. (For a discussion of the applicability of the concept of "social world," see Grathoff 1978: 129.) Our sketch, a set of constructed types derived from fifteen years of research, is an interpretive scheme that describes the subculture's basic organizational forms. Actual empirical cases may fit one of the constructed types very well or may include characteristics of more than one type (McKinney 1966).

FLYING SAUCER ORGANIZATIONS

The Flying Saucer World is a collectivity of small, mostly urban, groups and perhaps 10,000 persons, mostly in rural areas, who are not active participants in these groups. Within this collectivity are five organizational styles. First is the *neo-theosophical group*. Using psychic techniques they receive messages from "space brothers," contemplate them, and in some instances distribute printed copies by mail to subscribers. The messages are an elaboration of an obscure dogma in which a cosmic hierarchy is understood to be assisting earthlings in their spiritual evolution. Patriotic and politically conservative commentary is sometimes included. Publications are identifiable by the use of violet print and references to certain "masters" long known to Theosophists (see Braden 1949: 221-56, and Judah 1967: 92-145). Flying saucers are epiphenomenal for this population because, in effect, space brothers have merely replaced (terrestrial and extraterrestrial) theosophical masters as demigods.

A second organizational form is the *urban discussion group*. These are open-ended organizations marked by intellectual curiosity. Membership consists of friendship circles with little formal organization, the focus of attention being the "channel," a medium who psychically communicates with the space brothers. Many of these groups have more than one channel, each a source of information for discussion. Prestige is partially determined by knowledge of various occult systems and while the manifest interest is flying saucers, the discussion of popular occultisms inevitably becomes the dominant preoccupation. These tendencies encourage organizational fragmentation and instability. Consequently, these groups often have no names (Schutz 1973 offers a comprehensive sociological study).

A third organizational form is the *religious cult*. In this variation a contactee establishes a monopoly over access to the space brothers, thereby legitimating his or her position. These groups characteristically are goal-oriented and pursue a series of missions set out by the space brothers. The construction of machines to achieve these goals is common. Membership typically is stratified into 1) true believers who devote an enormous amount of time meeting the leader's demand for participation, 2) sympathizers who belong to several groups and either hide whatever cynicism they have about the leader, or more commonly, reconcile their multiple involvements with a pluralistic world view, and 3) curiosity seekers prompted by one or more motives, none of which is satisfied by prolonged participation. Ironically, curiosity seekers provide financial assistance, an audience for the contactee, and justification for his or her mission (Wallis 1974).

A fourth organizational form is the *isolated hobbyist and collector*. Unlike participants in flying saucer groups, these consumers show little interest in the occult and report having few psychic experiences of their own. Basically, this is a remnant of persons, mostly male and working class, who in the 1950s were caught up in the excitement surrounding the colorful stories of the contactees and continued their interest by purchasing literature from specialty publishing houses and mail-order

merchandisers. There is an adventure-story bias here, an interest in tales about beings from outer space, lost continents, the hollow earth, and sinister men in black who harass those who "know too much" about flying saucers. This audience specifically does not recommend popular occultisms such as satanism, witchcraft, yoga, and astrology (Stupple and Dashti 1977).

A fifth organizational form is the *civilian UFO research organization*. Here, in a new generation of enthusiasts, can be found closet occultists in the company of trained scientific researchers. In the 1950s and early 1960s civilian research groups avoided association with contactees and occultists, gradually, since the mid-1960s when the abduction case of Betty and Barney Hill (Fuller 1966) received positive public reaction, the contactee and the occult both became respectable—if treated scientifically, of course. Prestige within civilian research organizations is awarded first on a basis of academic credentials and second on research achievements. Annual conventions are quasireligious events that celebrate guest speakers. A "middle ufology," composed of persons without pedigrees, also exists. Middle ufologists have close links to the publishers and distributors of the expanded flying-saucer folklore and tend to take themselves less seriously.

Astral travel, the first popular explanatory system to develop, has been popular with America's psychic community for more than a century. Anti-gravity flowered with the explosion of UFO reports in the 1950s and with the contactees who claimed face-to-face meetings with space people. Both schemes are expressions of popular occult thought. Certain features of this world-view can be listed: 1) the universe is seen as a Newtonian world-machine: general laws govern everybody in the solar system, 2) the cosmos is arranged in a hierarchy: a chain-of-being unifies everything that could be created, from God through the inanimate world, and 3) all of the natural world follows immutable laws of evolution, both material and moral—to think and act ethically is to advance one's self in an interplanetary system (Leventhal 1976: 1–10, Randall 1940: 253–81, Whitehead 1967: 41–56).

Colonial Americans commonly assumed that other planets were inhabited. In France, Voltaire penned a story about a visit to Earth by inhabitants of Saturn and Sirius, the latter visitor having nearly a thousand different senses. In Germany, Immanuel Kant speculated that the further away a planet was from the sun, the more highly advanced its inhabitants were (Leventhal 1976: 247). In Sweden, Emanuel Swedenborg wrote about beings on the moon and life on other planets—stories that might be metaphorical for a spiritual inner-space world that, according to his reasoning, corresponded to the physical world (Swedenborg 1912a, 1912b, Spence 1960: 395). In any instance, the 18th-century universe was believed to be infinite and inhabited, and it was assumed this was as God wanted it (Woods and Woods 1979).

From a sociological perspective we can see in such folk theories an accumulation of objective culture, in this instance, belief systems. But while objective culture accumulates indefinitely, from the standpoint of the individual these belief systems are merely a stock of available knowledge open to transformation. To follow the transformation of the ideas of astral travel and anti-gravity we shall outline the im-

pact of the contactees on this process. By limiting our inquiry this way we oversimplify, but I argue that it is valid to see the contactees as causal agents because they established themselves as legitimate sources of sacred information. Thus with some reservations in mind, let us begin.

THEOSOPHY

Once upon a time there was Helena Petrovna Blavatsky (1831-1891) "HPB," as she called herself, was the dynamic force upon which the Theosophical Society was formed in 1876 (see Goldfarb 1972 and Williams 1946). A Russian-born medium and psychic, she allegedly played with invisible friends as a child and was to claim meeting the Master "M" on the "physical plane" in London when she was twenty years old. Later, in India, she is said to have contacted the Great White Brotherhood, a band of adepts who revealed to her secret lost literature. In 1877, with an ascended master occupying her body, she wrote *Isis Unveiled* (1877), later followed by its sequel, *The Secret Doctrine* (1888). Both books are long, obscure, polemical, and still a source of wonder. They are the sources from which Theosophists developed an ingenious cosmology that mapped both the heavens and the world within—and tied both together with the theory of astral travel.

Although *Isis* and other books were claimed to be divinely inspired, the masters always remained in the background. Not beings to worship, they were the source of bureaucratic legitimacy for the leaders of the Theosophical Society. HPB, for example, received messages from the masters in the form of letters from M (El Morya) and K H (Koot Humi). Henry Olcott (1832-1907), cofounder of the Theosophical Society, received letters from the Brotherhood of Luxor (Braden 1949: 233). Annie Besant, who was to be Blavatsky's chosen successor, had a more dramatic encounter with the masters—in 1889 she claimed that she was "awakened in the night, startled to find the air in the room thrown into pulsing electrical waves, and there appeared the radiant *astral* figure of the Master, visible to my physical eyes" (emphasis added) (Williams 1931: 197). When Blavatsky died, Besant was at sea and Olcott, president of the Society, was in India. This left William Judge (1852-1896), the Society's counsel, in control over Theosophy in America (see Williams 1931). Judge, in an attempt to exploit the situation, cabled Besant in London requesting her not to do anything until he got there.

After he arrived, things began to happen. Among other things, Besant found messages written with crayons on rice paper endorsing Judge as being "honest," and "right," and to "follow him and stick."

Besant, impressed with those messages, joined forces with Judge, and together they dominated Theosophy in England and America, leaving Olcott virtually isolated. But Olcott decided to challenge Judge for power and the struggle lasted until 1894 when Olcott successfully challenged the authenticity of Judge's messages. He also sent a messenger to Besant to tell her of the wonders of India. Judge, uneasy

about these developments, was unable to stop Besant from joining Olcott. Eventually Judge's messages were discredited and he died in 1896 after convincing the 6,000-member American branch to secede. Olcott died in 1907 leaving Besant the Society's most powerful leader.

The theosophical movement underwent substantial changes as the 20th century unfolded. Its dogma crystallized in the hands of a man, one Charles Leadbeater (1847–1920), who was later to break with the Society to form his own organization, the Liberal Catholic Church. Leadbeater was a spiritualist and a curate in an English country village until age 37 when he met HPB, shed his old life, and sailed with her to India. He did five years of missionary work in Ceylon, specializing in the education of boys. In 1889 he was persuaded by prominent English theosophist A. P. Sinnett to return to England where he became a sensation as a medium bringing forth messages from the masters Morya and Koot Humi to Sinnett. This so impressed Besant that she persuaded him to join her household of Chelas (students) and within a year Besant had fully and without reservation accepted him. Together they refined the concept of color aura. In 1904, however, Leadbeater was accused of homosexual relations with several boys. Called before a committee, he was asked to resign from the Society. Besant responded by disassociating herself from Leadbeater but in 1907 she made overtures toward reconciliation.

Leadbeater, for all his difficulties, was an important contributor to Theosophy. He was the main exponent of the themes of authority and hierarchies, he emphasized ranks and grades along the mystic path, and he was the main influence in ritualizing the dogma. In 1912 he published *The Handbook of Theosophy*, in which he refined ideas available in Blavatsky's works only by tortuous examination (Leadbeater 1912). He outlined the interplanetary chain of evolution and the evolution of "root races" on Earth. He argued that ten "schemes of evolution" presently exist in our solar system but only seven schemes have planets in the physical world. Included is the unrecognized planet Vulcan and two unnamed planets beyond Neptune's orbit. Mars and Mercury are in the same chain as Earth, he explained, and each has human, animal, and vegetable life. Furthermore, he explained that some Earthlings are advanced beings from the moon and although moon men are of a higher race than, say, Aryans (who are the highest native race), all the races on Earth can, through reincarnation, eventually catch up with the moon men. Of greater importance than the moon men was the visitation by certain Venusians known as Lords of the Flame. These adepts hold the highest ranks in the Great White Brotherhood and are said to have had a wondrous effect on our evolution—mostly in the advancement of our intellect. Since Venus is in a higher order of evolution than Earth, members of the Brotherhood only help and advise, they do not associate freely with Earthlings. Among the seven root races on Earth are seven subraces, which in turn, are divided into seven branch races. The Aryan race, Leadbeater reported, was a little more than halfway through the evolutionary ladder.

Paralleling this external structure was an internal structure known as the "subtle world." Life exists at seven levels according to this doctrine, each occupying *the*

same time and space These levels are the divine, the monadic, the spiritual, the intuitional, the mental, the astral (or emotional), and the physical Not only does each world have its own type of matter, each has a different degree of density due to its rate of molecular vibration Obviously, Leadbeater modeled his cosmology after the science of physics as he understood it and this was in keeping with the stated aims of Theosophy to encourage the comparative study of religion, philosophy, and science

Leadbeater codified the work of HPB but the theory of astral travel goes back much further in time Herbert Spencer, a contemporary of HPB and a person she held in contempt, traced what he found to be the history of the theory of astral travel (although he did not use that term), in the history of primitive religion Beginning with the mysteries of echoes and shadows, the primitives, Spencer concluded, developed the explanation that each person has a double In dreams the double leaves the body and becomes a discarnate spirit in a world of other spirits (Spencer 1896 132–50) In fact, this understanding is close to HPB's understanding of the *astral world*, which she felt was inhabited by doubles, demons, angels, and good and bad elementals including fairies which are just beyond the powers of human vision This (as in Spencer's analysis) is a region into which man passes upon death, staying for a period of time determined by his state of development—the primitive savage staying a relatively short time, the civilized man relatively longer Both primitive and civilized man also enter into the astral world during dreams and altered states of consciousness In the astral world the civilized man may do work helpful for humanity (Spence 1960 41–42)

That the theosophical concept of the astral world was available in the 1880s is documented by an occult novel, *A Dweller on Two Planets* (Oliver 1905) This book allegedly was written backwards by automatic writing by Frederick Spencer Oliver, an 18-year-old boy who was the instrument of an adept named Phyllos the Thibetan In *Dweller* the hero has adventures in and out of the astral world As "Zailm" he visited Atlantis where he became Secretary of Records In a later reincarnation, he became Walter Pierson and traveled to Mount Shasta in California where he met a master dressed in a turban Introduced as Mol Lang of Pertroz, the master took Walter to Venus On Venus, Pierson (then in his astral body) was impressed by the highly evolved Venusians

What splendid physiques these people have, how graceful and perfect every line, it is as if the body were molded upon the soul, and perfect in its every physical contact

And he learned why Earthling astronomers have not and never will observe life on Venus Venusians vibrate at a frequency too high for the human senses to notice

A Dweller on Two Planets, together with Leadbeater's *Handbook of Theosophy*, demonstrate that Theosophists had developed the basic explanatory scheme for what was to become flying saucer folklore But soon to come were revisionists who, while

effective organizationally, abandoned the elegance of early theosophical thought Leadbeater's Liberal Catholic Sect, open only to males, was hardly a model for success More effective were Katherine Tingley, Alice Bailey, Earlyne Chaney, and Guy Ballard

LATER OCCULT GROUPS

Katherine Tingley (1847–1929), a professional spiritualist medium, succeeded Judge as head of the American branch (Judah 1967, Greenwalt 1955) Earlier she had organized the Society of Mercy to visit prisoners and hospitals, supporting the work by dramatic recitals and spiritualist readings In 1896 she led a theosophical crusade throughout the world and was successful in spite of resistance by Besant's forces After successes in Europe she went head-to-head with Besant in India Tingley fared less well in India where Besant enjoyed popular support but she did use the occasion to seek out and meet a master Gone for two days she returned enraptured, speaking of an anonymous master clothed in Tibetan garb and "whittling a yoke peg while one of his chelas plowed a nearby field" Gaining the sanction of a master, she returned to America where she involved many Theosophists in social work

The early 1900s was a time of fragmentation for Theosophy and decline of the mother organization Tingley's most important contribution to this fragmentation was her "Point Loma experiment" in which she saw the realization of her desire to build a utopian community Point Loma, a regal structure in San Diego, was a sensation inside its Roman and Egyptian gates rose massive buildings with domes of leaded glass as if in a fairy tale It housed a school for children from elementary grades through college and graduated thousands of students before its failure in the 1940s Tingley died in an automobile accident in 1929, a time when many wealthy Theosophists were withdrawing their financial support for the project

Alice Bailey's (1880–1949) contributions were less splendid but her influence on clerics and lay persons broke through the boundaries of her own sect, the Arcane School, much as the Unity Church has made Christian Science thought accessible to the general public (Judah 1967, Bailey 1951) Born in a well-to-do English family Bailey had private tutors and lived in relative seclusion She had attempted suicide three times when, at age 15, she was visited by a tall stranger in European clothes and a turban Later identified as "Koot Hoomi," the stranger told her that she had important work to do on Earth In 1919, by then a Theosophist for two years, she claimed to have contacted the master known as the Tibetan, who asked her to write books for him She refused but changed her mind upon advice from Koot Hoomi Bailey founded the Arcane School in 1923, offering correspondence courses in occult philosophy The Arcane School was successful because it 1) included Jesus Christ in the ranks of the masters, and 2) did not demand complete loyalty of its students Especially attractive has been the Arcane School's doctrine of the second

coming of Christ. In sum, the Arcane School became a revitalizing movement within Theosophy and a bridge to Christianity. Subsequent Theosophists, including those involved in the flying saucer social world, were also to reserve a high place for Jesus Christ in their systems.

Earlyne Chaney (1917–) continued the process of outreach—making theosophical ideas available to the general public (Judah 1967, Chaney 1974). Like HPB, she had invisible playmates. And like HPB, Leadbeater, and Alice Bailey, she met Koot Hoomi (this time spelled Kut-Hu-Mi). Materializing at a woodpile when Earlyne was age thirteen, Kut-Hu-Mi prophesied that she would lose someone dear and that this would change her life. Thus, at age twenty-two when her fiancé died, Earlyne fasted until her master reappeared. And reappear Kut-Hu-Mi did, this time giving her a formula for unfolding her psychic powers and a suggestion to study spiritualism. Kut-Hu-Mi must have been convincing because she gave up a successful career as a Hollywood actress and went searching for spiritual comfort. Eventually she met like-minded Robert Chaney and together they founded Astara in 1951. Astara [in 1980] is a successful correspondence school that combines Theosophy and Christianity. It has an active healing ministry patterned after evangelical Christian churches but with an occult methodology. For two dollars a lesson subscribers can take a series of lessons that lead to initiation in the Great White Brotherhood. (Acceptance into the GWB, however, remains a matter of ultimate decision by the masters who are the current members.) Astara literature explains that the lessons will

Instruct you in a secret mental technique of spiritual unfoldment
Reveal how to use your “Third Eye,” the all-seeing eye of the
superconscious

Illuminate mystic teachings of the Silver cord, the Seed atoms, the
Etheric Web, the Kundalini Fire, the Chakras [Chaney and Chaney
1961]

The Chaney's claim to have had several initiations inside Mount Shasta where they entered in their *astral bodies*. In these initiations and in other revelations they learned ideas basic to the flying saucer folklore of the 1950s. Jesus himself told of grave conditions on Earth, the coming holocaust, and that the planet was entering a new expansion with higher vibrations that many would not be able to stand. Therefore, Astarians would have to leave Earth for another planet where they can continue their evolution without disruption. Religious scholars will find in Astara several items that were taken from the IAM sect: a chart illustrating the octaves of the body, the magical manipulation of the masters by the contactees in healing rites, and the story of meeting masters in Mount Shasta. The Chaney's are not alone in their use of I AM dogma; there are many such groups influenced by this most successful and influential of all the theosophical sects.

The I AM was the creation of Guy and Edna Ballard, two practicing occultists who most observers feel were frauds. In 1932 Guy Ballard claimed to have met the

ascended master Saint Germain on a mountainside in the Grand Tetons (Ballard 1934) Saint Germain allegedly took Ballard inside the mountain and by "projected consciousness" took him through several reincarnations Among other things he learned that he used to be George Washington, that Edna had been Joan of Arc, and their son Donald had been Lafayette Inside the mountain Ballard also met twelve "tall masters from Venus" including Sanat Kumara (also known as Jesus Christ) After some meetings in their Chicago home, the Ballards took their message on the road Crusading around the country they were fabulously successful and by the time of Guy Ballard's death in 1939 they had hundreds of thousands of followers

Their extraordinary success was partly due to their cooptation of key figures in William Pelley's fascist Silver Shirt movement but more important were their theatrical presentations in which they inverted the traditional relationship between masters and contactees By using "decrees" they mobilized great audiences to shout loud commands to the Mighty Ones to intervene on their behalf Destroy the Labor Union Beast! Destroy the Communist Beast! they demanded The Ballards gained several fortunes and much notoriety for their trouble There were lawsuits, internal feuds—and high-living Guy Ballard's death in 1939 (he had claimed to be immortal) discouraged many A conviction for mail fraud publicly discredited Edna and Donald in 1940

Contrary to popular thought and some histories of religion, the I AM still prospers today, greatly reduced in size but full of fight Members are friendly, but cautious with strangers My personal research suggests that young and middle-aged male applicants for membership are not encouraged but attractive young females fare much better.

Since the I AM went into relative seclusion in the 1940s several splinter groups formed The most successful of these by far is the Summit Lighthouse, which by now has its own educational institution (Summit University) in California reminiscent of Point Loma While the I AM has maintained a garbled cosmology tightly controlled by the autocratic Ballards, the Summit Lighthouse has a sophisticated, intellectual approach Much of its literature, for instance, is footnoted and referenced (Stupple 1975)

The Ballards made contributions to the changing theosophical dogma By the 1930s the idea of the *astral world* was in decline and virtually passed out of Theosophy while still being alive in Rosicrucian thought and some systems of magick (as opposed to "magic") The Ballards took a peculiar approach to this subject, an approach that becomes clear when contrasted to traditional Theosophy In traditional theosophical thought Evil is downplayed, and the long cycle of evolution is automatic and can only be speeded up, the Ballards, on the other hand, made a contest between Good and Evil and made evolution problematic They and the Ascended Masters obviously were "good," but where in the system was the locus of Evil? Well, it was not in the astral world where nefarious elementals and demons are at work—at least, not exactly The Ballards' theory was that Black Magicians are the negative force and they operate on the physical plane In December 1964, after I had

gained acceptance by group members while conducting covert research on the IAM in Kansas City, Missouri, a long-time member, eager to inform me about the Black Magicians, gave me an account that I reconstruct from my fieldnotes

Many years ago a group of adepts learned that they could use their thought power for evil as well as good. They could call upon the mighty words "I AM" and evoke whatever they wanted. They brought much destructive activity to the world before the Ascended Masters revoked their cloak of invisibility and put an end to their work. Much of the discarnate energy they created is still in the world today and is the source of all evil.

As protection against Black Magicians and other agents of evil, the Ballards created a convoluted version of the theosophical notion that there are seven levels to the self, each vibrating at a different octave. In the IAM formulation, members can call upon their "Higher Mental" body (which is between twelve and fifty feet above the head depending on one's state of grace) to come down and form a force field (or "tube of light") to protect the physical body from danger. This is accomplished by special decrees shouted by the individual student or by a group. In this formulation the Causal Body is a circle of colors surrounding the Higher Mental Body. The colors are determined by a person's past lives. An Electronic Body is above the Higher Mental Body and it is into this finer substance that the Higher Mental Body moves upon reaching the goal of the IAM true believer—ascension to the level of the masters. In IAM literature, when mentioned at all, the astral body is referred to as the emotional body and is to be "overcome" just as the physical body is to be protected. Other occultists, however, have made a distinction between the "higher astral" and the "lower astral." Among participants in the flying saucer social world that I have met, several displayed a dread of psychic contacts because of the chance that the actual contact might be with elementals in the lower astral plane.

Although the idea of the astral *world* is alive today only in a modified sense, the idea of astral *travel* is very popular. But for the most part, astral travel has been vulgarized to simply mean out-of-the-body-experiences (OOBEs). It is possible to buy paperback manuals that teach astral projection (Crookall 1970, Fox 1962). And occult magazines like *Fate* carry advertisements promising instruction in this once arcane art. Finally, astral projection has become a subject for scientific study, or what might better be called protoscientific study.

UFOs AND THE OCCULT

Anomalies are the cornerstone of the occult. Therefore, when UFOs entered American popular culture in 1947 they were incorporated into theosophical lore. UFOs fit easily—ascended masters simply became space pilots. In one exotic story, Maurice

Doreal, founder of the Brotherhood of the White Temple in Sedalia, Colorado, explained that there are three different types of flying saucers, including one piloted by serpent-people who once were icebound in Siberia only to defrost and overtake the Communist regime. Compared to the Brotherhood of the White Temple, most neo-theosophical groups are routine stuff. Following the correspondence-school model of Alice Bailey and Earlyne Chaney, they recruit members from the readership of occult magazines where they appear to aim their advertisements at I AM dropouts. These groups, while built upon a stable population, remained unknown to the average person caught up in the flying saucer movement in the late 1940s and early 1950s.

The difference between the neo-theosophists and the average UFO enthusiast is ontological. The theosophical view of the world is organic. UFOs are understood to be part of a living, self-regulating, goal-directed being. But to most observers, UFOs appear to be physical things, suggesting a *mechanistic* model. For instance, the civilian research groups, most notably the National Investigation Committee on Aerial Phenomena (NICAP) led by Major Donald Keyhoe, assumed what has been called a "nuts and bolts" approach advocating the Extraterrestrial Hypothesis (ETH). Among the early contactees (those that came to the public's attention in 1952–1954), some were obviously schooled in Theosophy but it was their stories of face-to-face contact with space people that caught the public's imagination. In addition to the ETH (which they affirmed) they also evoked two other mechanistic explanations: telepathy and anti-gravity.

Most major contactees claimed that some, if not all, of their communications with space people were telepathic; telepathy, they argued, is possible because the brain is electronic. George Adamski, the first and most important of the contactees, sold a correspondence school course in telepathy in which he compared the brain to a radio that both broadcasts and receives (Adamski 1958). His close associate George Hunt Williamson published a book based on messages received by radio (Williamson 1963). Daniel Fry spoke with a spaceman named A-Lan by "direct modulation of the auditory nerve" while he was in "attunement with the vibratory frequency of the communicating intelligence" (Fry 1954). George Van Tassel explained that "ESP is the resonance of one person in resonance with another person" (Van Tassel 1976: 101).

Anti-gravity and its correlate free energy are the most common explanations for space travel among enthusiasts with a mechanistic orientation. Leonard Cramp's 1955 text *Space, Gravity, and the Flying Saucer* contains the most ambitious attempt to explain how anti-gravity works (Cramp 1955). Cramp concluded that flying saucers create and control their own gravitational field, placing it "fore and aft as desired [so that] the saucer would accelerate in that direction so long as the field was there and the acceleration would entirely depend on the strength of the field." Also in 1955, Morris Jessup, an astronomer, published *The Case for the UFO*, one of the most controversial of flying saucer books (Jessup 1955). Jessup argued that flying saucers are living things that inhabit the atmosphere in the region of the "earth-

sun-moon gravitational neutral " Anti-gravity, Jessup speculated, was not only the energy source for flying saucers but also was the cause of the disappearance of ships at sea Jessup is best known for 1) his suicide, and 2) the appearance of a copy of *The Case for the UFO* that allegedly was annotated by three unknown persons who had inside information about UFOs This "Varo" edition, and the claims made about it, is one of the enduring mysteries in flying saucer folklore

Perhaps the best known exponent of free energy was George Van Tassel, the inventor of the Integratron, a "rejuvenation machine" for which he solicited funds until his death in 1979 Updating the I AM vocabulary somewhat, Van Tassel advanced a theory similar to the Ballards' "tube of light" theory

When the balance between the physical reflection of the real body reaches a state of equilibrium, these vortices will be of the same length diameter, and speed of rotation If you are concerned with the welfare of others you automatically establish a "field" of protection around you because the Universal Mind compensates directly, and equally, for every thought and action [Van Tassel 1958 76]

In 1958 Otis T Carr attempted to use the theory of anti-gravity in the first human-made space ship Carr had gained publicity from his appearances on the Long John Nebel radio show He claimed that his OTC-X1 Circular Foil Spacecraft featured an Ultron Electrical Accumulator and utilized the "geometry of space in fourth dimensional physical form " It promised to fly to the moon in five and one half hours and after its maiden flight, it would be commercially available for \$20 million Unfortunately, when the media arrived in Oklahoma City, the site of the proposed launching, they found the OTC-X1 in shambles and Otis Carr in a hospital (Nebel 1962 58-76)

CONTACTEES IN DETROIT

Adamski, Williamson, Van Tassel, Fry, Jessup, and Howard Menger (a New Jersey contactee who invented a "perpetual motion" machine) appeared in Detroit¹ in the 1950s, lecturing to various UFO clubs George Adamski, the first to arrive, drew 4,700 people to a lecture at the Detroit Masonic Temple Audience members were given a questionnaire which asked if they would be interested in joining a study group to further investigate the flying saucer mystery Many did and this became the basis for the movement in Detroit At first, members of a central committee met with the groups to discuss the contactees' claims but, my informants tell me, most people quickly grew tired of contactee stories and desired something more relevant to their

¹ Professor Stuppel, who was at Eastern Michigan University, did much of his fieldwork in the Detroit area, and so based much of the following discussion on the insights he gained from these experiences and observations

own lives. The result was the spontaneous development of contactees within these groups who channeled their own messages from the space brothers. In 1958, fourteen of these groups met together in a convention.

The 1960s was a period of rapid decline of interest in the contactees but three Detroit-area groups took root and remained active in 1980. The first was the Interplanetary Space Center led by Laura Mundo, one of the original group of people who brought Adamski to Detroit. Although she attempted to solidify her position with stories about meeting "Orthon," a Venusian that Adamski had met on a California desert, Mundo had little success keeping her organization going and advancing her own theory of accelerating sun spots. Nevertheless Mundo remained an important figure in the contactee world in the early 1980s, supported by only a handful of coworkers. Her prominence was due to a newsletter she sent internationally to UFO buffs. A network of similar newsletters formed a communications net spreading fact, rumor, and gossip throughout the flying saucer social world.

A second group is the Aetherius Society headed by George King, a British contactee in Los Angeles. As the "Voice of Interplanetary Parliament," King claimed a monopoly over contact with the space beings and the terrestrial masters of the Great White Brotherhood. Throughout the 1970s the Aetherius Society in Detroit maintained a membership of twenty-five, second only to the forty members in Los Angeles. The Aetherius Society is a direct descendent of the IAM and has freely adapted some of the I AM's literature. By manipulating the space brothers, King, like Guy Ballard before him, is more magician than priest—a mystagogue to use Max Weber's term. He keeps members busy with pilgrimages to holy mountains and with related missions. In 1975 I attended a lecture in Detroit where he revealed a "spiritual battery" that he allegedly used to prevent international disasters. Later I watched as local members attempted to charge a battery of their own "prayer teams" ran up, two-by-two, to a shoebox-shaped object while a circle of members surrounded them and amplified the power by chanting a mantra.

A third group is the Institute for Cosmic Research led by Gordon (a pseudonym), a charismatic contactee who claims to be in touch with space brothers from Io, a satellite of Jupiter that Gordon preferred to call a planet. Gordon attended the 1950s contactee lectures in Detroit and went to Oklahoma City to attend the launching of the OTC-X1. In his version, Gordon destroyed the OTC-X1 because Carr was being "mislead by shysters" and then went to the hospital where Carr was staying and healed the surprised would-be space pilot. In 1967 Gordon began building a one-person anti-gravity machine (flying saucer) which, he claimed, would fly into the stratosphere using the "free energy of space" available in vortices created where "lines of force" cross. This, he said, was the "stairway to the stars." One hundred people were members of this project, fifty of which were active. The saucer was no toy, it was, by any measure, impressive. Forty feet in diameter, it had twin flanges that promised to rotate in opposite directions, creating free energy. Its construction was possible because a community of machine-shop owners worked on the parts in their shops. Folks like me with few mechanical skills did the "dirty work" like lying

on one's back underneath the ship gluing wooden parts together Dissension finally brought a halt to the construction of the space ship but Gordon was able to convince some of his followers to work on new machines to convert water into pure energy Of course the space brothers backed this new plan (McNeece 1974)

Another group was active in Detroit beginning with the early 1970s This is the informally named "Tuesday Night Group" that met once a week to meditate and channel messages from the space brothers Four of the eight members were channelers Typically, sessions ended when the channel who was speaking that evening announced that a certain space brother would contact another channeler the following week The messages (which have been tape recorded and transcribed) can be described as uplifting, vague, and personal Very little turnover in personnel occurred in eight years of the group's existence New members were not sought because they tended to be disruptive Older members recall being part of similar groups in the 1950s

The late 1960s and the 1970s saw a change in the flying saucer movement the return to the organic metaphor The revived organic view can be seen in the belief systems of the four Detroit groups And what is this new orthodoxy? That UFOs are fourth-dimensional, and that they move in and out of the world of human perception making themselves invisible at will Funny thing, HPB and company would have said the same thing 100 years ago

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Research Note

FURTHER QUANTIFICATION OF DISTANCE-RELATED EFFECTS IN THE TRANS-EN-PROVENCE CASE

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ABSTRACT In a previous report known as "the Trans-en-Provence case," a number of biochemical changes were detected in plants following the landing of an unidentified object. Some relationships were discerned when the concentrations of cellular compounds were plotted against the distances from the epicenter of the phenomenon, but this point deserved further examination. This paper aims to verify and clarify the preliminary results within the limits of the possible. Revised algebraic treatments of the data allowed a number of relationships to appear as behaving with the reciprocal of the square of the distance. This was the case for some photosynthetic pigments, for glucose and for two amino acids. Uncertainty remains because of the natural biological variability and the lack of a controlled experimental design (a limitation imposed by the constraints of ecological conditions at the site). However, these results might strengthen the hypothesis of the involvement of electromagnetic and/or gravitational forces in the observed events, provided linear relations could be ascertained between the corresponding metabolic functions and these forces in similar conditions. Suggestions are made for experimental designs that could allow quantitative results to be obtained from other trace cases.

The Trans-en-Provence, France, case remains one of the most challenging reports of a UFO landing, since the witness' observations have been supported by a thorough biochemical analysis of traces left on plants in the area (Bounias 1990). The witness saw a silent saucer-shaped object (Figure 1), matte metallic-colored and about 3 meters in diameter, land near his house, take off vertically to a height of 20-30 meters, and disappear at extreme horizontal velocity (GEPAN 1983)(see Figure 2). A circular trace remained visible at the landing site (Figure 3), and a police investigation was begun on January 9, 1981, the day after the incident. Samples of plants and soil were collected by local police on the fourth and fifteenth days afterwards, both from the trace site and from 20 meters away as a control. Then, following the author's instructions, five specimens of a wild variety of alfalfa, *Medicago minima*, were collected at day 40 from the epicenter of the trace, from the periphery of the trace (1.5 meters away), and 2.1 meters, 3.8 meters, and 10.0 meters from the epicenter. Two years later during the same season (day 730), new specimens of the same plants were taken under similar conditions, in order to serve as a posteriori controls.

The first data to be published showed significant variations of photosynthetic pigments, carbohydrates, and amino acids correlated to distance at day 40, but not at day 730. This, among other observations, indicated that something had occurred

where the object was seen to land and that these effects disappeared with time (Bounias 1990) The aim of this article is to provide a more accurate description of the mathematical function of the distance-related variations of some of the components of the plants collected at day 40

MATHEMATICAL BACKGROUND

Let $f(D) = \frac{h}{D^n}$ be a parametric function decreasing with exponent, n , of the distance, D , of the source in each geometric dimension Then, the response should be infinity for $D = 0$ However, according to the properties of the involved system, the actual response, R , may range from an initial value, R_0 , at the lower distance to a limit, R_{lim} , at theoretically infinite distances (Figure 4) Now, let $D = (a + d)$ the appropriate shift allowing R to be $R = R_0$ at $d = 0$ Then, provided h is an amplification coefficient, the variations of R versus d become

$$R - R_{lim} = \frac{h}{(a + d)^n} \quad (1)$$

$$\text{that is, since } R = R_0 \text{ when } d = 0, (a)^n = \alpha = \frac{h}{(R_0 - R_{lim})} \quad (2)$$

The response therefore behaves as follows

$$R_i = R_{lim} + \frac{(R_0 - R_{lim})}{\left(1 + \frac{d_i}{a}\right)^n} \quad (3)$$

Unfortunately, as it stands, this equation does not allow an exact determination of coefficient n , as well as parameters a and h from pairs of experimental values R_i obtained at d_i Note that, additionally, values of d_i cannot usually be chosen by the observer, but rather depend on the practical features encountered in the area Two major problems emerge from this situation one is the determination of the limit R_{lim} and the second is the determination of n Neither can be estimated unless the other is known

Let us develop the denominator of Eq (3)

$$\left(1 + \frac{d_i}{a}\right)^n = \sum_{k=0}^n C_k^n \left(\frac{d_i}{a}\right)^k, \text{ that is in the considered case}$$

$$\left(1 + \frac{d_i}{a}\right)^n = 1 + \frac{nd_i}{a} + \dots + n \left(\frac{d_i}{a}\right)^{n-1} + \left(\frac{d_i}{a}\right)^n \quad (4)$$

The four terms explicated in Eq (4) are fairly consistent with the case of exponent

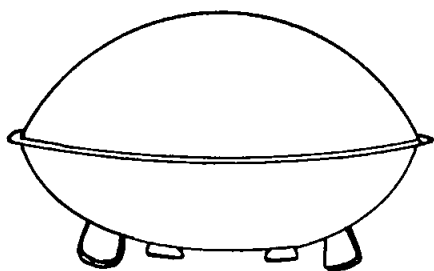


Fig 1 The object as described by the witness, "Renato Colini" (pseudonym given in the original report) (GEPAN 1983)

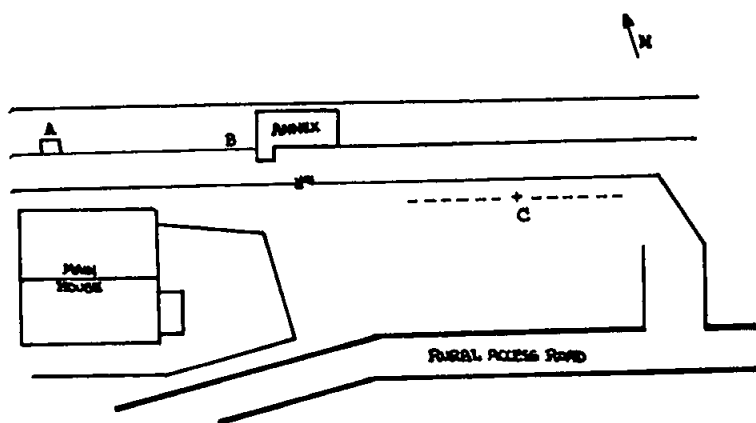


Fig 2 The layout of the landing site. The site is on several levels from the area just in front of the house to the level of the forest above the annex. Colini witnessed the object from points A and B. The trace was found at point C on the dirt access path (dotted line) between the road entrance and the house.

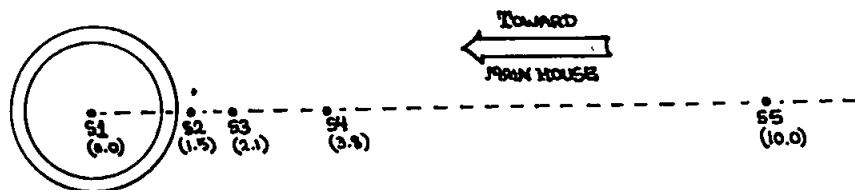


Fig 3 The trace was a nearly perfect circle lying on the dirt access path. It had about a 1.2-meter radius. Samples of the sparsely occurring vegetation were collected at available intervals along the path for the testing described in this article.

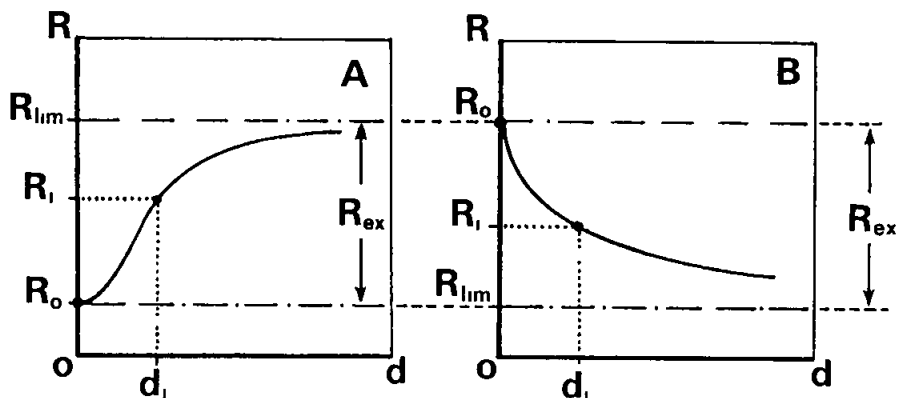


Fig 4 Theoretical examples of increasing (A) or decreasing (B) relations, showing the respective positions of their common parameters (R_0 , R_{lim} , R_{ex}), with respect to the observed pairs (R_i , d_i)

coefficients that are practically situated below $n = 4$. Since this system is largely unsolvable, a first attempt was made, discarding the terms of intermediate exponents, that is from Eqs (3) and (4)

$$\frac{(R_i - R_{lim})}{(R_0 - R_{lim})} = \frac{1}{1 + \left(\frac{nd_i}{a}\right) + \left(\frac{d_i}{a}\right)^n} \quad (5)$$

$$Q_i = \left(\frac{(R_0 - R_{lim})}{(R_i - R_{lim})} - 1 \right) = \left(\frac{nd_i}{a}\right) + \left(\frac{d_i}{a}\right)^n \quad (6)$$

Now let us choose $d_j = kd_i$, where k is a known value, then

$$Q_j = \left(\frac{nd_j}{a}\right) + \left(\frac{d_j}{a}\right)^n = \left(\frac{knd_i}{a}\right) + \left(\frac{kd_i}{a}\right)^n \quad (7)$$

Let both members of Eq (6) be multiplied by k , then

$$kQ_i = \left(\frac{knd_i}{a}\right) + k\left(\frac{d_i}{a}\right)^n \quad (7-1)$$

Subtracting kQ_i to Q_j (or the converse, so that the result remains positive) eliminates the term $\left(\frac{knd_i}{a}\right)$ and, taking the logarithmic form, allows n to be isolated

$$\ln(kQ_j - Q_i) = n(\ln d_i) + \ln(n) \left(\frac{k - k^n}{a^n}\right) \quad (8)$$

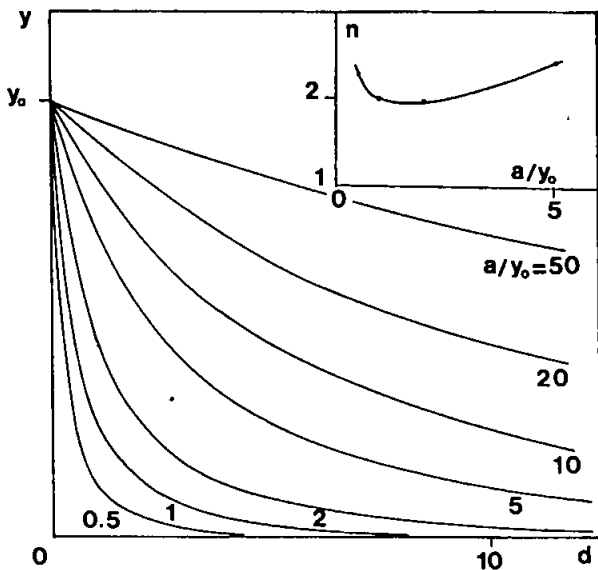


Fig 5 Model of Eq (1) reduced to $y_i = \frac{y_0}{\left(1 + \frac{d}{a}\right)^n}$, with $y_i = R_i - R_{lim}$, and $y_0 = R_0 - R_{lim}$. The graphs represent a family of curves corresponding to a series of chosen values of a and y_0 , all other parameters remaining unchanged. For simplification, the ratio $\frac{a}{y_0}$ is given as representative of a set of values of a for $y_0 = 1$. The values of n that can be practically estimated using Eq (8) are given in the upper frame.

As shown in Figure 5, this solution may provide quite satisfactory results from accurate data when the numerical value of parameter a is not far from once to twice the value of $R_0 - R_{lim}$.

A second attempt was made by keeping only the extreme terms in Eq (4), that is $\left(1 + \frac{d}{a}\right)^n$ is approximated by $\left(1 + \left(\frac{d}{a}\right)^n\right)$. This makes the problem similar to a well-known one in enzymology, which has now been extended to various other biological phenomena (Bounias 1989, 1993).

In numerous cases, the ascending or decreasing relations could be represented by an equation of the following type:

$$|R_i - R_0| = \frac{|R_{lim} - R_0|}{\left(1 + \left(\frac{D_{50}}{d_i}\right)^n\right)} \quad (9)$$

with R_i the biochemical response at distance d_i , R_0 at $d = 0$, and R_{lim} the asymptotic limit. Then, $R_i, R_{lim} > R_0$ in ascending curves and conversely, $R_i, R_{lim} < R_0$ in decreasing ones. Parameter n is a coefficient of shape related to the variance of the distribution of the deviations, and D_{50} is the distance at which the response reaches half of its maximum range (called R_{ex} , for simplification, in Figure 4). All three descriptive parameters can be calculated using appropriate methods (Bounias 1989). Since $R_{lim} - R_0 = \pm R_{ex}$, a constant for each class of compounds, a decreasing curve will become

$$R_i = R_{lim} + \frac{R_{ex}}{\left(1 + \left(\frac{d_i}{D_{50}}\right)^n\right)} \quad (10)$$

Eq (1) can be replotted in the following form

$$\left(\frac{R_{ex}}{R_i - R_0}\right) - 1 = \left(\frac{d_i}{D_{50}}\right)^n \quad (11)$$

which equals by logarithmic transformation

$$\ln\left(\left(\frac{R_{ex}}{R_i - R_{lim}}\right) - 1\right) = n (\ln d_i) - n (\ln D_{50}) \quad (12)$$

Parameters n and D_{50} can therefore be calculated by linear regression, with n the regression slope and $\ln D_{50}$ the horizontal intercept. The standard deviation of the slope n is given by the following relation

$$\sigma_n = \sqrt{\frac{\left(\left(\frac{n}{\rho}\right)^2 - n^2\right)}{N - 2}} \quad (13)$$

where ρ is the correlation coefficient and N the number of pairs of data. Then, the significance of the correlation coefficient is given from the distribution of Student's t table, at $v = (N - 2)$ degrees of freedom by the expression

$$t(\rho) = \sqrt{\frac{v\rho^2}{1 - \rho^2}} \quad (14)$$

Remaining is the problem of the estimation of the asymptotic limit, R_{lim} , in experimental conditions that do not allow the values of d_i to be appropriately chosen for an exact calculation. This, indeed, is precisely the case for a UFO landing

For a decreasing curve, Eq (11) can be replotted in an alternative form where $y_i = (R_0 - R_i)$ and $n > 1$

$$d_i^n = R_{ex} \left(\frac{d_i}{y_i} \right)^n - D_{50}^n, \quad \text{or} \quad (15)$$

$$d_i = R_{ex} \left(\frac{d_i}{y_i} \right) - \frac{D_{50}^n}{(d_i)^{n-1}} \quad (16)$$

A plot of d_i versus $\left(\frac{d_i}{y_i} \right)$ therefore tends to the value R_{ex} when d_i becomes large enough to allow the second term to be neglected (clearly, $\frac{D_{50}^n}{(d_i)^{n-1}} \rightarrow 0$ when $d_i \rightarrow \infty$)

Then, the asymptote becomes

$$d_i = R_{ex} \left(\frac{d_i}{y_i} \right) \quad (17)$$

Figure 6 illustrates a family of such plots obtained for various values of parameter n , in the case of $n > 1$.

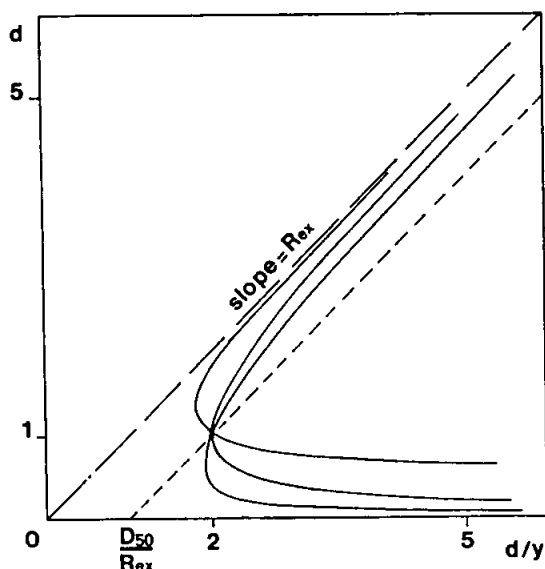


Fig 6 Model of Eq (17) replotted in the reduced form $d_i = R_{ex} \left(\frac{d_i}{y_i} \right) + \left(\frac{D_{50}^n}{y_i^{n-1}} \right)$ using different chosen values of $n > 1$. The curves clearly show that the slope rapidly tends to R_{ex} at high values of $\left(\frac{d_i}{y_i} \right)$. Other parameters are $y_i = \pm R_i - R_{lim}$, and $R_{ex} = R_0 - R_{lim}$, as in text

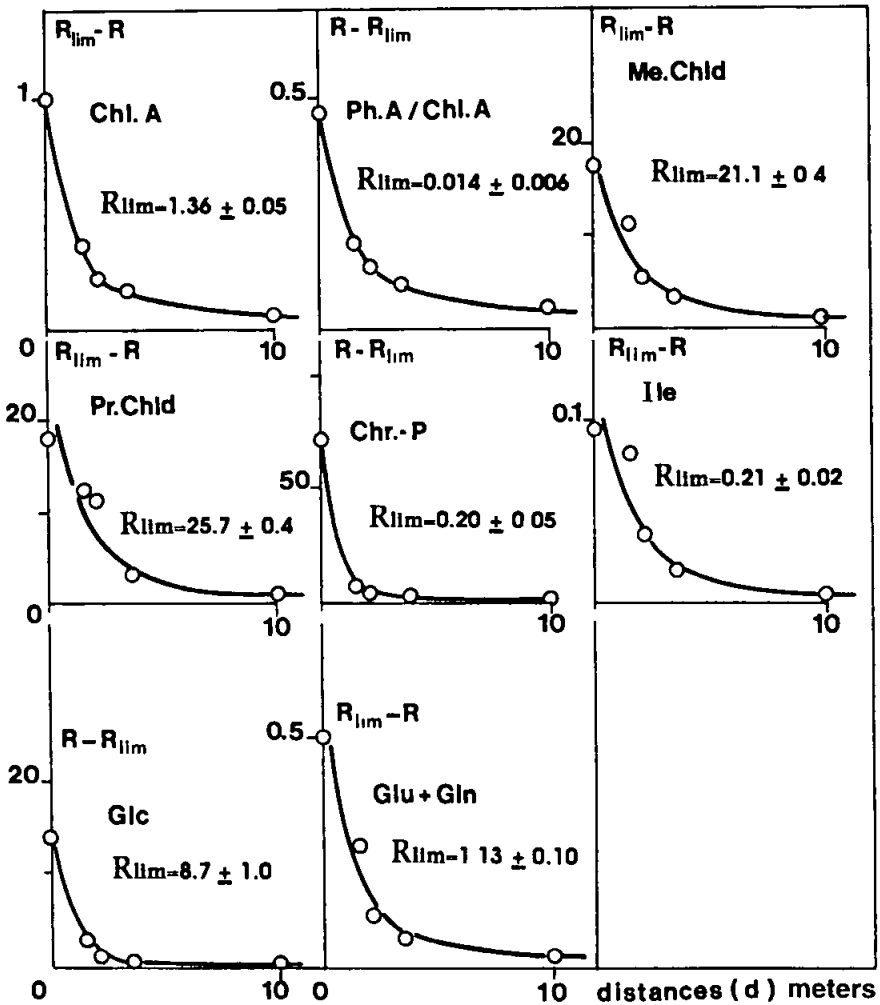


Fig 7 Variations of the concentrations of biochemical parameters against distances, d
Abbreviations

Ph A / Chl A = ratio of concentrations of phaeophytin a and chlorophyll a

Chl A + B = sum of concentrations of chlorophylls a and b

Chr-P = chromenol-phosphate

Me-Chld = methylchlorophyllide

Pr-Chld = protochlorophyllide

Glc = glucose

Glu + Gln = glutamic acid + glutamine

Ile = isoleucine

Concentrations are expressed in nanomole/mg fresh weight. Coefficients of variation range from 0.090 to 0.14 from triplicate determinations. Data are expressed as $R - R_{lim}$ or $R_{lim} - R$ from the calculated values of R_{lim} given on each graph.

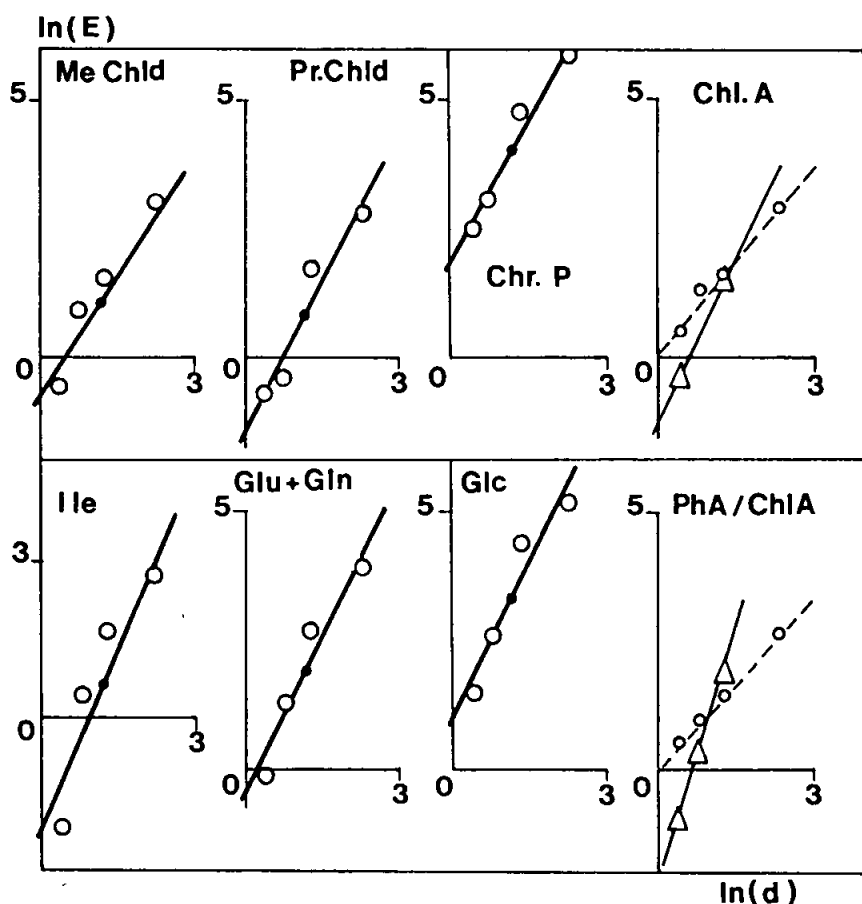


Fig 8 Bilogarithmic plots of distance-response relationships according to Eq (12) (O) or Eq (18) (Δ). Symbols are as in Fig 7. Statistical parameters are given in Table 1.

This allows a satisfactory estimation of R_{ex} from the values of the response obtained at the higher values of d_i , and allows in turn the application of Eqs (8) and (12) for further estimation of parameter n . This plot has been widely used in enzymology as the "Hanes plot" S versus S/v , where S = substrate concentrations and v = enzyme velocity, in so-called Michaelian cases, that is, for $n = 1$. Here it is shown that it can be generalized to non-Michaelian or "cooperative" cases, that is, for $n > 1$ (the same property holds for $n < 1$, that is, in Eq (16) d_i^{n-1} would just become d_i^{1-n}). Indeed, this representation might appear in the reverse form, but in this case the slope would tend to $\frac{1}{R_{ex}}$ and the calculation would just become longer for a quite similar

result, since both direct and reverse slopes basically share very similar values when the determination coefficient ρ^2 is high, as is the case here (see Fig. 8). In other less favorable conditions, for better accuracy the converse slope $\frac{1}{R_{ex}}$ could preferably be calculated together with its corresponding standard deviation, which would in turn give its coefficient of variation CV . Then the standard deviation of R_{ex} could be estimated as the product $CV(R_{ex})$.

RESULTS AND DISCUSSION

Figure 7 illustrates a set of results expressed as the functions of distances d from the epicenter of the phenomenon. The model parameters calculated from these data are given in Table 1. It is interesting to note that in most cases, the exponential coefficients are not statistically different from 2 ($P \gg 0.1$). The bilogarithmic plots according to Eq. (12) are illustrated by Figure 8. The cases of chlorophyll a and the ratio phaeophytin a / chlorophyll a (the more fragile among the studied molecules) differ. However, using Eq. (8) with the available values finally arrives at a fairly analogous result. This discrepancy likely comes from the influence of the choice of the type of approximation on the behavior of the data, as might be expected, particularly for experiments carried out in uncontrolled conditions and therefore subject to variability in samples.

The obtained data are therefore consistent with a phenomenon that behaves as an approximation of the reciprocal of the square of the distance. The two expressions used for this calculation belong to the same class, except that $(a + d)^n$ contains a suite of additional terms, namely a polynomial of degree $(n - 1)$ which does not appear in $(1 + d^n)$. Nevertheless, this term may prove negligible and allow the relation to remain structurally unchanged, and the calculated values of n to be validated.

It should be emphasized, however, that several other conditions must be satisfied before it can be assumed that the phenomenon obeys a physical law behaving as the reciprocal of the square of the distance to the source. First, it should be ascertained that the biochemical effects are linearly connected to the involved energy, and also that this relation is conserved in the effects remaining after a 40-day delay. Because of these theoretical and practical difficulties in the analysis just completed, it must be said that the obtained relationships *suggest* but do not necessarily determine absolutely the algebraic shape of the variations. Mathematically, the relationship of the biological response to the forces applied at a specific time t might be approximated as follows:

Let R be the response given by the biological system at time, t , under the action of level F of the involved physical force. Then, one can roughly write

$$R = R_0 \pm K(t)F_m, \text{ with } m < 0 \text{ or } > 0, \text{ thus} \quad (18)$$

Table 1. Calculated Values of the Exponential Coefficient Affecting the Distances

Compound Classes	(n) ± σ _n	Equation used	ρ	P(ρ)
Phaeophytin <i>a</i> / Chl <i>a</i>	1 74 ± 0 2	(8)		*
	1 18 ± 0 17	(12)	0 998 (P = 0 02)	
Chlorophyll <i>a</i>	2 01 ± SD	(8)		**
	1 14 ± 0 05	(12)	0 983 (P < 0 01)	
Chromenol-phosphate	1 85 ± 0 34	(12)	0 997 (P < 0 05)	
Methylchlorophyllide	1 47 ± 0 37	(12)	0 940 (P < 0 06)	
Protochlorophyllide	1 93 ± 0 40	(12)	0 952 (P < 0 04)	
Glucose	1 94 ± 0 51	(12)	0 937 (P < 0 07)	
Glutamic acid + glutamine	2 00 ± 0 43	(12)	0 957 (P < 0 06)	
Isoleucine	2 31 ± 0 79	(12)	0 900 (P = 0 10)	
Average of the series	1 76 ± 0 38 (N=10)			

NOTE The standard deviations (σ_n) are given from the regression parameters for *n* = 4 pairs of data

Correlation coefficients (ρ) are given with their significance levels

Chl = chlorophyll

* *d*₁ = 1 5 / *k* = 2 33, *d*₁ = 3 5 / *k* = 2 86, *d*₁ = 1 5 / *k* = 1 4, *d*₁ = 2 1 / *k* = 1 67

** *d*₁ = 1 5 / *k* = 2 33, *d*₁ = 3 5 / *k* = 2 86 — SD not calculable from 2 pairs of 2 points

$$R = R_0 \pm \frac{K(t)F_0^m}{(1 + \frac{d}{a})^{(n+m)}}, \text{ with} \quad (19)$$

K(t) a function in exp *t* and in *t*^τ

Additionally, experiments made in natural (and generally unexpected) conditions will hardly favor the verification of any relation in the three dimensions of our usual geometric space. Only models using well-defined molecular targets could therefore help in getting a more accurate validation of the observed results.

Traces left on the ground by a well-described though unidentified object have not often been thoroughly analyzed. A more recent case (Bounias and Daurade-LeVagueresse, unpublished) has led to significant conclusions, but the object could not be clearly depicted. Swords (1991) has reviewed a number of studies made in the Delphos case, showing differences in the mineral composition of the soil on the traces as compared with controls, and a retardation of plant growth within the ring area. Differences in organic composition were observed, as well as changes in the aspect of microbial species. It seems to me very important to organize the sampling procedure in such a way that the data can be eventually plotted against distances from the epicenter of the landing zone. In effect, provided a defined relationship can be assumed between the putative cause and the considered parameters, then the results would yield significant information on the nature of the physical forces involved and allow laboratory models to be established. In the Trans-en-Provence case, the finding that effects behaved as the reciprocal of the square of the distance suggests that the

involved energy might be of either electromagnetic or gravitational origin. Unfortunately, the author's proposals of conducting further research on the action of electromagnetic waves on biological systems (whole plants and molecular systems, such as enzymes, for instance) have not been funded, although they would have been suitable as subjects for doctoral research, and supported by the many results obtained in this field (Bounias 1984, 1987; Chernoff, et al 1992, Markov, et al 1992, Grundler, et al 1992). Even reports minimizing the possible effects of weak electric and magnetic fields (Oak Ridge 1992) have been severely criticized by other researchers (Savitz 1992, Tenforde 1992).

Concerning the problem of the alteration of growth of various organisms in the soil near the traces, quantitative work might also be done, for instance as follows

- 1 Ground samples would be collected at increasing distances from the epicenter, along a radial axis. Several radii should be selected according to biotope availabilities, in order to assess the spatial homogeneity of the effects. Note that in the Trans-en-Provence case, the sun exposure and the presence of a stone wall did not allow homogeneous sampling to be used along more than a single axis.
- 2 Samples in which significant changes in biological parameters had been quantified would then be diluted with control soil by factors of 2, 4, 8, 16, etc. Then, studying again the biological parameters in living organisms grown from these new samples would show the eventual quenching of the effects by decrements in the levels of causal agents.

Such laboratory work, by contrast with the effects studied on living organisms naturally present near the trace area, could be conducted in well-controlled conditions, and therefore allow a much greater reliability. Rather simple biological parameters, such as germination and growth of selected plants, might be sufficient in a first approach to allow accurate quantification of the studied phenomena (Bounias 1993) before more sophisticated research is conducted. My aim was limited to trying to take the best advantage of the situation and perhaps to help other researchers involved in future cases.

The author wishes to acknowledge Paris-X university mathematician Dr. André Bonaly for helpful discussions and revision of the manuscript.

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Comments and Responses

This section permits readers to critique or comment on works previously published in the Journal of UFO Studies. The original authors and other readers are encouraged to respond. Readers are also encouraged to critique or comment on the works appearing in this issue.

E.T. AND THE AIR FORCE

(Comment on Michael D. Swords, 1992, Astronomers, the Extraterrestrial Hypothesis, and the United States Air Force, *Journal of UFO Studies*, Vol. 4 79–129)

Dr. Swords's treatise regarding the influence of the extraterrestrial hypothesis on the U.S. Air Force's approach to the investigation and analysis of reported sightings of unidentified aerial phenomena is extremely well done and exhumes many different views from other works addressing similar subject areas. It is obvious that this paper did not suffer from a lack of dedication, nor an overdose of proprioception. However, there are several issues that I believe require discussion.

I am sure you are aware of the semantic difficulties associated with the discussion of unidentified phenomena. Most people understand the words "flying saucer" to mean a type of space vehicle, directly or indirectly intelligently controlled, and of extraterrestrial origin, an "unidentified flying object" is just that—an object seen in the air by observers with no preconception as to its origin or identity. "Flying saucer" assumes identification, with origin and purpose unknown, while "UFO" suggests that *nothing* is known about the phenomenon. Many people use these terms interchangeably, and this seems to be the view you have taken in your article. During my tour as director of the Air Force's aerial phenomena program I attempted, with Allen Hynek's concurrence, to have clear definitions for terms used in conjunction with the program, but usage outside of the Air Force, and to some extent within the military services as well, proved too difficult to control.

During the 1958–1962 time period we were acutely aware of the increased significance of the extraterrestrial hypothesis, particularly as it pertained to Mars and Venus, and we took positive steps to expand our interest and our ability to respond in this area. To this end the panel of specialists we formed to review the more significant cases was expanded to include an astronomer and a physicist from the staff, both with doctorates. In addition, on several occasions Dr. Donald H. Menzel spent weeks at Wright-Patterson Air Force Base reviewing both old and new cases, and he made many useful suggestions. Dr. Hynek also solicited opinions from some of his

colleagues, and this information also served to condition our thinking. We allowed this increased emphasis although we were aware of the danger of improperly vectoring the thinking of investigators and analysts alike, thereby destroying the objectivity that we felt was so important to the program. Another result, to our point of view undesirable, was the possible accusation by outsiders that we had some concrete evidence that called for this emphasis, a conclusion that would have been without foundation.

Recognizing that even the most open-minded person may have induced prejudices—misjudgments prompted by personal experiences and educational inclinations—we discussed many subjects with our panel in an effort to minimize this factor. We discussed the probability of life elsewhere in the universe and accepted the popular view that, with in excess of 100 billion stars in the Milky Way galaxy alone, some form of life was extremely likely. Our talks also addressed the prospects of interstellar travel, the other major consideration of the space-travel hypothesis, and it was uniformly endorsed that it would take a civilization centuries ahead of us in science and technology to deal with the major problems associated with this type of travel. While accepting the probability of extraterrestrial life and the feasibility of interstellar travel, we still had no unequivocal evidence from any sighting reported to us that would prompt any scientist to conclude definitely that our planet was being visited by vehicles directly or indirectly under the control of intelligent beings alien to earth.

This in no way denied the existence of extraterrestrial life or of visitations by space vehicles, but rather was a straightforward statement of the fact that we had no evidence to support such a conclusion. Frankly, those of us associated with the program hoped that we could uncover some such evidence. It is easy to postulate a conclusion to a receptive audience, for then the conclusion lends credence to the premise. On the other hand, to have an intrusive audience accept a conclusion that is not supported by plausible evidence is almost impossible.

As you well know, the primary effort of the Air Force in this program was to determine if the subjects of the sightings were in any way a threat to the security of the United States. It is a common belief that the service pursued this objective to the exclusion of all other interests. Nothing could be further from the truth. We were interested in determining signature data from these sightings to preclude the exercising of our defense forces needlessly. As our defense response system leaned more to automation as our technology advanced, this information increased in importance. Also, there was the investigation for genuinely new scientific information. Admittedly, this did have the lowest priority. During my tour with the project we prepared two staff studies recommending that either NASA or one of the scientific organizations of the Air Force assume responsibility for this phase in order to ensure increased emphasis in this area. It so happened that about that time planning was already underway to move the entire foreign technology function into the research and development organization.

During World War II intelligence detachments, staffed with intelligence, scien-

tific and technical specialists, were strategically located across the United States. The primary mission of these detachments in wartime and shortly thereafter was to search out, neutralize, and evaluate Japanese balloon bombs. Following the Kenneth Arnold sighting in 1947 these detachments found their mission extended to the investigation of sightings of unidentified phenomena that came to the attention of our military sources. This outstanding capability was available until the downsizing and reorganization of the services following the Korean conflict resulted in the closing of these detachments. Forced to relocate organizationally the responsibility for the investigation of UFO sightings, we turned to the Air Force bases. Though there were more bases than the lost detachments to support this function, there were shortcomings. The skills of the investigators no longer reflected the homogeneity that was so evident among detachment personnel. The reason for this is that a base with an operational flying mission requires personnel having different skills from a base with a research and development mission. Until the bases developed their capability for dealing with this new responsibility, there was a greater need for support from the Wright-Patterson office, reducing the number of cases to which we could give extensive attention.

To help with these analyses we turned to many agencies outside the Air Force. We used the FBI, the Food and Drug Administration, the Weather Bureau, the National Photographic Interpretation Center, and contractor laboratories. The importance of this new assignment to the bases was also mirrored in the fact that it was made a special subject of inspections by the Office of the Inspector General.

There were other programs that showed the Air Force's interest in space science, some of them more or less spinoffs of the aerial phenomena program. An attempt to communicate with extraterrestrial intelligent life, SETI (Search for Extraterrestrial Intelligence) was conducted. This project was much along the same lines as the current program of the same name but without the advantages that come with more than 35 years of improved technology. One program, Project Twinkle, to study fireball-like objects sighted in New Mexico, was conducted by the Cambridge Research Laboratory. A statistical study of the unidentified aerial phenomena reported to the Air Force was completed by a contractor in an effort to code them by color, shape, size, and any other common factors. The results of this study were published in Project Blue Book's *Special Report No. 14* (1955).

I hope this comment provides you with sufficient information regarding the Air Force's overall view of this program, its outlook on the space-traveler hypothesis, and some of the problems associated with carrying out this assignment.

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AN UPDATE ON *SECRET LIFE*

Secret Life was published in early 1992 and represented the state of my knowledge of the abduction phenomenon until that time. The book has enjoyed success in that it has sold nearly all the hardbacks that were published and is now in a trade softcover edition. It was the product of 27 years of UFO research and more than 300 hypnotic regression sessions with abductees. It is the first book devoted to UFO research that Simon and Schuster has published.

In March 1992 Simon and Schuster mounted a publicity tour for the book in New York, Philadelphia, Boston, Washington, San Francisco, and Los Angeles. During a two-week period, I appeared on about fifty radio and television shows. But, as all UFO researchers know, the subject is fascinating to the media, especially during "sweeps" ratings periods, and requests for appearances quickly escalated. Ultimately, I appeared on more than 250 radio and television shows in eighteen months. As can be expected, the hosts' level of knowledge of abductions was fairly low. If I was able to talk to the host before the show I would give my standard warning that skepticism was acceptable but ridicule was not. That seemed to work in most instances. Without the warning some, like Geraldo Rivera, found ridicule to be irresistible. In the main, however, the hosts of most interview shows were fair and a few even privately admitted to having had abduction experiences. The same was true in the print media. As all researchers know, no matter how sincere a writer seems to be, one never knows whether the article will be accurate or take the subject seriously.

The public reception of the book has been extremely positive. I included a note in the last chapter asking people who think that they might have had abduction related experiences to write to me. By the end of 1993, I had received more than 1,500 letters. The volume of mail was far greater than I had expected, but I wanted to respond to all the writers. I began by answering the letters as they trickled in. Then I fell a few behind, and before I knew it I was a hundred letters behind, then two hundred, and so on. It was obvious that I would not be able to answer the letters as I had wanted to. After about eighteen months of struggling with them, I hired a part-time worker to help me. I managed to answer about half of the letters before my assistant, for personal reasons, was unable to work anymore. The mail has continued to come in at about two or three letters each day and I am now trying to get help again, but answering them is an enormous task.

The letters, are, in Budd Hopkins's words, "a daily dose of pain." Many are extremely wrenching accounts of abductions and their sometimes catastrophic effect on the victims' lives. The writers' ages range from twelve years old to eighty-one, accurately reflecting the lifelong quality of the phenomenon. They describe the full range of procedures that I discussed in *Secret Life*: examinations, mental procedures, machines, staring, mindscan, child presentations, sexual activity, breathing pools, and so forth. Although few writers remember complete abduction events, many remember bits and pieces of them. Some detail a lifelong range of extraordinary phenomena closely associated with unremembered abduction events and have

come to suspect that they are abductees but want to find out for sure. Many writers are relieved to have finally recognized the origin of the unexplained phenomena that have been happening to them since they were children. They can now comprehend their experiences and they realize that they are neither mentally ill nor alone. Others are grateful to write about their experiences to someone who understands what they have been going through. Often they were loath to tell even their spouses about their "secret lives" without suffering ridicule and disbelief. Many writers characteristically state: "This is the most difficult letter I have ever had to write" or "I am going to write this letter and send it off immediately, because I know that if I wait I won't send it." Just as often they say that writing the letter has made them feel better because of its cathartic effect.

Many of the letters express fear for their families, especially their children. Parents feel powerless to stop their children's abductions. They plead for help and desperately want me to tell them what to do to bring an end to the phenomenon. These are the most heart-rending letters and I read them with despair at their plight and with frustration that all I can offer them in the way of protection is words of sympathy and encouragement. Sometimes children and adolescents write about their experiences. On the same day in December 1993, I received two letters from a boy, 12, in Iowa, and a girl, 14, in California. Each of these letters, profoundly moving, was an articulate description of a young life intertwined with abductions and the anguished victim struggling to come to terms with the situation. The clarity and depth of their writing was unusual, but their circumstances were not. The fourteen-year old girl wrote of having inordinate fears of being in the dark, missing-time experiences, unexplained vaginal problems, loss of romantic interest in boys, and inexplicably wanting to help aliens "with their studies." The thirteen-year old Iowa boy talked about seeing aliens, missing-time experiences, anomalous scars, and having a mother who has had experiences. "I wish it would all stop!" he wrote.

The volume of letters has once again shown the enormous scope of the abduction phenomenon. The Robert T. Bigelow Foundation's Roper Organization poll correctly pointed out the potential number of abductees in America. The letters, phone messages, and the response from books and media appearances that both Budd Hopkins and I have received simply reaffirm what we already know—great numbers of people have had abduction experiences. Budd Hopkins alone has received more than 8,000 letters since 1981. This number is obviously the tip of the iceberg. There are formulas to determine the number of nonwriting people each letter might represent, but suffice it to say it is large. The Roper Poll's finding that at least 2% of Americans have had experiences consistent with the experiences that abductees have had is accurate and perhaps even conservative.

One of the great problems with abduction research is that there are so few practitioners who are competent in it or who can give aid and comfort to its victims. Furthermore, the methodology employed in abduction research is not standardized and at times it can be haphazard. Incompetent hypnosis, confabulation, false memories, implanted "memories," and a hypnotist's personal agenda can all come into

play when trying to understand a person's abduction experiences. To help remedy this situation, the Robert Bigelow Foundation has generously sponsored a series of free workshops for mental health professionals designed to introduce them to the abduction phenomenon, teach them methods in hypnosis, and suggest therapeutic strategies for abductees. These have been very successful with more than 400 therapists participating in workshops in Los Angeles, New York, Atlanta, Chicago, and San Francisco. Budd Hopkins, John Carpenter and I have participated in them along with Michael Swords, and John Mack at times. By doing this we have been building a list of therapists to whom we can refer potential abductees for therapy and research. This is extremely important and more work will be done in this area.

The therapists' response to *Secret Life* has been very gratifying. They, and many abduction researchers, have used it as a navigation guide through the abduction experience and have continually replicated my findings. In addition, I have done a considerable amount of private consultation with therapists around the country who are unable to attend the workshops, and who are either interested in the subject or who have clients who are victims of it. I have also worked with abductees who have been referred by therapists who read my book and realized that they had little expertise in this area. Some of them have been present at the regression sessions of their clients in order to learn hypnotic techniques along with the root of their clients' difficulties. Of course, I and other abduction researchers have helped mental health professionals who are themselves abductees understand and cope with their own experiences.

The verification of the information in *Secret Life* by members of the research community has been extremely encouraging. I have been in the happy position of having nearly all that I have written in my book about the nature and structure of the abduction phenomenon verified by others. Abduction research is an area where knowledge is growing at a tremendously rapid rate. If I were to write the book now, I might put in more events and new facts, and I would highlight others that might not have been emphasized enough, but I would not change the structure of the abduction or the procedures as I have organized them, nor would I rescind what I have written. Of course, there are those who do not agree with this assessment. As can be expected, professional debunkers, CSICOPers, and nervous amateur UFO researchers have mounted superficial attacks on it. Their commonality is that they are not primary abduction investigators and know virtually nothing about the intricacies of abduction hypnosis, research, or therapy. It has been satisfying to know that those researchers who are engaged in the hypnotic investigation of abductions and serious research in the subject have applauded the book. They have told me that they have used it as a guide to help them decipher the abduction accounts that they have been researching.

It is also important to understand that the abduction phenomenon is not restricted to the United States. I have received letters from researchers and possible abductees from around the world including Europe, the Middle East, Asia, and Africa. If my mail is any indication, one can expect to find the abduction phenomenon every-

where There is good evidence to believe that the accounts are the same regardless of upbringing or culture Recently I worked with a young man born in the West African country of Mali who has had typical abduction experiences It is testimony to the international nature of the phenomenon and interest in it that *Secret Life* is currently being published in England, France, Spain, Germany, the Czech Republic, Taiwan, South Korea, and Japan It will be interesting to see readers' response from these countries I suspect that it will further confirm its global extent

Finally, I would like to take this opportunity to thank all the readers, researchers, academics, and mental health professionals who have made it a point to send me words of encouragement and gratitude I am now beginning work on a new book that will extend my findings and also be a bit more speculative about the meaning of the abduction phenomenon Perhaps I will even attempt a hypothesis about what direction it is headed and why Publishing in this area is a very risky business for an academic, and Temple University has not been overly receptive It is always comforting to know that there are those who sympathize with the problems of writing a book on this subject and who have gone out of their way to be supportive In these contentious times I am heartened by the knowledge that there are many kind and caring people in this field

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ABDUCTION RESEARCH ENCOURAGED

(Response to Jacobs)

The *Journal of UFO Studies* thanks Dr Jacobs for his informative letter documenting his research odyssey of the past two years We applaud his dedication to the phenomenon, his commitment to caring for human beings expressing great troubles to him, and his call for more competent researchers and therapists to join in the exploration and clarification of this mystery The *Journal's* role in all these UFO-related matters is to welcome competent research and writing from all schools of thought on these controversial issues, and to serve as an academic-oriented repository of the best information and thinking related to the field As such, it must properly be both open and academically conservative at the same time This difficult, perhaps impossible, task is nevertheless one which *JUFOS* has taken on, and will continue to attempt to do

The relevance of this editorial policy to the abduction phenomenon is obvious, at least to the editor That is, all responsibly researched and reasoned positions are potentially publishable Secondly, all published studies and materials will be cri-

tiqued by the conservative standards of academia, and that critique related to the data, assumptions, and reasoning in the published piece itself. Such a policy unfortunately can manifest itself in cool assessments of some works, for example, the Roper Poll. This is not a happy circumstance, but it is a professional one, and while apologizing for the discomfort this may cause, the *Journal* wishes to call upon the creativity and commitment of many researchers to do the necessary work and writing, each in their different styles, ultimately to clarify these fascinating and important matters. This is Science—a community of many diverse thinkers working, publishing, debating, from whom is distilled a more accurate conception of the world.

David Jacobs, a friend who is often nettled with me, understandably, has certainly done his part to contribute to the necessary mix from which this more accurate conception will emerge. His level of involvement is not excelled by anyone. Regardless of what each researcher, armchair or active, may think of the details of his CE4 scenario, it is the product of great commitment and should be examined closely by others willing to make a significant commitment of their own. In this spirit, *JUFOS* would like to express its desire to receive academic-quality articles on UFO abduction research from both established and first-time authors.

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Book Review

Demons, Doctors, and Aliens By James Pontolillo Arlington, Va The International Fortean Organization, Occasional Paper #2, March 1993. 22 pp

A mere 22 pages seem too confined a space to argue for a radical new perception of the UFO abduction phenomenon James Pontolillo attempts just such a feat in his booklet, *Demons, Doctors, and Aliens*, where he undertakes "an exploration into the relationships among witch trial evidence, sexual-medical traditions, and alien abductions "

This exploration includes a refutation of the case for literal abductions as well as two constructive arguments, one to link abductions to older beliefs, and another to identify the cultural motivation for all these traditions Ambitious as his goals may be, he raises important questions about the abduction phenomenon and builds a provocative case for a psychological point of view Yet this report is ultimately a failure The many strengths of the argument conceal critical weaknesses, and the reasons why it fails in both its destructive and constructive aims make up an exemplary lesson in scholarship gone astray

His thesis is that abduction reports tell us nothing about aliens, but say a great deal about negative Western attitudes toward women This surprising notion makes sense to him because abductions are about sexual and bodily victimization, sometimes of children but most often of women, and this theme has a long history in Western culture The same idea characterizes the rich traditions of witchcraft and demonology during medieval and Reformation times These traditions grew out of a perception of women as sexual and evil by nature that appears in early Jewish texts, then develops to the full in Christian writings Nineteenth-century medicine and psychiatry joined religious institutions in the cause of repressing female desire, while expressions of its danger pervade the art, literature, and attitudes of 19th- and 20th-century secular society Narratives of involuntary medical and sexual abuse are neither new nor unique to abductions The growth of the abduction phenomenon marks a new and active variant of an old tradition of cultural misogyny Aliens take up the work of supernatural beings, but the medical and sexual focus proves that the message remains the same

Pontolillo builds his case on the ruins of a literal interpretation of abduction He rejects any claim of alien visitations because it lacks persuasive demonstrations in its favor, while too many likenesses to other cultural phenomena stand against it He dismisses physical evidence on the grounds that none of the scars, implants, or resi-

dues attributed to alien activity have borne out claims of extraterrestrial origin

The source of abductions is psychosocial in his view, but not in any simplistic sense. A mixture of factors contributes to this complex phenomenon. Content derives in part from genuine experience, perhaps half-remembered medical procedures or actual sexual abuse, with strong emotional reactions still attached. Other key story elements come from exposure to abduction accounts in the media, where books, articles, movies, and television equip narrators with all the plots and content they need. The personality of narrators plays a part when an ability to fantasize helps the story along, and personal needs motivate people to claim abduction experiences. A vital ingredient in the formation of abduction stories is the interaction between subject and investigator. In fact much of the blame lies with abduction investigators who confabulate the UFO fantasy they want to hear by leading phant witnesses under hypnosis to tell the appropriate story.

Pontolillo marshals an imposing list of objections from the critical and skeptical literature to demolish the credibility of the extraterrestrial hypothesis. His positive argument builds on the similarities between UFO abductions and older traditions of kidnap and assault by supernatural powers. Here again he amasses many parallels. For example, the devil left a mark on witches just as aliens scar abductees during the examination procedure. Traditions of sexual contact between demons and humans closely resemble the sperm-taking and impregnating activities of abducting aliens, while stories of offspring from the mating of demons and fairies with humans correspond with modern accounts of hybrids. These parallels confirm the relationship of new tradition with old, and underscore the need to seek their meaning in a common theme. He finds that theme in the prevailing misogyny of Western culture.

Concise, forceful, and scholarly in tone, the report covers a lot of territory in its brief space. Pontolillo seeks out the many vulnerabilities of the proponents' position with a keen eye and scores the shaky case for an objective phenomenon with a sharp pen. No one can deny the painful elusiveness of physical evidence, or its necessity to establish the reality of such a bizarre claim as alien abduction. Many are the claims, but no proof has been forthcoming. The assertions trail off into discouraging uncertainty. He is also quite right to question the techniques of investigators, and to warn that confabulation under hypnosis can produce vivid false memories despite the good intentions of the hypnotist. Recognition of the multiple possible cultural and psychological contributions to the story comes as a relief after hearing so many critics beat the drum for their one favorite solution. His efforts to relate abductions to witchcraft, demonology, and a broader pattern of cultural misogyny are more speculative, but he touches on enough similarities of content, structure, and theme to demonstrate that the concept is not so far-fetched as it first might seem. He raises a strong voice for psychology as the answer to the abduction mystery.

Why then does this report ultimately prove unsatisfying, at least to this reader? The answer lies less with the claims of the argument than with the ways the author builds it. An initial impression of a tight case gives way on closer examination to awareness of all too many gaps and flaws.

Pontolillo never misses a weakness in the arguments favoring an objective phenomenon, but at the same time he never acknowledges one iota of puzzlement or mystery in the reports. He treats the skeptical position as a "done deal," as if proponents cling to their beliefs without a shred of evidence or counterargument to support them. An inexperienced reader might well regard Pontolillo's refutation of abduction credibility as a devastating and definitive critique. These appearances are deceiving. The refutation works only through a consistently one-sided argument that never gives the opposing viewpoint a hearing. Well-known skeptics may set up a straw-man argument to knock down, but here even the dummy takes a vacation. In these pages the positive case simply does not exist.

One disturbing hallmark of the report is a biased selectivity of evidence. Old arguments long laid to rest now rise from the grave as if alive and well. Experiments with hypnotized non-abductees produce reports "that differ in no substantive manner from the reports of those who claim to have been abducted," so we read. What we do not see is the fact that the entities described in these experiments differ from subject to subject, and the body of descriptions in no way duplicates the frequency of types found among abductee reports. Philip Klass proposed a relationship between investigator personality and the nature of the abduction experience each investigator finds. This proposal becomes supporting evidence that the investigator rather than the abductee determines the quality of the experience. In reality the same investigator typically hears abductees with both positive and negative reactions, but the reader of Pontolillo's report never learns of this counterbalancing evidence.

Running debates in the literature run on just one side of the street when taken into this report. A recurrent consistency in the event sequence of abduction reports strikes me as unusual, while Martin Kottmeyer and Dennis Stillings argue that narratives naturally assume a dramatic structure. In this assertion they are quite right. Jokes and urban legends, even accounts of personal experience do indeed take on a spontaneous structure with the story elements ordered for dramatic effect. Pontolillo correctly reports the statements of these critics, but at the same time he misses the point. Repeated narratives like urban legends maintain a dramatic structure even as they vary the elements that fill the structural positions. The first narrator emphasizes one event as climax while the same event becomes subsidiary for a second, event substitutions may change the entire story, and all these variations happen within the framework of dramatic structure. The peculiarity of abduction reports is their tendency to keep the same events in the same order even when no logical necessity demands it, and when the personal differences in abductees and investigators should upset this consistency. Occurrences counter to expectation usually merit attention in any fair evaluation, but none is forthcoming here.

The litany of criticisms offered as absolute and unscrutinized goes on and on. Multiple-witness cases count for nothing, for all we find in these pages. Hypnosis is responsible for much of the consistency in abduction reports, never mind that people tell essentially the same story with or without hypnosis. Media influence gives people the ideas for abductions, yet for some reason abductees select the same elements

over and over, despite the countless more imaginative ideas Hollywood has placed at the storyteller's disposal. Investigators lead their witnesses to tell the same story, of course, but it never strikes the author as odd that the investigators lead with a curious singleness of purpose. No less than 50 investigators furnished a sample of 103 high-information, high-reliability reports in my comparative study of abductions, yet this sample showed considerable consistency in event sequence and content despite inevitable differences in expectations, beliefs, and wishes among the investigators. The reward of ignoring this evidence is simplicity, but the price is a serious distortion of the facts.

A suspicion that the author is only selectively familiar with the abduction literature also creeps in to stay. He cites a limited number of famous cases and never mentions the landmark reports of Charles Hickson or Travis Walton, both publicized in the national press and important in any argument for media influence. Pontolillo raises the issue of fantasy proneness as a probable contributing factor, but disregards studies such as those by Kenneth Ring that tested for this trait and failed to find it. Budd Hopkins introduced the "skeleton key effect" at the 1988 MUFON UFO Symposium, proposing hidden abduction as the answer to some of the phobias and odd behaviors of abductees. David Jacobs elaborated a similar idea of postabduction syndrome at the same meeting. Pontolillo raises these proposals and disparages them without citing the sources from which they came, relying instead on a secondhand source and emphasizing only the most extreme consequences as if they represented the norm. The author shortchanges his own position when he overlooks serious doubts about missing-fetus cases arising from the research of Ann Druffel and Dr. Richard Neal. These omissions cannot inspire an aware reader with confidence that the author is in full command of the material he criticizes.

An uncharitable picture of abduction investigators and researchers fills Pontolillo's report. They are human and suffer from human failings, but surely the declaration that "pseudoreligious beliefs, conscious fraud, scholarly incompetence and the promise of monetary gain have all been rampant subsidiary influences in modern ufology" is unnecessarily harsh. Does it seem too implausible that investigators who have spent many hours listening to abductees might become convinced by what they hear? These investigators may reach wrong conclusions. They may even be guilty of unwittingly leading the witness. But honest error or a will to believe some very convincing stories need not be equated with moral turpitude. Worse still, the persistent one-sidedness in exposition of the abduction arguments and evidence makes proponents look like gullible fools. They appear to hold their position without a reasonable leg to stand on. They are neither knaves nor fools, and to imply otherwise even by omission demonstrates why theologians identify some forms of omission as a sin.

The art of rhetorical slanting turns up time and again in this report. We read that Dr. John Mack is "one of the few professional psychiatrists to espouse a pro-abduction stand." This statement is true as it stands, but it would stand more forthrightly in a context that acknowledged him as one of the few psychiatrists who has actually

listened to abductees Pontolillo makes much—and rightly so—of scientific conclusions that memories recovered through hypnosis are prone to gross distortion. Any defense of abduction data must answer this concern, and I attempted to do so by testing hypotheses derived from the theories. A comparison of various investigators as well as of reports obtained with and without hypnosis revealed no apparent difference in sequence or content. Hypnosis failed to leave its predicted marks. Pontolillo states that my response to the criticisms of hypnosis “consists of shallow contradictions unsupported by clinical evidence.” I willingly admit that my study left much to be desired and no one should read too much into it, but my effort took a basic step in scientific methodology, and gave me some evidence for ammunition. Pontolillo offers no tests, just assertions, then implies that I am the one who merely shouted no.

If Pontolillo is right to refute the literal truth of abduction, he is right for the wrong reasons. His argument misrepresents the current state of knowledge and debate, but to what purpose? Do we sense doubts that his case could stand its ground in a fair fight? Simply closing an eye to positive evidence and genuine puzzles is bad strategy. It cannot make them go away. Any informed reader knows they are there and wants to see a confrontation with facts, all the facts, and not with fictions contrived for unfair advantage.

Readers hoping for these faults to mend themselves in the constructive arguments are again doomed to disappointment. The author sustains his case for abductions as modern variants of old traditions on the same unbalanced diet of selections and rejections that failed before.

An assertion that “alien abductions are the continuation of an ancient, ongoing cycle of religio-mythic beliefs” recurs throughout the paper. This claim marks an important proposition—the relationship between abduction and older traditions is genetic or organic, rather than circumstantial or superficial. How modern narrators retell an old story in a new guise comes out in the conclusion. Medical experiences, sleep-related imagery, and life traumas provide the raw materials, which are then “reconstructed in accordance with historical and cultural precedents,” and the influences of both media and abduction investigators. Abduction stories are not primarily the result of some mysterious teleology, but conventional products of lines of transmission linking the present with the past.

A persuasive argument that traditions bear a genetic relationship requires one of two demonstrations—that one tradition has an undeniable historical continuity with another, or the traditions share too complex a pattern of sequence and content for chance to explain. Pontolillo attempts both arguments, invoking a history of cultural misogyny to support one point and the content of witchcraft, demonology, and fairy lore to uphold the other. Neither effort is very persuasive. To begin with the attempt to relate abduction content to earlier traditions, we see a procession of comparisons more self-serving of the argument than self-evident.

Alarm bells ought to ring whenever we read of witchcraft treated as one monolithic tradition. On the contrary, medieval and Reformation witchcraft ideas belong to a complex tangle of traditions, some driven by folk beliefs, some by learned

notions, with extensive geographical and temporal differences that cannot compress into one tidy package without serious damage to the contents. The same oversimplification spoils Pontolillo's treatment of fairies. He refers only to the human types without indicating that concurrent traditions included the small and the large, the beautiful and the ugly, the godlike and the bestial, the humanoid and the monstrous. At one point he notices that the monk Guazzo "relat[ed] several conflicting case histories" about witchcraft, but never follows up this clue to plural traditions and claims the whole mass of associated motifs for his usage without ever asking which witch is which.

An undeniable convenience results from this laxity. It throws open a cultural unabridged dictionary full of so many descriptions that comparisons come easily. No coherent traditions get in the way and motifs broken off from their context pass for satisfactory units of comparison. The genital equipment of demons may be animal as well as human, the penis huge and scaly or sinuous and pointed, sensations associated with the sex act burning or cold and pleasant or painful. Sex acts with aliens or genital examinations involving instruments so closely resemble these same descriptions, we read, that "the parallels with purported demonic sexual relations are inescapable." Let's see—large or small, hot or cold, pleasant or painful, rough or smooth. Is any possibility left out? If this sort of evidence proves the continuity of traditions, no rapist need fear conviction ever again. Now instead of pleading the devil made him do it, he can cite the victim's description, whatever it may be, as proof positive that the devil did it in person.

The truth is that slipshod comparisons prove nothing at all about the relatedness of two traditions. When the pool of possible comparisons becomes too large, some matches are inevitable by chance alone. The matter grows worse when the author forces a union between motifs with all too little in common. In one (unique?) example from the annals of demonology a woman claimed that an incubus performed sexual acts with her through a hole just below her navel. This instance must mean that the needle-in-the-navel or abdomen from abductees has the same origin. The incubus sometimes had a bifurcate penis able to abuse two orifices at once, and the reader is supposed to see the obvious relationship between this description and an abductee who had tubes in her throat and urethra at the same time. Surely these matches amount to an academic shotgun wedding.

Double standards for evidence are commonplace. The bifurcate penis reappears as alien examiners inserted two thin probes up an abductee's nose. According to a psychoanalytic theory, the nose is an erogenous zone, so the probe becomes a penile symbol. Yet we also read that most examination features are not symbolic, but distorted memories of genuine medical or sexual experiences. Elements of the abduction story seem to be real or symbolic as circumstances require. Another example centers around Pontolillo's claim that abductions are about women. When David Jacobs relates the sexual ordeals of women abductees, his findings deserve close attention because "the many correspondences between the events comprising both abductions and demonic liaisons lead inevitably to one conclusion." Male experi-

ences “apparently are of insufficient importance to warrant critical examination or classification in any manner,” thereby confirming the focus on women. Then we read in another paragraph, “as an adjunct to his efforts at unifying and casting a pseudo-medical air over abduction experiences, Jacobs relates a stereotypical sequence said to occur among the male abductees he has worked with.” We are supposed to understand that pattern in women’s experiences proves Pontolillo’s argument, while pattern in men’s experiences either does not exist, or proves only that investigators strive to make something out of nothing. The investigators simply can’t win, their critic can never lose.

Blaming coercive investigators or media exposure makes sense, at least if we ignore the number of investigators involved and the variety presented by the media. Yet Pontolillo’s argument only rarely calls on these reasons, and instead relies on assertions harder to justify. The bifurcate penis occurs now and then in demonological lore, and bifurcate instruments occasionally appear in abduction reports. He sees enough similarities to cite this detail as proof of kinship between the two traditions. What are we readers to make of that claim? Most people are not well versed in the minutiae of witchcraft testimony, and the bifurcate form only rarely appears in either donor or receiver tradition, so a case for “historical precedent” seems strained beyond the breaking point. Analogies are easy to find if the terms are abstract enough and limited to superficial similarities of form or function. The result is also quite empty of meaning. The argument best demonstrates just how implausible the supposed chain of transmission really is.

Neither details nor larger patterns support the proposed relationship. All too many so-called similarities hang on one favorable aspect, while a multitude of differences go unnoticed. The “supernatural lapse of time in fairyland” is like abduction time lapse only insofar as they share an apparent breakdown in the usual flow of time. In fact the two are very different, with real time running much slower in fairyland than on earth, while the abductee’s time lapse is only a memory gap or block. Fairies are supposed to be like aliens in disposition because both types of entities may be “malevolent, kind or indifferent.” Again, what else could they be? They differ a great deal in appearance and activities. Pontolillo admits that the traditions he wishes to compare may intersect only here and there when he says that

the only explicit abduction-related activities that have been consistently attributed to fairies are 1) the semi-coerced taking of humans to visit the fairy realm and 2) the practice of substituting a fairy “changing” for a human child.

The same admission recurs when he says that the abduction medical scenario is a modern addition, and

in none of the voluminous witch trial documentation or in the more limited fairy lore is any mention to be found of medical procedures

practiced by these supernatural entities upon their human consorts

Differences do indeed separate abduction reports from older traditions. The torture of witches at the hands of inquisitors parallels the examination of abductees, but legal proceedings carried out by humans differ from medical procedures carried out by aliens in terms of actors, locations, activities, and purposes. Inconsistencies also dog any attempt to equate old and new traditions of hybrid offspring. Demons could cause false pregnancy, impregnate women with semen stolen from men, or breed monstrous hybrids. Fairies were capable of breeding with humans, or stole human babies and replaced them with changelings (usually aged fairies only apparently infantile). According to abduction reports, pregnancies vanish not because the aliens deceive a woman, but because they steal the fetus in its early development. A child made from human semen and egg would not be hybrid but fully human. Monstrous births and changelings do not figure into the abduction tradition at all. Only the combination of alien and human—a genuine hybrid—appears in abduction reports, a sameness that compares with only one strand of earlier traditions. Their diversity contrasts with the consistency of abductions to set these traditions apart, not to support their alleged similarity.

The case for abductions continuing older traditions succeeds only in a neverland of distorted evidence and contrived argument. If the abduction examination doesn't fit the past, it becomes a modern addition. If the revelry of the witches' sabbat has no modern parallels, it disappears from consideration. Details chosen at random and blurred by abstraction, a piece from this tradition or that come together in an ad hoc assemblage devoid of persuasive power. The fable of the blind men and the elephant comes to mind. The man who felt only the tail decided that he held a rope, the man who felt only the legs concluded that they were pillars. An emphasis on details and a dedication to preconceptions convinced each man that he was right, but the elephant got away. Some parts of abduction reports are analogous with some elements of older traditions, but this finding by no means establishes homologies between past and present. The ties that bind these traditions into a continuity have the tensile strength of cobwebs.

The historical identification of abduction as an expression of cultural misogyny fares no better than the previous arguments. While it is true that one strand of scholarly opinion traces the witch-trial epidemics of the 16th and 17th centuries to outbreaks of an underlying hostility toward women, this notion reflects a minority viewpoint and just one of many interpretations of equal or greater merit. Pontolillo addresses little of the extensive witchcraft scholarship. Had he consulted a prominent study like Norman Cohn's book, *Europe's Inner Demons*, he would find the subject more nuanced and less simplistic than his interpretation allows. Cohn distinguishes between folk witchcraft on the local level and learned theories promulgated by social institutions. Accusers in the local village usually blamed the witch for *maleficia*, acts of malicious harm like sickness, hailstorms, or injury to livestock. These accusations grew out of interpersonal conflicts and the vicissitudes of daily life. The

concept of witchcraft driving church and civil authorities to persecute witches on a large scale centered around the dread of a vast conspiracy of Satan-worshippers, a powerful, well-organized cadre committed to act as the enemy of order, goodness, and humanity itself. In practice the two ideas often merged. Women *were* frequently the victims, and regarded as more susceptible to Satan's lure. Yet these facts fall far short of proving that a special brand of hatred drove the witchcraft holocaust.

The case for continuation of a misogynistic tradition into the 19th and 20th centuries depends on some eccentricities of medical history and certain artistic representations. Some appalling misapplications of medicine and psychiatry did indeed take place, and yes, fear, suspicion, and hatred of women do appear in art. The question is, how broad a generalization can we make from these examples? I suggest we can't go very far. A common error indexes an entire culture by the actions of its cultural elite. These intellectuals most often leave their thoughts behind, but their impact as well as their ideas may be quite marginal to the vast population that lives and dies under the influence of folk and mass culture. Popular representations of women are often adulatory, or even show them as human, with everyday strengths and weaknesses. The rabid misogyny of some intellectuals has never displaced these more tolerant popular images, and even the "vamp" or "fatal attraction" depictions of women in the movies amount to a small minority of all the facets portrayed. If we are to believe that media influences are such potent shapers of our opinion, we must then accept that their cultural message on women is more benign than misogynistic on the whole.

Why choose to compare abductions with witchcraft, anyway? Its popular associations with women make it the comparison of choice for finding misogyny, but that basis for selection amounts to begging the question. We never read a word about other relevant traditions. Tours of heaven and hell, or shamanic initiations, also have a long history and resemble abductions in many respects. Virulent outbreaks of persecution, propaganda, and atrocity stories aimed at Jews, Catholics, Freemasons, and Communists have legitimate claims as heirs of the witch hunts, but go unmentioned in this paper. Are they excluded before the fact because males are too prominent? A misogynistic interpretation for abductions seems to work only by limitation of the comparisons to a select and convenient few.

Even if misogyny infected Western culture to the extent Pontolillo asserts, the best reason to doubt this explanation for abductions is the number of males involved. In my catalogue of some 300 abduction cases appearing in the literature to about 1985, men outnumbered women by 64% to 36%. A new catalogue of cases since 1985 breaks down to almost exactly fifty-fifty. A survey of abduction investigators growing out of the 1992 MIT conference brought responses from thirteen investigators. Seven of them found a 60%-40% preponderance of females, one other an even higher proportion. Three respondents found half and half, while two found a disproportionate number of males. The proportions differ from investigator to investigator, and the slight overall leaning toward the female side amounts to underwhelming evidence that women are the real subject of abductions.

In all fairness Pontolillo could not know most of these findings. The authority he cites for exact figures is David Jacobs, who reports that 56% of his subjects are women. When balanced against the figure of 80% that Pontolillo cites as the portion of female victims in witch trials, the reader can only wonder how a 6% majority of women is supposed to compare to a 30% majority. This unusual arithmetic adds up to a difference between the two traditions, rather than the alleged similarity.

Neither comparison nor history support the proposition that abduction reports continue older traditions. If no chain of transmission is apparent, another possibility is the re-creation of a past belief as long as its basic theme survives. There is of course a strand of misogyny in Western culture as in many others, so the rebirth of a new tradition of victimization much like the old and without any direct ties to the past remains a possibility. Yet the proposition that abductions are expressions of misogyny needs lots of women to earn any credibility. They simply are not present in the records, and this argument falls to pieces on that hard fact alone.

All in all this paper is a serious disappointment. Not one of the three points of argument receives anything like satisfactory proof, but this failure is only part of the letdown. A more serious concern is the pervasive lapse of basic scholarly standards. The author is clearly learned and capable of an evenhanded weighing of evidence pro and con, yet fails to make the effort. What we read is an argument cast in black and white, no shades of gray allowed. Conclusions do not follow from evidence, but determine the selection and slanting of evidence until the result is a caricature of scholarship, an exercise in propaganda and polemic masquerading as scholarship. The bad buries the good until the reader loses sight even of the legitimate criticisms and reasonable proposals. Nothing like the careful historical research of Martin Kottmeyer or the good humor of Phil Klass enlivens the text, and it certainly does not sum up the present state of debate in an accurate and informative manner. I cannot recommend this paper as a way to learn anything about the abduction phenomenon, except the lengths to which some critics go to discredit it.

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Literature Reviews

SETI/ETI AND UFOs

The following literature review is not comprehensive or anything close to it. The Extraterrestrial Intelligence (ETI) literature, even in book format, is too vast. What I have done here is choose a set of volumes which, if owned, would constitute an excellent nucleus for a reader's understanding of the topic and how the state of thinking has developed. Other books cry out for inclusion, especially when one moves past overviews such as these into deeply technical studies, but in my opinion these are the best choices.

Life on Other Worlds By H. Spencer Jones. New York: MacMillan, 1940. 299 pp.

It is from this volume that the great afflux of scientific ETI literature finds its origin. Its author was prestigious: Sir Harold Spencer Jones, Britain's Astronomer Royal. Its timing was inspired, perhaps, by a smattering of spectroscopic information coming in from Venus and Mars and the blossoming of an expanded concept of the cosmos. It was also a time of a paradigm shift in planetary formation theory, and the first inklings of learning how simple molecules might combine to make biological ones. A strong intellectual stew was brewing and Jones had the courage and adventurousness to step up and tackle the speculations directly.

The result was this book of ten chapters which treated (not in his words) the cosmos, the possibility of life arising, science's tools, the development of the Earth, solar system bodies without atmospheres, the giant gas planets, Venus, Mars, the origin of solar systems, and possibilities beyond the solar system. This array of topics set the ground rules for the debate on ETI, which was beginning in earnest. As time went on, astronomers such as Frank Drake, director of the Arecibo radio telescope observatory of the National Astronomy and Ionosphere Center, refined and reordered the concerns of these topics into the famous Drake Equation for organizing debate on the factors involved in ETI development. But men like Jones defined the debate's initial focus.

The first several chapters do what almost all ETI books do: set a respectable scientific tone for the more speculative material at the book's end. Almost all academics construct books this way. It is called "rationally building one's case" or "politically protecting one's reputation." Simply jumping into the fun of it all just won't do. Jones finally starts enjoying himself in Chapter 7, "Venus the Earth's Twin Sister." Here he squeezes every bit of tenuous data on Venus dry. And he finds that despite the views of others (including Harvard astronomers Fred Whipple and Donald

Menzel), Venus is dry and no fit place for life. Unfortunately for the romance of it all, he was right and Whipple and Menzel were wrong.

The next chapter is wonderfully entitled "Mars, the Planet of Ebbing Life." He spends some time on Lowell's "canals" and ETI hypothesis, and the debate that raged across the Atlantic between opposing camps. The subject is treated with respect, and Jones tentatively opts for the canals as real but natural formations (and not so many as in Lowell's maps). He suggests an atmosphere with no oxygen but with the probability of life, albeit struggling and dying away. H. G. Wells it is not, but certainly close enough to light the imagination.

After discussing the new theory that solar systems are common rather than rare, Jones's eye is led inevitably to the vastness of deep space. Here, in a short conservative chapter, is where his heart, like those of many SETIans to come, really lies. Far out in the darkness, safely separated from ourselves and sober scientific scrutiny, the "others" surely are. Maybe, he says, they are our equals, then, in a moment thunderstruck with awareness, perhaps our superiors. Inspired by the melancholy that often plagues the educated, "we may hope [they] have managed their affairs better than Man has managed his." But 'tis all speculation. "It must remain forever a sealed book," for never shall we meet.

The UFO phenomenon doubtless came as a disagreeable shock to the good doctor. I doubt whether Jones would ever have written the book if the UFO wave had begun in 1940. The book, or at least its ideas in the Mars and deep space chapters, can be seen as providing a reasonable foundation for the ETH explanation for UFOs. As UFOs intruded upon his intellectual life, he responded as most of his tribe have, in public anyway, by hurling insults and wholly negative commentary. In this way he became the father of the schizoid SETIans whose doctrine would be: They're everywhere but here.

Intelligent Life in the Universe. By I. S. Shklovskii and Carl Sagan. San Francisco: Holden-Day, 1966. 509 pp. Translation and revision of Shklovskii's *Vselennaiia, zhizn', razum*, Moscow, 1963.

Through the late 1950s and the 1960s, our knowledge of the atmospheres of Mars and Venus increased greatly, and the likelihood of other intelligent life forms birthed in our solar system became very slim. This distanced ETI even more effectively and made academic debates safer. Great scientists like Harlow Shapley were willing to speculate about millions of deep-space civilizations. And, as long as they conveniently ignored any insights from the history of technology, the professors could comfortably engage in all sorts of discussion about life way out there not able to get here. Into this brew came an imaginative risk-taking young man named Carl Sagan, who hadn't yet learned how narrow were the mean streets of elitist academia. And he started saying all sorts of interesting things about the possibilities of extraterrestrial intelligence. By serendipitous coincidence, he sent some of these ideas to an imaginative old man from a risk-taking country, who was just completing a popular

book on the topic. A congenial synthesis of data and creativity was born, and thus, the book we have before us.

The volume is partly the pop-science book of Shklovskii and partly added material, mostly by Sagan. Both authors approved of every part of the final product. It has 35 chapters, broken into a short introduction and three sections of eleven chapters each. These sections follow the typical trend of ETI books: information on cosmic development, information on biological development, and ideas about ETI.

The introductory Chapter Two is interesting to students of ufology, and needs a little background. Shklovskii's original book contained some fascinating speculations about what has now come to be known as the "ancient astronaut hypothesis." The ideas of Soviet ethnologist M. M. Agrest struck a chord with Shklovskii, and Sagan's partner-to-be included discussions of the Bible, the Tassili Saharan frescoes, etc., in the original book. These were seriously considered as candidates for ancient astronaut influence. This idea obviously rang the young American's bell as well, and he decided to include and augment it in their joint book. But this was surely risky with all the talk of UFOs. Sagan's solution, I believe, was Chapter Two.

In 1961, with a shiny new Ph.D. degree and impressive research credentials from the University of Chicago, Sagan began his professional career at the University of California at Berkeley. There at the tender age of 27 he came face-to-face with the worst aspects of ufology. In the fall of 1961, alleged contactee Reinhold Schmidt was on trial for allegedly conning widows out of their cash by various nefarious schemes (involving quartz and cancer and valuable mines and titillations). It was all quite an appalling legal circus. Into this fiasco stepped our intrepid adventurer as a surprise scientific witness to the possibility that Schmidt's contacts from Saturn really might be able to live there. This was not as wide an audience as the Johnny Carson show, but was apparently just as entertaining and unpredictable. Sagan, with an appropriately dry, nutty humor, retells this goofiness in Chapter Two. In many ways it is vintage CSICOP: a wonderfully effective laugh curtain smearing the whole field. This provided an excellent shield for his speculations on ancient astronauts. It is worth the read.

Sagan did not let the issue lie following his retelling of the "Helmut Winckler" trial, though. He went on to fight a few other fights (the problem of saucer cults, the problem of believing in God, and the problem of sincere people believing in UFOs and causing the scientific community problems... since the scientists are always being bugged by and misquoted by the enthusiasts). His discussion of these things tended to counter the negative effect of his trial room yuk at the saucerers. In fact his penultimate paragraph reflects on the ancient astronaut chapter to come and surmises "perhaps we have not been cautious enough [in our speculations], but I rather suspect that the opposite is the case."

The eleven chapters that follow in the section entitled "The Universe" are what all such chapters are in the ETI literature: condensed astronomy books aimed at convincing the readers that there's a whole lot of universe out there, acting exactly like ours. The second eleven-chapter sequence, "Life in the Universe," is somewhat more

remarkable, due in large part to Sagan's frontier knowledge of the formation of amino acids from primordial gases, and related subjects concerning the beginnings of life. Mars is examined in depth and in interesting detail, holding out hopes for simple life and pleading to NASA and the Soviets not to contaminate the surface with unsterilized spacecraft. The Moon is surveyed, with hope held out for fossil microbial life from the early days. Transient lunar phenomena are mentioned but only in reference to active subterranean geology. Mercury and Venus are almost, but not completely, dispensed with as life candidates. The outer planets and their moons are little discussed, but the possibility of "life in the Jupiterian clouds" is hinted.

Carbonaceous chondrites (meteorites containing a surprisingly high percentage of organized carbon compounds) are given some thorough analysis. The controversial Orgueil meteorite, that allegedly contained evidence of spores similar to terrestrial pollen, gets a workout (negative on the ET-pollen hypothesis). Lastly, life in other solar systems is examined. Here, after the astronomy, Sagan takes on exobiology, oscillating between statements that acknowledge certain restrictions placed upon life forms and those that see no reason why all sorts of strange things might occur. As his career progressed, he seems to have tended more toward the latter. It does, after all, make for a fuller, livelier universe.

The final eleven chapters constitute the pioneering "Intelligent Life in the Universe" section. The first big risk is a chapter on the Moons of Mars. Here, mainly through the driving interest of Shklovskii, we are treated to a powerful set of arguments that Phobos is an artificial structure. Sagan apparently agrees. This is getting pretty close to UFOs from Mars, methinks. But, in case you hadn't heard, this hypothesis has been discarded, based as it was on some bad assumptions. However, at the time, such a statement by academics wouldn't have been viewed as conservative, or helpful as far as the U S Air Force (USAF) was concerned. This unhelpfulness may have had something to do with the good doctor being asked to serve on the USAF's ad hoc Scientific Board meeting in 1966 (the "O'Brian Committee"), wherein he had explained to him the problems of the UFO phenomenon via the Robertson Panel Report and Major Quintanilla of Blue Book. Shortly afterward, Sagan wrote an almost wholly negative report on UFOs for the *Encyclopedia Americana* which the Air Force incorporated into its yearly Project Blue Book Reports (1967, 1968).

The ideas of radio and optical contact with extraterrestrial civilizations are cast about in a normal fashion, and there is an analysis of the newly formatted Drake Equation. Sagan's answer: Between 50 thousand and one million current civilizations are out there somewhere. Radio signal format is discussed, as well as the possibility of interstellar probes and direct contact. Then comes the second big risk: Chapter 33, "Possible Consequences of Direct Contact."

This is the chapter that broaches the ideas of M. M. Agrest on ancient astronauts (it seems that Sagan owes both of his big risks to Shklovskii). Some back-of-the-envelope calculations suggest an interstellar-visitation time sequence of perhaps every 100,000 years. One such visit could have occurred, therefore, in relatively recent times. Is there any evidence for it? Agrest, as stated earlier, suggested Biblical

stories like Sodom and Gomorrah, or the central Saharan rock paintings of Tassili. Getting fully in the mood, Sagan offered a more intriguing possibility—the creation and flood myths of ancient Mesopotamia. These stories *are* interesting when read in this light, but I doubt that Sagan wanted himself memorialized as the intellectual grandfather of Erich von Däniken and Zecharia Sitchin. Nevertheless, he spends six whole pages teasing this hypothesis out, and leaving it with only the mildest conservative caveat—a caveat immediately followed with a statement that I'm sure he now wishes he hadn't made.

It seems possible that the Earth has been visited by various Galactic civilizations many times (possibly 10, during geological time). It is not out of the question that artifacts of these visits still exist—although none have been found to date—or even that some kind of base is maintained within the solar system to provide continuity for successive expeditions.

It is a thought worth meditating upon. And it is a thought almost entirely abhorrent to the astronomy establishment that fanatically maintains: They can't get here from there. The young doctor was obviously not meditating enough on the closed-mindedness and irrational fears of academic science when he wrote these words, and even less when he decided to say them at the 1966 meeting of the American Astronautical Society in Washington, D.C. The headlines read, "Scientist feels superspacemen have visited Earth many times." No one but Sagan himself knows what inner-circle price he's had to pay for that public display of honest, childlike zest for discovery, but since that time his public demeanor toward UFOs has been brutally and constantly negative. It is my opinion that he has always remained true to his natural curiosity about this topic, but now must remain strictly in the closet. Others of his close SETI associates are the same. Science's elite should carefully note this shameful quashing of *joie de vivre* and innovative thinking.

Intelligence in the Universe By Roger A. MacGowan and Frederick I. Ordway
Englewood Cliffs, N.J.: Prentice-Hall, 1966. 406 pp.

Almost at the same time as the Shklovskii-Sagan book, the second of the mid-1960's classics of the ETI field appeared. Its authors are identified as working for the Army Missile Command's Computation Center (MacGowan), and London's General Astronautics Research Corporation (Ordway). Frederick Ordway was simultaneously working as scientific consultant to Stanley Kubrick's famous film, *2001: A Space Odyssey*. Ordway's interests range across the usual ETI territory, MacGowan's are more specialized on the issue of what constitutes intelligence.

Their book contains seventeen chapters, with the beginning and end largely Ordway's and the middle MacGowan's. The beginning and end are classic ETI fare: cosmology, biological origins and evolution, SETI within and without the solar sys-

tem The middle is unusual It contains several chapters on intelligence and thinking, and reveals an approach one would expect from a computer expert The reader interested in speculations about robots with humanlike intelligence will find this stimulating, while followers of the Roger Penrose school (such as myself) will be less entranced

The key viewpoints on the probable evolution of extraterrestrial life are in many ways covered better in this book than the previous one, and the bibliography is far better This could be in part because Sagan was hampered by an already existing book, and these authors could start from scratch (Ordway had already surveyed much of the field in a book one year earlier) Sagan is much better at the origin of useful biochemical molecules (his research strength), but in some of the speculative areas, like whether the moons of Mars are artificial, he is perhaps socially limited by the fact that it is Shklovskii's interest, and so MacGowan and Ordway do a much better job Surprisingly, the Russian mystery of Tunguska plays no role in Shklovskii and Sagan but is discussed thoroughly here

Sagan's ancient astronaut concerns are briefly mentioned by MacGowan and Ordway to which they add the mystery of the vitrified forts (they say "round towers") of the British Isles and the Salzburg cube From their descriptions they obviously know little about either, unfortunately Regarding our favorite topic, they rely entirely on two resources Menzel and Boyd's *The World of Flying Saucers* ("without question the most authoritative to appear on the subject") and an official statement from the USAF, which is quoted at length Naturally given this background, UFO research does not fare well A second UFO work is referenced in the bibliography Jacques Vallee's *Anatomy of a Phenomenon* Apparently, they paid no attention to it whatever, as it certainly should have moderated their views Vallee's book was originally in the Air Force's list of further readings on UFOs, but was dropped fairly quickly from the Blue Book Project Reports (another stellar example of truth-seeking objectivity)

Even though our subject is not treated with any scholarship or professionalism, most topics in the rest of the book are It is generally a good read, and its strengths and coverage blend well with the Sagan-Shklovskii text This is why both volumes are recommended to cover this middle ground between Jones's icebreaker and the books of the 1980s and today

Worlds Beyond By Ian Ridpath New York Harper and Row, 1976 176 pp

Ian Ridpath is a clever science journalist with a specialty in outer space and astronomy The job of this oversized paperback is to boil down the heady intellectual broth of the professors to something slightly more concrete and readable (although none of the books reviewed here are particularly complex or technical) Ridpath succeeds in producing a nice introduction to the ideas surrounding SETI, and, since this field moves slowly if at all, it is worth owning and perusing

The book contains nine chapters, generally covering the usual ETI fare Cosmol-

ogy, as usual, gets Chapter One, followed by three chapters on life and human development (although in eccentric order). These chapters are pretty straightforward, with a little imaginative flying taking place with the ideas of Freeman Dyson in Chapter Three.

The origin of life in our solar system encompasses the fifth and sixth chapters. Again, an intelligent overview of relatively standard stuff, including the ideas of Su-Shu Huang and Stephen Dole on planets and habitability, and Carl Sagan's Jovian atmospheric life speculation. Establishment SETI dominates Chapter Seven, largely based on the Soviet-sponsored Byurakan Conference and NASA's dream, Project Cyclops. A hint of concern that we should *not* broadcast our presence is broached here.

Chapter Eight is a little refreshing for most SETI books in that it takes seriously the notion that humanity may one day build starships to travel interstellar space. Theory, such as relativistic time compression, is briefly discussed, but also such nuts-and-bolts topics as Bussard ramjets, Daedalus starprobes, and the geography of nearby space as a set of stellar steppingstones. Most SETIian books refuse to consider human starflight for two very sound strategic reasons. It opens the doors for visitation in the other direction (for example, UFOs), and it allows the argument that SETIian money might be better spent looking for visitors than far distant telephone pals.

The final chapter, "Probes from the Skies," broaches the inevitable. Could they get here, would they get here, are they here? Ridpath covers theories and Bracewell probes, the Duncan Lunan "starprobe" from Epsilon Bootes, and finally to UFOs. Once there, we are again eased in strategically via the AAAS 1969 Symposium (mostly negative, but the positive elements are given fair stage time here), and then on to Allen Hynek and Peter Sturrock and the AIAA. Ridpath signs off respectfully but conservatively, leaving the reader with a surprising and welcome feeling of openness to the mystery. In this, Ian Ridpath is a *rara avis*. Perhaps it is because he has no tenure committee to waltz around.

The Search for Life in the Universe By Donald Goldsmith and Tobias Owen. Menlo Park, Calif.: Benjamin/Cummings, 1980. 436 pp.

The authors of this addition to the SETI literature are both astronomers and associated (in different ways) with the State University of New York at Buffalo. Donald Goldsmith is a media-oriented educator and a writer of textbooks. Tobias Owen is a professor-researcher with wide interests on the frontiers of the planetary sciences. Their collaboration has produced this excellent volume, which is, unusually enough, a textbook rather than an expository monograph like most of the rest.

The textbook has twenty-two chapters broken into five recognizable sections: an Introduction ("Why Do We Search?"), a section on cosmic development ("The Universe at Large"), a section on biological origins and development ("Life"), one on our local space ("The Search for Life in the Solar System"), and—what we really

want to know—a look at deep space (“The Search for Extraterrestrial Intelligence”) The chapter counts of these sections (1, 6, 4, 5, 6) show the astronomical (vs biological) emphasis of the authors The volume has excellent references, illustrations, and index

Goldsmith and Owen succeed above all other ETI writers in their discussions of esoteric but important elements in our thinking about extraterrestrial life Some examples of this cogency of fact and insight include the vast superiority of liquid water as a life solvent, how worlds completely covered with water might promote evolution, the inevitability of intelligence, the importance of the Gaia principle in maintaining environmental balance, the indefensibility of silicon-polymer life, and the (appropriately in my opinion) liberal interpretation of sizes of habitable zones around stars Although all of these treatments are brief, they stand out from other works in that they are there at all, and they are well reasoned and explained Many of these viewpoints are refreshingly liberal in their stance, something often seen in this literature but in a most piecemeal, selective way SETIans are by nature adventurers and risktakers, but I was still unprepared for the authors’ statement that the coloration of the Saturnian moon, Iapetus, might be considered an artificial effect left as an alien signpost! To this exploratory romanticism, I say, hurrah and congratulations You could almost hear a little of the academic glacier melting and the sun come out

Chapter Eighteen covers one of the subjects near and dear to me the Drake Equation and the estimation of the current number of galactic civilizations It is a bit amazing and amusing how many varieties of the Drake Equation exist Every SETIian seems to like their own variety Goldsmith and Owen are no exceptions and they produce a Drake that is a modified, Saganesque version (This equation starts with the total number of stars in the galaxy rather than the rate of star formation The reason for this is showmanship and public relations Beginning with the number of stars allows one to say the word “billions” regularly, and we all know how much Sagan enjoys doing that) Someone not well-versed in the Drake Equation game might be astonished at how routinely all these different equations come to the same conclusion—that the number of technical civilizations in the galaxy today is approximately equal to the number of years one would expect them on average to survive However, it is my suspicion that it is more a question of all of us reading the same literature and having the same optimistic turn of mind (mildly suppressed by academic conservatism) Everybody then rigs their estimations (with appropriate areas of conservatism—not always in the same places) to come out with about the same result Goldsmith and Owen add something special to the discussion, however They bracket their best guess with reasonable most and least favorable extremes This is interesting in that it shows that the most favorable extreme is only slightly higher than the best guess, whereas the least favorable extreme is depressingly, vastly lower However, if one leans toward a vision of the universe that involves anthropic principles, this (the best guess being close to the most favorable extreme) is exactly what one would expect The authors leave us with a feeling of millions of civilizations in our galaxy, and millions of billions in the universe, and

visions of O'Neill cylinders, Dyson spheres, and Kardashev civilizations dancing in our imaginations

In Chapter Twenty-one the ufologists finally come to our world. The treatment is, as always, a disappointment, but not as much as we are accustomed to enduring. This is despite the statement that the authors are deeply indebted to Philip Klass, described as "one of the most thorough [shudder] investigators of UFO sightings." Despite this, Goldsmith and Owen present a longer, more detailed and serious treatment of the subject than one is likely to find elsewhere in this literature. The discussion begins with a listing of four categories of possible evidence (an actual extraterrestrial, an actual spaceship or some significant elements of its technology, a detailed, instrumented record of a UFO event (pictures, traces), a set of reports from persons claiming to witness extraordinary phenomena and entities). The authors immediately discount categories one and two. (However, they note in passing that a few *individuals* say they have been abducted. The italics apparently are a subtle guard against having to take Betty and Barney Hill or any other multiple-witness event seriously.) Artifactual claims are relegated to charlatanry. Photographic evidence is rapidly dismissed, although the pictures of the McMinnville object are displayed, nearly without comment. This leaves them with only the academically safer category of witness reports to deal with.

Here the authors spend a more gratifying amount of time discussing some actual cases, although with great conservatism. The Chiles-Whitted case is recounted (with Kenneth Arnold and Charles Mantell mentioned as an introduction). Later sightings in 1968 are utilized to explain the event as a meteor. But the authors admit that they can never be sure. It is this unusual occasional candor which sets this book apart. The Lubbock Lights are discussed less satisfactorily as a much too casual agreement that they were reflections off birds. (This naiveté was probably occasioned by not reading the extent of the primary data. Who knows? Maybe they *were* birds, but one can't *race* to this conclusion given the number of observers and the photographs.) The planet Venus is trotted out to make its obligatory claim to vast numbers of reports (in this case, Georgia in the fall of 1967). Lastly, Socorro is described. Admiringly, the authors admit that this is an intriguing but unproven case. I believe, my friends, that this is the best we can ever expect in a standard SETI work.

The chapter continues, however, with more interesting discussion. In a footnote, Father Gill is dismissed as influencing the faithful to say what he believed about the Boianai case (a howlingly unsupported interpretation, but effective in disregarding a powerful, credible case). J. Allen Hynek's classification scheme is described, initially respectfully and responsibly, but then gigglingly smeared by a wisecrack by some smirking elitist. This light touch was cleverly effective, vintage CSICOP. It successfully relegates all CE4 witnesses to the realm of rather disturbed people, and thereby beneath scientific dignity. In discussing Hynek's strangeness criterion, the authors reach the conclusion they wish. Strangeness is inversely related to credibility. However, Hynek and others who actually research these things have often found exactly the opposite relationship.

Another interesting section follows arguments for (and against) the spacecraft hypothesis. Here one finds a reasonable, but extremely restricted and imagination-poor, discussion of why the UFO-ETI connection doesn't make sense. Included are impoverished speculations about alien interests, motives, behaviors, and erroneous facts about who sees UFOs and who doesn't. We will all be enlightened to learn that nobody in China sees them, for instance (despite reports to the contrary). It is another remarkable instance of the human ratchet-mind at work. "I talked to one group of Chinese scientists; they heard of no reports in China, that is what I want to hear, so that is true." My mental vision of preferred reality ratchets forward another notch toward self-manufactured certitude. Yes, I know—we *all* do this, including myself, Let's just all admit it and try to do better in the future.

Despite all this predictable dodging of UFO bullets, the authors *still* are worth reading, *still* to be commended for at least taking *some* effort to include all that they do. It is a giant step beyond the other works. They end their chapter blasting Erich von Daniken's nonsense, lies, and errors, and this too helps smear over into modern UFO research. But it is not Goldsmith's and Owen's fault in this case. Von Daniken is there, and he and others like him *do* smear us and make our work more difficult.

Two parenthetical observations: all UFO organizations and administrations should take more note of this guilt by association when they cavalierly allow dubiously speculative individuals to occupy time on their conference programs and take up pages in their journals, and then wonder why people might want to question some of the other things they say, believe, and support. Goldsmith and Owen are perfectly justified in approaching everything in ufology with extreme wariness when there are von Danikens about (and apparently welcomed). The second observation is more subtle and, perhaps, amusing. In their lambasting of von Daniken they feature things like Great Pyramid number symbolism, the Palenque astronaut, and the Nazca airfield—all appropriate areas of such horribly bad reasoning and "scholarship." They footnote a brief mention of Sumerian legends, the area where, as we have seen, Carl Sagan found reason to entertain a possible ET-effect on our cultures. It was most diplomatic of them not to bash this possibility, coming as it did from one of their own tribe. However, it would not have done to discuss it favorably either, given the goals of debunking—so, a vague footnote. Ah yes, it was neatly done, as Sidney Greenstreet would say.

The very last part of the chapter in some ways is the best and the most surprising: the list of books for further reading. I expected to see a couple of Klass's books, and maybe Menzel or Sheaffer—whatever was available from the skeptical school. One Klass volume, Martin Gardner's *Fads and Fallacies*, the mostly, but not completely, skeptical AAAS symposium, edited by Sagan and Page, and the January 1978 *Playboy* (a balanced pro/con discussion) are all listed. Hynek's *UFO Experience* and Vallee's *Anatomy of a Phenomenon* hold down the sympathetic UFO end. I was astonished at its balance and almost broke into applause. (Too bad that the tenor of the chapter itself didn't reflect it better, but this is about as far as academic conservatism and the SETIans can go—at least in public.) Long-time UFO researchers

won't be satisfied with the treatment of the field by a long shot, but given the biases and pressures and barriers to learning this material in depth, Goldsmith and Owen should be given some appreciation for their try. This, for many reasons, is the best book in this field.

The Search for Extraterrestrial Intelligence: Listening for Life in the Cosmos By Thomas R. McDonough. New York: John Wiley & Sons, 1987. 244 pp.

Thomas McDonough is a Ph.D. astrophysicist and lecturer. Here he has written a very readable volume, apparently intended more for the general public than the academic audience. The volume is composed of sixteen chapters, arranged in a somewhat eccentric topical fashion and of a usually nontechnical tone. There are adequate illustrations, amply salted with cartoons. There is a suggested readings list, with notes on the technicalness of each book, and an index. It is polished and professional, and the academic community's equivalent offering to the work by Ian Ridpath (as to level of difficulty).

The sixteen chapters only roughly follow the usual pattern of the field. An introduction is followed by extensive history and science fiction chapters. McDonough is apparently a science fiction fan, and it dominates the second largest chapter devoted to a single topic ("Search in the Solar System" is the largest). Cosmology and the origins of life are crammed into one medium-sized chapter. After asking the question whether other planetary systems exist, the book launches into seven chapters of SETI-oriented bits and pieces (everything from our own casual TV transmissions to politics). The volume ends with space travel, UFOs, ETI pessimists, and a look to the future. Chapters vary in length from six pages to twenty-eight. The UFO chapter rates only eight pages.

The text is not without interest and things to comment upon. As we search the solar system speculating on the possibilities of odd (noncarbon-based, nonwater-based) life, we learn that McDonough is open to all sorts of imaginative possibilities. Most oddly, he presents the phenomenon of quantum tunneling as a means whereby nature might violate our common sense and common laws, and produce cold evolution in impossible places. Since such views are a lot more speculative (even more bizarre) than the concept of extraterrestrial visitations to our solar system, one would expect a reasonably open-minded treatment of ufology later. One is disappointed in the latter, of course, leading us once more to cope with the question of why.

The pulsar phenomenon is treated historically and at some length. It apparently really interested the author, and he has striven mightily to imagine means to salvage the LGM hypothesis (that these pulsars are intelligent signals and not just natural processes). This also interested me, as I had thought that the LGM had been almost immediately discarded in favor of spinning, neutron-star lighthouse effects (LGM—Little Green Men—was the radioastronomers' nickname for the ETI hypothesis for the pulses).

Spacecraft for possible interstellar missions get some serious play. Writing in 1986–1987, McDonough had the advantage over the earlier authors of listening to the solarsail ideas of Robert Forward, which he includes. Forward's microwave-powered light-sails come in relatively small and simple robot probes (Starwisps) or large and complex human-carrying starships. His imaginative scenarios lessen the "it's just impossible" view of interstellar travel (especially unmanned probes). All this, of course, falls within the bounds of current science, though not current technology. By adding nanotechnology and very clever replication technology to the solarsail concept, a civilization might Starwisp itself in the form of information to other star systems and replicate all or part of it there (UFOs? research probes?) All such talk, of course, puts even more pressure on the "they can't get here" views of traditional astronomers.

One passing note before discussing the points of greatest UFO/ETH interest. In the section on origins of life, there is a frequently seen category of statement. Francis Crick is cited with reference to the concept of panspermia (that life originated elsewhere and has been deliberately seeded across space in the form of viruses or spores). The statement of interest is, "Crick is not easy to dismiss" (p. 68). Following this is the citation of his Nobel Prize for the structure of DNA. In this little pregnant moment one glimpses the root of many of the problems with current scientific thinking. Having no desire to denigrate Professor Crick in any way (he didn't make the dismissal statement, after all), I will make it generic. Dr. Humplefeather has just been elected to the National Academy of Sciences due to his work in nuclear physics. He has now decided to write a paper on abortion. Well, read it if you wish, but I would suggest that he is terribly easy to dismiss in that arena. Let's push it further. He decides better and changes his topic to UFOs or SETI or parapsychology. With no evidence to suggest his depth of familiarity with the data and literature, he is terribly easy to dismiss. Now push it a further step. He gets really conscientious, admits that he doesn't know what he's talking about, and changes his topic to nuclear physics. Well, better. What is he writing about, precisely? Theoretical or experimental physics? High energy or low? Fusion, fission, electromagnetism? Did what he picked match his knowledge? The point I hope is obvious. Crick's brilliance in the molecular biology of DNA may not make him any sort of authority on panspermia, the survival of biological spores in deep-space conditions, or even how rapidly one would expect the genesis of life forms on a new, sterile planet. He is simply out of his field. He is brilliant, yes, but he cannot claim to know what he's talking about just because he is brilliant. Knowledge and intelligence are two entirely different things.

I wouldn't have bothered with this foray into intellectual street wisdom if there was no need to ram the point home. No one is hard to dismiss simply because he has credentials, honors, or proven intelligence if he isn't smart about what he is writing or talking about. That goes for nearly every writer in the SETI literature when they begin to talk about UFOs. Not one shows any in-depth familiarity with UFO research or literature, yet they feel empowered somehow to talk about it. Carl Sagan,

who probably knows more about it than others of his tribe (Frank Drake is a possible exception), rarely writes about UFOs in the SETI literature, but even he, when doing so, displays a very off-center selectivity in what he allows to come into his consciousness and affect his views. No one is hard to dismiss simply because he is a sharp guy with Dr in front of his name. McDonough, however, has paid homage to this mythology of the all-brilliant tribal elder in his comment about Crick. Although an understandable foible of our scientific civilization, we need to recognize it for what it is. Unfortunately, this same myth has cost ufology dearly in the past. The 1953 Robertson Panel employed the myth to allow five ignorant but brilliant men to dismiss the UFO phenomenon in two days of casual deliberations.

The seventh McDonough chapter discusses SETI and the Drake Equation civilization-estimation game. McDonough uses Carl Sagan's form of the Drake (and in many ways the text seems a loyal follower of the Sagan school of thought on all this stuff). The statements on numbers of stars and fractions of stars with planets are brief and nearly contentless (as to any reasoning process). The factor for number of life-suited planets per system is conservative and dumps the romanticisms of quantum-tunneling and solarsailing. How often life actually develops gets a nearly thoughtless estimate of one out of ten, even for earthlike environments. McDonough is more aggressive and enthusiastic about life evolving to intelligence, however, but in his estimate caves in finally to extreme conservatism with a dizzying attitudinal change (only one in a hundred ecologies ever produce an intelligence). Going from intelligence to high technology produces a rather odd statement about civilizations "going different directions" (being great at crafts, or biology, but no mechanics difficult to envision), and this seems to be the basis for a one-in-ten estimate. He is now down to four million galactic locations where a hi-tech civilization might ever arise. His last factor (the "civilization-killer" the fraction of lifespan divided by total lifespan of the star or planet involved) occasions, as always, a short essay on optimism and pessimism. He adds speculation on vegetating civilizations filled with mind-buzzed pleasure addicts, and dead civilizations animated by their still functioning robots, all in good science-fiction style. He sides with optimism here, but an odd optimism apparently based on dead people and self-repairing machines. He imagines perhaps 4,000 current civilizations waiting now, out there.

So now we are down to the treatment of UFOs in Chapter Fourteen. He starts out brilliantly in academic safety with a somewhat sleazy quote about tabloids from J. Allen Hynek (which probably has him turning in his grave—talk about balanced selectivity!), and a Gary Larson cartoon. That plus a good debunking opening story by Philip Klass ("The Sherlock Holmes of UFO's" good grief!) has placed us squarely where any card-carrying SETIian wants to be: in laugh, giggle, and smirk-land. The following paragraph gives one Thomas McDonough's considered viewpoint on the subject:

To most SETI scientists, UFO's are a can of worms into which they do not want to stick their hands. So filled with fraud, misidentification

and pure lunacy is the UFO business, and so time-consuming is a proper investigation, that most scientists avoid it. SETI researchers already have the difficulty of often being confused in the public mind with the nuts who think we are being visited by little green men from Venus.

Now, my friends, how is that for well-balanced, intellectually honest commentary?

But there seems to be more to McDonough than his negativity suggests. In fact one suspects that privately there is some interest and sympathy here. I say this because, once he has effectively chuckled and red-flagged himself into an unassailably safe position, the text begins to moderate considerably. It is still very careful, but we become aware that the author knows some of the history of the field (Arnold, RAND), that it has a worldwide aspect, and that many astronomers consider it worthy of study (Sturrock's survey). The major moderation comes in the form of several pages of description of an interview (from 1984) with J. Allen Hynek. This is extremely good for us (ufology) and safe for McDonough (it is, after all, only done in the interests of a balanced viewpoint, and Allen can be blamed for anything that's said that the establishment doesn't like).

I would like to present some of this material verbatim as it is vintage Hynek and speaks poignantly to the status of our field as it interfaces with the academic establishment.

I don't talk about UFO's very much anymore. I talk about the UFO *phenomenon*. The phenomenon is the continual flow of reports, now from 140 countries—and our databank now has over 100,000 entries—and the patterns in those reports.

We're especially interested in the cases where there are independent witnesses when there's no possibility of collusion—witnesses who don't even *know* each other, yet they're reporting essentially the same thing. Now, *that's* the phenomenon.

I haven't the slightest idea what's behind it. It may be some aspect of our own psyche and intelligence. We might have to look for some possible interface between our physical dimension and another.

The public in general thinks that Hollywood has solved the problem. UFO's are spacecraft piloted by E.T. Fine, let's go and play bridge now or something else. The problem's solved. Well, it's *not* solved.

All it takes is one idiot to say he's gone to Venus in a flying saucer and have that published in the *National Enquirer* [or in one of our UFO journals], and it sets back serious work on UFO's tremendously. [One might consider our current casually "accepted" and published claims

about direct psychic and instrumental signalling and “calling down” of UFOs (as well as many other highly dubious enthusiasms) in the same light]

In science, there should always be room for honest disagreement. I would much rather talk with an intelligent skeptic than with a dyed-in-the-wool believer who will accept anything and buy the Brooklyn Bridge tomorrow. Pseudoscience is a very real danger in our society, no doubt about it. But we should look at the possibilities. The playing field of science should have some bleachers, where we can have some of these things on the fringes, not to be ridiculed, but to be looked at. To be looked at.

McDonough follows this presentation of Allen’s wisdom by his own personal statement. He begins with a sympathetic positive statement about science’s need to look at anomalies and translates that into an objective attitude toward UFOs. He acknowledges that we might “find some real phenomenon, not necessarily beings from another world.” But then he restrains himself and warns us of the lurking foolishness with an amusing advertising-plane anecdote courtesy of Klass. The warning is, of course, appropriate, but the inclusion of the story is defensive. The lengthy Hynek interview almost saves this book’s treatment of ufology from the appalling ignorance, bias, and superficiality of most other works in the SETI field, but the intensity of the negative front end of the chapter is too much to warrant the author much applause. This withholding of praise is further supported by one of the worst-balanced suggested-reading lists I’ve seen: Condon (without Saunders and Harkins), Gardner, two Klasses, Kusche, Oberg, Sheaffer, Story’s anti-von Daniken book, and *The Hynek UFO Report* (not even *The UFO Experience*). Brief editorial annotations to the bibliography project an intensely negative atmosphere.

The SETI field has an interesting literature that is important to the ufologist in many dimensions. Intelligent people author it, but they are largely ignorant of our business. They speak of UFOs and ancient astronauts apparently because they feel that they are forced to, but occasionally one catches glimpses of genuine interest. SETI is almost as multidisciplinary as ufology, and yet the authors bravely take on the many disciplines beyond their training to produce their texts. They seem well-read on all their fields of comment—except UFOs. It is another illuminating moment when one realizes that these authors feel the requirement to know what they’re talking about on every subject in their books, except UFOs. UFOs are not important enough to *really* bother knowing anything about. One reading of Klass is enough. UFOs have no standing, no one to be held accountable to. There is no UFO professional community and no active body of academic authors whose texts must be referred to and whose flow of critical journal articles and symposia talks need be feared if one writes in superficiality and error. The picture one takes from the SETI literature (and elsewhere) is that the academics who allege interest in the UFO field

have failed. They have not written enough books of academic quality that are academically accessible. They have not cooperated and formed a professional community. They do not take the time to *really* contribute with any consistency or at all to the continuance of the research and analysis process. As editor of this journal, I can attest to how few of the seeming myriad of Ph D consultants of alleged ufological interest ever actually *do* anything to further the acceptance of the field. Until this changes one can hardly blame the SETIans and their kin for their cavalier attitude to our academically impoverished field.

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CROP CIRCLES

Report on the Results of Project Argus: an Instrumented Study of the Physical Materials of Crop Circles. Edited by Michael Chorost. Durham, N C: Project Argus, 1993. 115 pp.

The major failure in crop circle literature has been a lack of properly presented scientific data. This failure is puzzling given the number of claims made and the readily available physical materials to test. Now, thanks largely to the energetic efforts of Michael Chorost, we have a report that might fill that gap.

Discussion of the report should begin with a statement of the nature of Project Argus. In the words of the report:

Project Argus was a joint effort initiated mainly by two organizations, the North American Circle (NAC) and the Centre for Crop Circle Studies (CCCS). The CCCS was established in 1990 in England with the aim of encouraging orderly scientific investigation. The NAC was established a year later in the United States, with the same goal. By the end of 1991, it was becoming clear that the two organizations had, between them, accumulated enough scientific expertise to design and carry out a joint research project. The CCCS gave the project \$2,000 as seed money and committed itself to logistical support, and a fundraising drive by the NAC brought the total funding to over \$34,000 by May 1992. Logistical needs such as housing and ground transportation were quickly arranged, and the project formally began its field work in England on July 9, 1992.

Observers of the crop circle scene got the impression that before Project Argus's

work, research in the area consisted largely of waiting for the report of a new artistic formation (and in a few nightwatch scenarios, fruitlessly staying up all evening) and then rushing to the scene to photograph it. This is more appropriate for art history or tourism than science, and veteran American ufologists were rolling their eyes and tearing their hair out, wondering where the scientific method was. Rumors existed of strange sounds, radiation effects, and botanical anomalies, but the level of vagueness was at least irritating if not appalling. Now most of these failings have been rectified. This is not to say that the current report has *proven* much, but its attempt to bring discipline to the subject deserves a round of applause.

The backbone of the report is the 32-page results section (coupled with some 40 pages of relevant appendices). The section is divided into eight minichapters, all but one presented in standard scientific language, making the reading of methodology and results easy to assess. Unfortunately, the sole exception to this contains a description of the work of Michigan biophysicist W. C. Levengood (not written by himself). This is unfortunate because this work, describing possible anomalies in affected plants, contains the most interesting results.

There are two characteristics that have become the siege guns warding off simplistic skepticism about crop circles. One of these is the elaborate interweaving of plant stems that veteran observers say cannot be duplicated by hoaxers. Well, since "hoaxers can't do it" arguments aren't very convincing (we've had 45 years of that in ufology), the second siege gun becomes more significant: the structural and cellular effects on the plants that are precisely Levengood's bailiwick.

The interested scientist wants to know: What exactly *are* the anomalies in the structure of the plants, especially the interweaving? Can these effects be produced by simple mechanical stress (crushing or wind effects)? Do we have any way to distinguish the effects of such forces? Do we have any valid, controlled studies? Simulations? Do the effects require heating? Electric fields? Microwaves? Have force simulations been conducted on any of the plant species affected for the correct times and seasons?

The Project Argus report does try to answer some of these questions. Cassandra McDonough's chapter on scanning electron microscopy is a worthwhile first step in unraveling the mystery, although much more sampling and comparative work needs to be done to determine how anomalous the observed features really are. I found the conclusion that "these samples revealed there may be some actual physical force that can create crop circles," either redundantly obvious (if read literally) or unsupported (if some unexpected extraordinary external agency was implied). However, the work is useful and apparently well done.

The same cannot be said for the chapter entitled "Electrophoresis (DNA Analysis)." Many anomalies occur here, but they are anomalies of scientific reporting. The first is that the laboratory at which the tests were done is never identified. The sampling seems crude (inside vs. outside—what about specific cells right at the bends?). The initial treatment was, by admission, crude (sun-drying) and impossible to standardize.

In fact, one worries about “grab-and-stuff” bag sampling all along in this research. Samples arrive in the United States in different states of moisture content (and who knows what else). The normal incubation period for biochemical extraction prior to electrophoresis was quadrupled for unspecified reasons. Consequently, excessively extracted pigments void the possibility of spectrophotometry (although I don’t understand this—all that should have been necessary was a second type of separation, or a simple dilution, depending on what moieties one wished to measure). The results are confusingly worded and poorly illustrated, and give no convincing evidence of any differences.

The most intriguing of the minichapters is the Levensgood chapter, written by Montague Keen. Keen leads us through the saga of these plant studies in a narrative format, alternating between quasiscientific commentary and various complaints about difficulties blocking the way. The near-results that keep winking in and out of his narrative are fascinating but completely unsatisfying to the scientific reader. We *do* learn that Levensgood has issued informal, unpublished reports to an apparently small group of crop-circle insiders since around the beginning of 1992. I received one of these (Lab Report 418) from Levensgood and his associate John Burke, and it is a significant improvement (in scientific orderliness and detail) over surrogate descriptions of his work. I can only wish that Levensgood and Burke find time to produce formal reports that can be more widely read and assessed. However, the unorthodox nature of this method of publication certainly does not help the advancement of science. I have no idea why such an informal dissemination is taking place, but I would not be surprised to find that mistreatment of honest explorers of anomalies by academia and the media plays a role. If that has anything to do with your informal publications, Dr. Levensgood, I understand completely, and all the serious members of ufology’s invisible college empathize with you. Whatever the stimulus, I am delighted that Levensgood and Burke immediately decided to send me their reports. In any case, we hope for a better world and a better way of distributing this information.

The other minichapters have to do with gamma-ray spectroscopy, magnetic surveys, electromagnetic field detection, and photographic surveillance. Nearly all of this is negative, but this is not to say it is worthless. Negative results tell us something and should always be reported. It is worth knowing that no lingering radiation exists and that radioactive materials are not involved. It is worth knowing that a photographer stayed up all of one night, saw and photographed nothing, yet a circle appeared in the morning and it was one with all the earmarks of a hoax.

The one intriguing possibility in these minichapters comes from magnetic mapping of the crop circle formations. Although not at all consistent, some of the structures seem to show magnetic anomalies. One wonders why the structures weren’t mapped at different times (say a month apart) with control areas. Much more work should be done here and there is no excuse not to do it (anyone with a magnetometer and the landowner’s permission could make a long-term study).

The rest of the report consists of some attempts at correlations and comparisons

(England and Canada), and Ralph Noyes's comments on luminosities associated with circles. In the notes and appendix, we learn what ufologists everywhere have discovered—that there is little or no reason to suspect that these beautiful artforms have anything to do with UFOs. Noyes and several prominent British ufologists agree.

However, there may be the occasional glowing ball of light that would interest the general anomalist. It has been remarkable to see the enthusiastic response of audiences to blurry videotapes of a white-colored, bird-sized object moving like an owl around the edges of the standing wheat of a circle after dusk. Is it the truth of the answer or the feel of the answer we cherish?

This review has been rough on this report at times, but that is the business of any objective reviewer. My overall feeling about this report is "Thank goodness for Michael Chorost and all the associated people who got this material out!"

This is a good thing—the first best step in turning the crop circle carnival into a real investigation. No one wants the fun to be over, but there comes a time when it would be nice to learn something for a change. Michael Chorost has opened the door. Now it's the research community's job to step up and square the facts of these circles for the serious ones among us at last. Lest we forget, some of the *best* fun comes with the discovery that something is *really* real after all.

The Crop Circle Enigma Edited by Ralph Noyes. Bath: Gateway Books, 1990. 192 pp.

If one wished to buy a single book as an introduction to the dimensions of this subject in a reasonably competent way, this would be it. In fact, it's almost the only option. It is fortunate, therefore, that it's well done. There is little objectivity in crop circle writing, and the individual chapters and authors in this volume are rarely an exception. *But*, because Ralph Noyes has collected such a splendid spectrum of varying believers, reading through the book paradoxically produces some objectivity (if one is of a mind to want it). There are thirteen chapters (fourteen if you count all of Ralph's small bits), written by eleven different authors. They are grouped into "Folktales and Fantasies," "Physics and Forces," "Meanings and Metaphysics," and "The Enigma." Most of the specifics of the chapters are extremely believe-it-or-not, and the science is woeful. Even Terence Meaden's material is surprisingly spotty in this regard. (We find that he believes in the efficacy of dowsing and that the forces that form the circles also explain Blessed Virgin Mary apparitions. As a Roman Catholic, my mind quickly raced through the incidents of Fatima, Lourdes, Guadalupe, and Medjugorje, and found very little comparison with any of this.)

The price of the book is practically made just by the interesting reading in Robert Rickard's chapter on fairies, folklore, and neat old stories. Meaden does present his own meteorological ideas, and this is a rare place to see them in print outside his own books and journal. John Michell presents his usual charming and uncritical telling of spirituality in nature and energies, and others chime in with all sorts of weird stuff (dowsing, channeling, kundalini, mantras and mandalas). If this doesn't

seem overly useful, I beg to differ. It gives one (isolated from the action as most of us are) the flavor of the multitheoretical wildness within which this seemingly beautiful but simple phenomenon is occurring. Remembering that these ideas are rocking around in the minds of the investigators and book and journal writers of this subject, may give one a perspective on the question of how much cool objectivity we are getting on this subject. And if you're interested, you will learn essentially nothing about UFOs in this volume. But it does seem to be a grand old time.

Crop Circles: A Mystery Solved By Jenny Randles and Paul Fuller. London: Robert Hale, 1990. 250 pp.

This is a book suffering from multiple personality disorder. It can't make up its mind whether it wants to report on the crop circles phenomenon or go to war against the extraterrestrial and most other forms of hypothesis in ufology. For many reasons, I wish that it had contented itself with the former and forgotten the latter. Building on the 1989 BUFORA publication, *Controversy of the Circles*, the crop circle and Meaden vortex portions of the book are quite good. This is, in fact, perhaps the best place to read the Meaden vortex theory in an understandable way. It is also perhaps the best researched book of any on the market. All of this is expected of two of Britain's accomplished UFO researchers and writers.

Amazingly, the authors become so enamored of the idea of glowing, swirling, low-energy plasma vortices that, despite the fact that these are yet unknown in nature, they decide that the majority of UFO cases are now explained by them. Déjà vu, Donald Menzel, Déjà vu, Phil Klass. The mind reels as one is treated to explanations of cases like the Trinitade Island photos, the Coyne helicopter encounter, and the Levelland, Texas, vehicle interference cases as (short-term? low-energy? unknown?) Meaden vortices. You walk away from this material wondering if it's a case of theoretical boredom setting in, rather than anything serious. Perhaps this thinking has been moderated three years later, but in this book it flaws a work badly needed in the crop circles field. By drawing attention toward UFOs, it lost a chance of forcing some solid research and scholarship on an area that obviously needs it.

Interestingly, it remained for the UFO research community in the United States, allied with Michael Chorost, to bring an order that Randles and Fuller almost did, but who decided to fight a battle on a different front.

1991: Scientific Evidence for the Crop Circle Phenomenon By Montague Keen. Norwich: Elvery Dowers, 1992. 47 pp.

This little monograph is worth having, especially if you don't have Chorost's report on Project Argus or Levengood's private research bulletins. This is because it is within these few pages that one can read of the botanical testing of the Pinelandia Biophysical Laboratory (which is interestingly anomalous) as well as the radionuclide work of Marshall Dudley (which may no longer be). Keen is a farmer

and an editor, who supports the multi-viewpoint CCCS research group in Britain and rejects Terence Meaden's CERES investigation rather vociferously. Keen's knowledge of Levengood's work made him the obvious choice to write the Levengood chapter in Chorost's report, so one can read its predecessor here. Professor Archie Roy, Glasgow University physicist and longtime observer of the anomalies scene, writes the preface.

Crop Circle Classification By Pat Palgrave-Moore. Norwich: Elvery Dowers, 1991. 47 pp.

This is another small monograph making a first step toward pattern-finding, and as such is worth reading. However, this is only true because of the woeful lack of better attempts in other books. The author's classification scheme is the simplest form of lumping raw shapes together with no assessment of the quality of the data (possibility of a hoax, reliability of sources, subtle information regarding swirl types, vegetation types, etc.). Where is the U.K. equivalent of Ted Phillips when we need him the most?

Circular Evidence By Pat Delgado and Colin Andrews. London: Bloomsbury, 1989. 190 pp.

Crop Circles: The Latest Evidence By Pat Delgado and Colin Andrews. London: Bloomsbury, 1990. 80 pp.

These two volumes are what one would call coffee-table books—filled with beautiful pictures and almost totally devoid of scientific or scholarly content. The 1990 book is precisely that: no index, no references, no scholarship, no science. It appears to be a vehicle for selling the beautiful pictures to that part of the public that wants nothing else. Well, that's fine, it's what makes capitalism great. The 1989 book is a little better on the surface (no index, some brief and poor references—one to the Billy Meier case, no science of any real sort—simple descriptions and vague measurement technique comments, and no scholarship). A chapter entitled "The Evidence" is just pictures. Either this use of evidence is meant to mean that merely looking at the pictures is all the evidence that the authors need (for whatever deductions), or that's all they feel that the readership cares about. Either way, it's a bit appalling. Having met one of the authors, and finding him to be a fine fellow, I would be slow to criticize such obviously made-for-public-sale books if it weren't for the liberal use of the word "evidence" in the titles and elsewhere. This isn't evidence. It's just pretty pictures attached to some vague caveat emptor claims, none of which are in any formal way presented or documented. Evidence can't be treated this lightly. Evidence comes from more than wonder-filled exploration and back-of-the-neck thrills. Purveying loose stories, carefully selected anecdotal claims, and undocumented facts as evidence is exactly what we don't need in the study of anomalies. Every critical thinker immediately recognizes the difference, and previously

uncommitted serious researchers take their interests elsewhere. Once again, as in most of ufology, we find little patience, little unromantic gruntwork, and almost no theory-cobbling reserve. Yes, it's hard work, yes, it's not as much fun, and it doesn't sell the books to the impatient, romantic, speculation-gobbling public. But just once in a while it would be nice to see authors who are serious when they use the words "science" and "evidence."

The Circle Makers By Andrew Collins. Lea-on-Sea, Essex: ABC Books, 1992. 351 pp.

Collins has been described as a former punk rocker, ufologist, anomalist, and psychic questor. Out of this background comes a unique answer to the circles problem. He dedicates the volume to Wilhelm Reich, which should prepare you. Yes, it is orgone energy that makes the circles. Meaden, Randles, and Fuller were close after all, but instead of plasmas the agents are orgone bioforms nesting on the ground. Well, why not? At least there's an index and a bibliography.

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JOURNAL OF SCIENTIFIC EXPLORATION

ISSN 0892-3310



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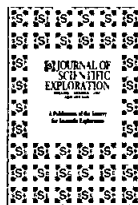
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